



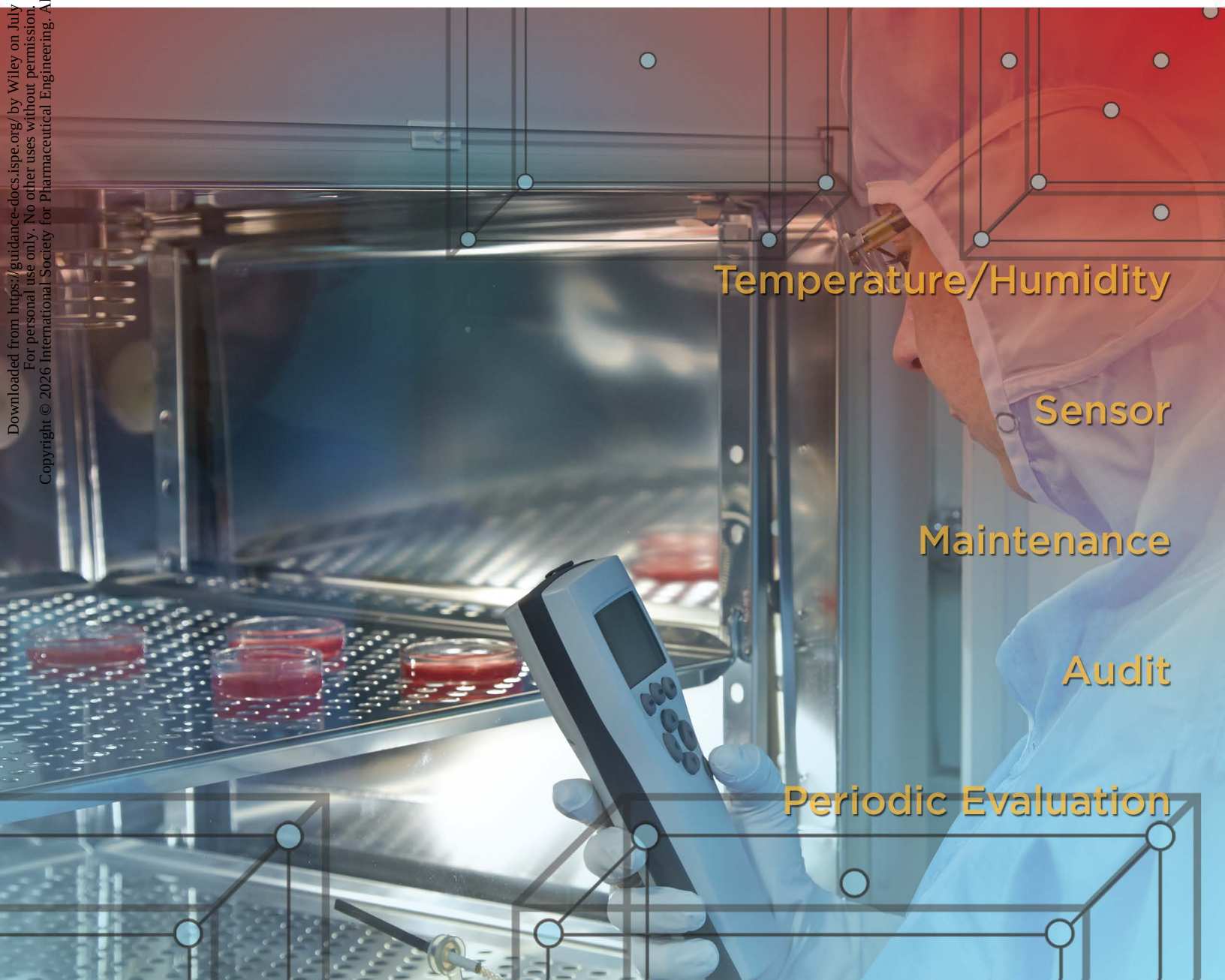
GOOD PRACTICE GUIDE:

Controlled Temperature Chambers

Commissioning and Qualification, Mapping and Monitoring

Second Edition

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Temperature/Humidity

Sensor

Maintenance

Audit

Periodic Evaluation



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Disclaimer:

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Preface

Global distribution of medicines requiring controlled temperatures has only increased in complexity since the first edition of this Guide was published. In order to meet changing industry practices, the second edition of this *ISPE Good Practice Guide: Controlled Temperature Chambers – Commissioning and Qualification, Mapping and Monitoring* was undertaken.

The primary purpose of this revision was to align with the *ISPE Baseline® Guide: Volume 5 – Commissioning and Qualification (Second Edition)*. Revising the Guide provided the opportunity to consider a periodic review frequency based on risk of failure to maintain uniform temperature and criticality of product stored rather than a standard, one-size-fits-all time span. In addition, the Guide was expanded to include ultra-low freezers, decommissioning a unit, and a discussion on using ocean reefers.

Together these improvements present a cost-effective way of demonstrating and maintaining compliance.

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Company affiliations are as of the final draft of the Guide.

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1 Introduction

A Controlled Temperature Chamber (CTC) is defined as a system, unit, equipment, or room in which the environmental conditions (typically temperature, sometimes temperature and humidity) of a chamber are controlled to meet specific user requirements.

ISPE Good Practice Guides are reviewed periodically to ensure alignment with industry best practices and other ISPE guidance documents. This *ISPE Good Practice Guide: Controlled Temperature Chamber – Commissioning and Qualification, Mapping and Monitoring* has been updated to align with the *ISPE Baseline® Guide: Volume 5 – Commissioning and Qualification (Second Edition)* [1].

The most significant changes are:

- Improvements to the commissioning and qualification activities resulting in more efficient processes
- Determining the periodic review frequency based on risk of failure to maintain uniform temperature and criticality of product stored

Additional revisions include:

- Expansion of freezer types to include ultra-low freezers and duration of storage needs
- Discussion on how materials of construction may impact cold rooms
- GxP storage facilities and warehouses descriptions
- Enhanced descriptions of monitoring systems
- High-level overview of decommissioning a unit
- Considerations for using ocean reefers (air-conditioned containers)

These revisions and other edits ensure that the Guide reflects current best practice and provides relevant guidance to industry, presenting a cost effective way of demonstrating and maintaining compliance.

Topics covered in this Guide include:

- Commissioning and Qualification
- Test Strategies
- Acceptance Criteria
- Qualification Approaches
- System Monitoring
- Periodic Evaluation
- Maintenance

- Decommissioning
- Packaged Transport Units

1.1 Purpose and Objectives

With the increasing complexity of global distribution for medicines requiring controlled temperatures, the *ISPE Good Practice Guide: Controlled Temperature Chamber – Commissioning and Qualification, Mapping and Monitoring* provides guidance on CTCs that operate under Good Manufacturing Practices (GMP). Topics include the definition of requirements (producing a user requirements document), design, purchase, commissioning (including temperature mapping), qualification, and maintenance of CTCs used to store and transport raw material, Active Pharmaceutical Ingredient (API), Work in Progress (WIP), and finished product. The approach presented is consistent with that described in the *ISPE Good Practice Guide: Cold Chain Management* [2].

1.2 Scope

This Guide provides guidance on CTCs operating under GMPs including:

- Purchased commercial off-the-shelf items, e.g., freezers, ultra-low freezers, refrigerators, refrigerated delivery vehicles, incubators, stability chambers, etc.
- Custom-built controlled temperature units, such as warehouses, walk-in cold rooms, walk-in freezers, etc.

This Guide does not include systems designed for heating/cooling of material such as blast freezers, although some of the concepts described may be useful when commissioning/qualifying these systems.

1.3 Benefits

There is an increased regulatory interest in cold chain driven by the growing number of products requiring controlled temperature shipping and storage, the complexity of the distribution network for the products, and governmental requirements for distribution of medicines.

This is reflected in the recent revisions to the related regulatory guidance (see Chapter 2). Utilizing this Guide provides a number of benefits in an area where there is little authoritative guidance, including:

- Provision of industry good practice for the temperature mapping of CTCs, considering the impact of load, and the use of the data to determine worst-case locations and number of monitoring sensors
- Development of test acceptance criteria
- A risk-based approach to practices for periodic review of system performance

1.4 Key Terms

Generally recognized temperature ranges are found in national codes such as USP <659> [3] and ICH Q1A [4].

Commissioning

A well planned, documented, and managed engineering approach to the start-up and turnover of facilities, systems, and equipment to the end user, that results in a safe and functional environment that meets established design requirements and stakeholder expectations. [1]

Design Authority

Architect, engineering company, specialist supplier

Mapping versus Monitoring Probes

Mapping probes are located in the storage locations to confirm that all locations remain in the defined limits during operations.

Monitoring probes are the independent means of verifying that the system remains operating within the qualified range.

Qualification

Per the *ISPE Baseline Guide: C&Q (Second Edition)* [1], a process to demonstrate and document that the critical manufacturing facilities, systems, utilities, and equipment are suitable for the intended purpose.

Rounding of Temperature Values

This Guide follows USP General Notices Section 7.20 Rounding Rules [5] when considering temperature values:

“When rounding is required, consider only one digit in the decimal place to the right of the last place in the limit expression. If this digit is smaller than 5, it is eliminated and the preceding digit is unchanged. If this digit is equal to or greater than 5, it is eliminated and the preceding digit is increased by 1.”

Temperature Mapping

Temperature mapping is the process of measuring and documenting temperatures at multiple predefined locations within the storage location over a specified study duration to verify the capability of a controlled temperature chamber to maintain predetermined temperature specifications.

2 Regulatory and Industry Guidance and Standards

Regulatory guidance covering CTCs include:

- Health Products Regulatory Authority (HPRA)
 - Guide to Good Distribution Practice of Medicinal Products for Human Use [6]
 - Guide to Control and Monitoring of Storage and Transportation Temperature Conditions for Medicinal Products and Active Substances [7]
- European Medicines Agency (EMA), Guidelines of 5 November 2013 on Good Distribution practice of medicinal products for human use [8]
- Health Canada, Guidelines for environmental control of drug products during storage and transportation [9]

Information on calibration and traceability is available from national standards organizations.

Additional guidance includes:

- PIC/S Guide PE 011-1 to Good Distribution Practice for Medicinal Products [10]
- WHO:
 - WHO Technical Report Series No. 961 [11]
 - > Annex 9: Model guidance for the storage and transport of time and temperature-sensitive pharmaceutical products
 - > Annex 5: Technical supplements to Model guidance for the storage and transport of time and temperature-sensitive pharmaceutical products (WHO Technical Report Series, No. 961, 2011), Annex 9
 - > WHO Technical Supplement, Calibration of temperature control and monitoring devices
 - > WHO Technical Supplement, Qualification of temperature-controlled road vehicles
 - > WHO Technical Supplement, Qualification of temperature-controlled storage areas
 - > WHO Technical Supplement, Temperature and humidity monitoring systems for transport operations
 - WHO, How to Temperature Map Cold Chain Equipment and Storage Areas WHO/IVB/19.09 [12]
 - WHO Evidence Brief, Cold room temperature mapping studies [13]
- United States Pharmacopeia (USP):
 - <659> Packaging and Storage Requirements [3]
 - <1079> Risks and Mitigation Strategies for the Storage and Transportation of Finished Drug Products [14]
 - <1118> Monitoring Devices – Time, Temperature and Humidity [15]

- ISO 17025 – General requirements for the competence of testing and calibration laboratories [16]
- Taylor, J.L., Recommendations on the Control and Monitoring of Storage and Transportation Temperatures of Medicinal Products [17]
- Seevers, R. et al., The use of mean kinetic temperature (mkt) in the handling, storage and distribution of temperature sensitive pharmaceuticals [18]

Regulatory guidance for Good Distribution Practice (GDP)¹ is updated on a regular basis. There is normally a requirement to demonstrate that product is being stored in the appropriate conditions. This can be accomplished through mapping the parameters in the storage areas to confirm that all storage locations used in the CTC remain within the defined conditions for these parameters.

There are no standards written specifically for CTCs as used in the pharmaceutical and biotechnology industry. Available standards are written for environmental test chambers and stability chambers. Although the same principles may be applied, the details on how the systems are to be used in an operational environment should also be considered.

A list of the applicable standards known at the time of publication is provided below for reference:

Japan

- JTM K 01 Temperature and Humidity Chambers – Test and Indication Method for Performance [19]
- JTM K 03 Environmental Temperature and Humidity Rooms – Test and Indication Method for Performance [20]
- JTM K 05 High Temperature Chambers – Test and Indication Method for Performance [21]

France

- FD X15-140 Measurement of Air Moisture – Climatic and Thermostatic Chambers – Characterization and Verification [22]

Germany

- DIN 12880 Electrical Laboratory Devices – Heating Ovens and Incubators [23]

IEC

- IEC 60068 Environmental Testing Series [24]

CTCs used in the pharmaceutical and biotechnology industry are usually used for long-term storage (for testing, long-term environmental studies or stability) or for short-term Work in Progress (WIP)/distribution warehousing.

¹ China (PRC) and South Korea use Good Supply Practice (GSP) rather than GDP.

3 User Requirements Specification

A User Requirements Specification (URS) defines the user's functional requirements for equipment or systems. By specifying requirements in the URS but not how those requirements are met, the designer is given freedom to provide the optimum design solution to meet the requirements.

Aspects to consider in relation to stored material include:

- Unit capacity, i.e., quantity, nature of the material to be stored, and airflow within chamber
- Whether the unit is providing cooling or heating (i.e., preconditioning) of the stored material, and if so:
 - What criteria need to be met, e.g., maximum load size, specific heat capacity supply, temperature, heating/cooling time?
 - How long and how frequently is the door to load/unload the unit open?
 - Which environmental conditions are critical, e.g., light/temperature/humidity?
 - What is the condition the loading door is opening into, e.g., external conditions or conditioned space?

Qualification is intended to demonstrate through testing (verification) that the requirements specified in the URS have been met. Confirmation that the user requirements have been met should be traceable in the qualification documentation.

Appendix 1 provides a discussion of typical URS content and presents an example of a URS for a walk-in unit.

Note: Information included in specifications or the purchase order general requirements (such as manuals, training, spare parts) is not included in the URS to avoid duplication. The URS should be a simple concise definition of the user requirements supported by relevant company specifications and standards.

As stated in the *ISPE Baseline Guide: C&Q (Second Edition)* [1]:

"A statement of the URS may be contained in various forms, e.g., purchase request, functional specification, change request, or data sheet. For standard off-the-shelf or simple catalog systems, a statement of the URS may be a purchase order, vendor cut sheet, or catalog data."

It should be noted that if a purchase order or catalog cut sheet is used then compliance with company standards/specifications should be confirmed or deviations justified.

4 Types of Controlled Temperature Chambers

The equipment considered in this Guide is listed below and is not all inclusive:

- Refrigerators
- Freezers
- Ultra-low freezers
- Liquid nitrogen freezers
- Cold rooms
- Stability chambers
- Incubators
- Heated chambers

Transport reefer containers are discussed in Section 10.2.

Note: Generally recognized temperature ranges are found in national codes such as USP <659> [3] and ICH Q1A [4].

4.1 Freezers

Freezers are used, but not limited to, for the preservation of materials:

- Active Pharmaceutical Ingredients (APIs)
- Process intermediates
- Stability samples
- Laboratory materials
- Cell banks
- Vaccines
- Cell and gene materials

Freezers ranging from -25°C to -10°C [3] and ultra-low freezers with set points ranging from -20°C to -90°C are used. Lower values down to -140°C are required for cryopreservation.

Temperature uniformity in freezers is maintained within a qualified range to ensure product quality is preserved. Peak variation from set point could range from -7°C to +4°C but can vary from model to model.

Door opening recovery temperatures could be in the range of 10 to 30 minutes and are influenced by the provision of one or more additional inner doors, and physical configuration such as a chest freezer versus a front opening freezer. In general, the greater the difference between ambient temperature and the internal specified temperature, the longer the recovery time.

Peak temperature variation from set point, the door opening recovery temperature time, and temperature recovery time after a power failure should be defined in the user requirements.

Operation can be controlled via a temperature controller that maintains the locally configured temperature set point and provides a local audible/visual warning for out of range temperature excursions and power failures.

Chambers are usually equipped with the general freezer refrigeration components of:

- Compressor
- Receiver
- Condenser
- Evaporator
- Tubing
- Insulation
- Fan
- Control panel
- Structural support

The refrigerant compressors are usually hermetically sealed for integrity. The evaporator is typically a tube and fin design. The chamber control monitors interior temperature using a permanently installed thermocouple or Resistance Temperature Detector (RTD) (also known as a control probe). Typically, the RTD is adjusted at the factory based on the temperature at the geometric center of the chamber.

Temperature calibration requirements are specified in the user requirements so that the most suitable temperature detectors are selected.

The evaporator is usually located within the upper portion of the chamber interior and chamber temperature uniformity is obtained through natural (by convection or "passive") air circulation. There are several different designs available that include positive forced-air circulation to keep temperature uniformity stable in space and in time.

For units with an automatic defrost, the control actuates an automatic defrost cycle at a predetermined frequency (typically 6 to 72 hours) to minimize ice accumulation, or by activation of electrical heaters (in which case the evaporator fans are turned down).

The evaporator defrost generally occurs by redirecting the hot refrigerant gas into the evaporator coil for a short time to melt ice/frost accumulation. Depending on the manufacturer and model, the defrost cycle is terminated by:

- Elapsed time
- Evaporator coil temperature set point

- A combination of time and temperature

The timed defrost is adjustable to allow for the conditions of use (frequency of door opening, length of time for door opening, and exterior air humidity level at time of door opening), with the preference to have the defrost cycle occur outside of normal working hours. The defrost cycle can usually be adjusted.

Models with manual defrost are also available. The temperature uniformity in time offered by the two systems could be different.

The defrost cycle can also be initiated by a low suction pressure/super heating of the compressor.

These aspects relevant to defrosting apply also to equipment such as refrigerators and freezers.

For GMP applications these systems should be monitored. It can be difficult to calibrate the system control sensor, and the use of an independent system allows routine calibration and the flexibility to place the sensor in a location that measures the worst-case temperature. Location is determined from test results (see Section 5.4).

Remote connections could also be wireless.

Backup cooling systems using liquid nitrogen or carbon dioxide (CO₂) are available and could be included, if necessary, to increase the reliability of operation and mitigate temperature excursions during power failures.

4.1.1 Conditioning/Blast Freezers

Facilities may process large quantities of bulk product that need to be frozen from ambient conditions to a specified temperature, e.g., -70°C, prior to transfer into a -70°C storage unit. Dedicated conditioning freezers are needed to perform this function, as storage freezers are not capable of handling the large volume of product and significant temperature disparity without potentially compromising the temperature of the stored material. This type of equipment is outside the scope of this Guide.

4.1.2 Liquid Nitrogen Freezers

Liquid nitrogen freezers are used for storing material below -120°C, for example, cell banks. If materials are immersed into liquid nitrogen, they eventually assume the temperature of the liquid (-196°C). Materials stored in the vapor above the liquid become warmer as they are moved away from the liquid (up to -120°C).

It is thus important to define the purpose of the equipment. In the case of cryopreservation of animal cells or tissues, the required temperature will be lower than the glass transition temperature of water, that is, -130°C. The temperature of the vapor can be changed by varying the amount of liquid nitrogen in the chamber.

4.2 Refrigerators

Refrigerators are typically used to store material within a qualified range of 2°C to 8°C [3, 4]. The capability of the equipment to achieve temperature uniformity should be considered when assigning the chamber set point.

Operation is typically controlled via a temperature controller that is used to maintain the locally configured temperature set point, and may also provide a local alarm. The suggested set point is in the midrange at about +5°C to reduce the possibility of the refrigerator going out of range.

Refrigerators may be equipped with fans that cool via forced-air circulation and utilize a hermetically sealed compressor and refrigeration system. Alternatively, refrigerators may use a heat exchanger located in the storage space, relying on convection to maintain an even temperature. Refrigerators should have wire shelving or provisions to allow air circulation.

Refrigerators will usually automatically defrost the evaporator to maintain system performance. Evaporator defrost may be performed by redirecting the hot refrigerant gas into the evaporator coil for a short time to melt any ice/frost accumulation or by using electrical heaters.

4.3 Cold Rooms

Cold rooms are typically used for the storage of material within a qualified range of 2°C to 8°C [3]. Cold rooms should be equipped with a dedicated refrigeration compressor/condensing unit. The air-cooled condensing unit is usually located outside, for example, on the roof of the building. Building safety construction codes provide instructions and limitations when chillers are located indoors in mechanical rooms. Consider specifying redundant components (condensing units, compressors, etc.). Cold rooms may be provided with redundant condensing systems, each capable of handling the total system load.

There is a risk that water condensation may form on internal surfaces from the effect of air entering when the door is open. The level of risk is influenced by the amplitude of absolute humidity difference across the door and therefore, by the thermo-hygrometric conditions of the environment used to access to the cold room. In this case, the cold room could have, for example, a desiccant dehumidifier (usually based on a silica gel absorption process), or a low Relative Humidity (RH) airlock at the entry.

Walk-in cold rooms may be prefabricated, metal-clad structures with an insulated floor, walls, door, and roof panels, furnished and installed as self-contained units. These may include all essential refrigeration equipment, controls, air circulation, and other equipment necessary to reach the specified environmental conditions.

The interior of the system should be constructed of materials that are easily wiped down, will not shed, and are resistant to the cleaning chemicals used.

Depending how low the required temperature is inside the cold room and its location in the building (near exterior walls, slab on grade, adjacent to a walk-in freezer or walk-in incubator), the floor could range from an insulated concrete slab with urethane cement-based coating, with no slip, no porous properties to a floor with an insulation panel like the other five sides and a vapor barrier. The flooring could be required also to withstand loads of pallet traffic and portable cages.

The total thickness of the cold room floor could be an issue when revamping existing plants since access ramps may be needed. To mitigate this, solutions are available based on the use of high-density microporous insulation sealed under vacuum with optimized thermal performance, thus reducing the thickness of the floor required.

The water permeability of the floor and risk of frost heave under the floor can be solved by proper ventilation with heat using a low voltage electrical mat. A better option is to have a thin ventilation cavity (empty space) by PVC profiles or equivalent open to the cold room sides.

For negative temperature applications, to prevent the door seals from freezing due to air leaks from the external side, consider electrical tracing or equivalent methods.

Cold room control systems usually consist of a local Programmable Logic Controller (PLC) to monitor and control the temperature within the cold room. Within the main chamber, the temperature may be measured by temperature sensors placed in different locations. One sensor should be routed to the main controller for controlling interior temperature. The second sensor should be connected to a monitoring system for temperature trending and alarms; for larger units there may be multiple monitoring sensors.

Redundancy of temperature devices may be considered.

4.4 Stability Chambers, Incubators/Incubator Shakers

Stability chambers are designed to maintain a stable environment for incubating materials or holding samples by controlling air temperature near to ambient or warmer than ambient conditions [3, 4]. These may include cultures for microorganisms or cell culture and hybridization in production or quality control laboratories. The incubator may accommodate cell culture vessels from small tissue culture flasks to roller bottles and small spinner flasks.

A local controller should control system operation for process parameters such as temperature, humidity, and CO₂. Incubator shakers can also control shaker speed.

The technology adopted to obtain the heat transfer (refrigeration cycle, Peltier effect) will influence the temperature mapping criteria to adopt.

4.5 GxP Storage Facilities and Warehouses

This section refers to areas that may store API, bulk drug substance, bulk drug intermediate, bulk finished goods, combination products, packaged finished goods, and raw materials with temperature and/or humidity monitoring and control requirements. Long-term storage areas are where products or materials are stored continuously for 72 hours or longer.

Temperature and/or humidity mapping is performed to ensure that the facility can maintain the required temperature and humidity for the products/material being stored. Seasonal mapping is required for facilities that experience summer (hot) and winter (cold) temperatures affecting the room conditions or the operation of compressors/condensers if located outdoors.

For internal facilities that do not share walls or windows with external environments, seasonal mapping is not needed.

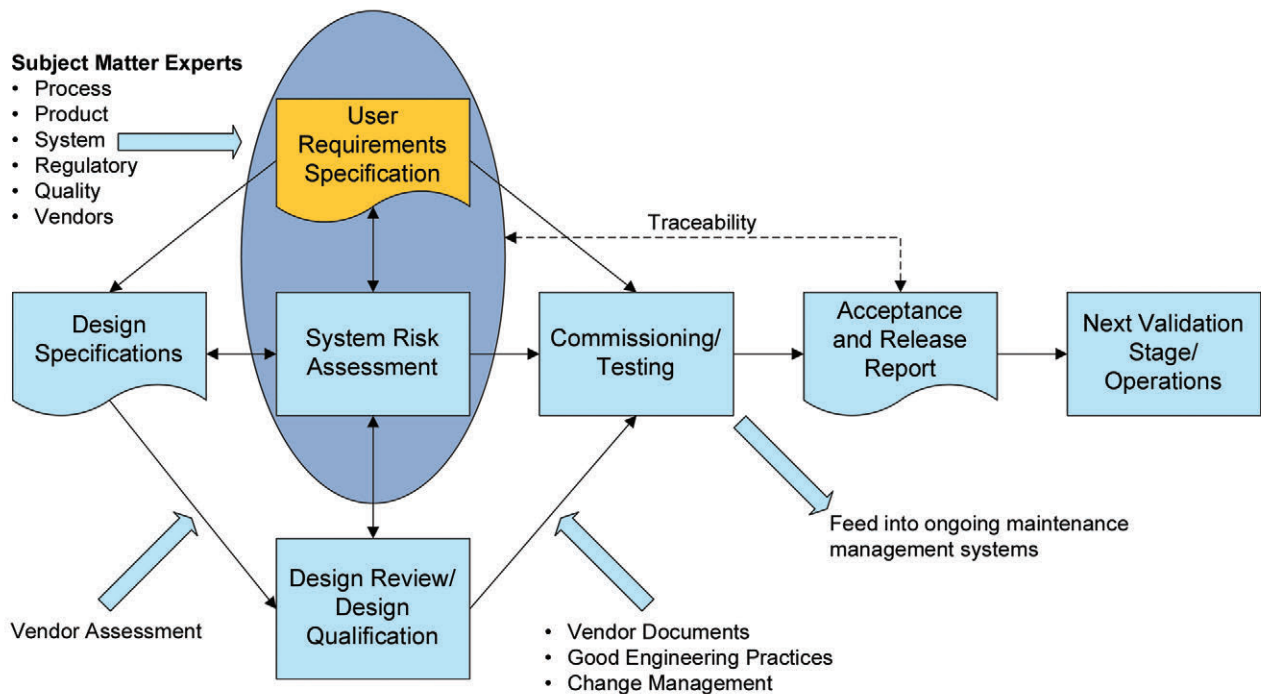
Facilities located in areas where the range of external temperatures is limited and the conditioning system provides only heating or cooling (so the air distribution pattern is consistent), mapping of only one season is necessary.

5 Commissioning and Qualification

This Guide describes a commissioning and qualification process aligned with the *ISPE Baseline Guide: C&Q (Second Edition)* [1]. This process is intended to be efficient, compliant, and ensures that the quality risks and risk controls associated with a system are understood. The process described (see Figure 5.1) is scalable to suit the complexity of the project and system.

Figure 5.1: Relational Overview of the Process [1]

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5.1 Commissioning Planning

The testing strategy and the acceptance criteria should be documented in a change control record, commissioning plan, or test strategy document prior to test execution. The objectives are to:

- Ensure consistency of approach and testing across the project
- Define the scope of the verification to be used to provide documented evidence that the system is installed and operates to meet specifications
- Provide clear and early definition of deliverables and handover requirements
- Ensure verification testing incorporates the Critical Design Elements (CDEs)/Critical Aspects (CAs) identified in the System Risk Assessment (SRA)
- Define roles and responsibilities
- Define safety considerations

- Describe the process to be used for the review and approval of test method statements
- Describe the process to be used to address test discrepancies
- Define the Engineering Change Management (ECM) system
- Define the acceptance criteria/expected results*
- Ensure full transparency on progress metrics (test execution and productivity)
- Define the progress reporting approach
- Define the training approach for maintenance users and end users
- Define witnessing requirements and associated test notification procedures
- Describe the process for handling and retaining raw data
- Describe the format and content of the commissioning report (e.g., is a particular documentation practice required?)
- Define responsibilities for operating and maintaining a facility during commissioning, until it is handed over/ released to the end user
- Identify who is responsible for maintaining the system and operating log from start-up to hand over/release to the end user
- Define how the system is handed over/released to the end user

* Acceptance criteria are “non-negotiable” test results. Expected results may be acceptable across a broader range, or are results taken for reference purposes.

CTCs typically use commercially available controllers, which are usually commissioned. The qualified independent monitoring system confirms the correct function of the system.

5.1.1 Roles and Responsibilities

Company structure, project size, and contract philosophy define the roles and responsibilities. It is important for a successful project that roles and responsibilities are discussed and defined with the final decisions documented and published to the team, usually through documents such as the commissioning or qualification plan.

Typical roles and the responsibilities involved in a small project are:

Construction Manager

The construction manager is responsible for delivering the project (which may be or include a CTC) usually to the stage of mechanical completion, meeting specifications and established quality standards. The scope may also include commissioning and qualification.

Commissioning Lead

The commissioning lead is responsible for coordinating and managing commissioning activities and allocating resources. This role takes ownership of the activity and ensures the supporting documentation is developed and completed to defined documentation standards to meet the schedule. The commissioning lead is responsible for the accuracy of the content. The commissioning lead is usually responsible for communication and coordination between functions, monitoring the walk down and punch list as well as the commissioning and qualification process.

System Owner

The system owner is responsible for the operation and maintenance of the system after commissioning and qualification. This role reviews and approves the commissioning and qualification plan, the verification documentation, qualification documentation, and associated summary reports. The system owner has decision-making authority and is accountable for compliance to the URS. During the design phase, the system owner should have oversight of the design process, and during construction they should determine the acceptance of the “as-built” system at mechanical completion, usually with the support of a technical lead.

The system owner may be a site representative, a member of the project team, or the role may be delegated to a Subject Matter Expert (SME).

Commissioning Staff

The commissioning staff are involved in the verification activities (execution, documentation, and expertise). Commissioning staff may be company employees or contractors. The work can be divided into two categories:

- Site activities: Verification of the completion of the systems presented by the construction team or manufacturer, and perform job safety analysis, installation and operational verification, submission of punch items, and training of the maintenance team
- Office activities: Preparation of the commissioning test documents, update construction documents as redlined during the commissioning phase, communicate executed protocols to the team, perform archiving

Owner Technical Services (SME) and/or Validation group

Technical support is provided to the system owner where necessary to ensure company specifications and standards are met, the design is robust, the installation is completed following specifications, and the technical content of the installation and operational verification are appropriate.

Validation Group

Although the role varies depending on the company organization, typically the validation group acts as a representative of Quality, ensuring compliance with company procedures, documentation standards, etc.

Training

This role trains the staff designated for specific roles who do not have the appropriate experience or training, based on the role requirements or specialist subcontractors used. Training may also be required for personnel conducting specific commissioning tests to ensure that the tests are carried out correctly and safely.

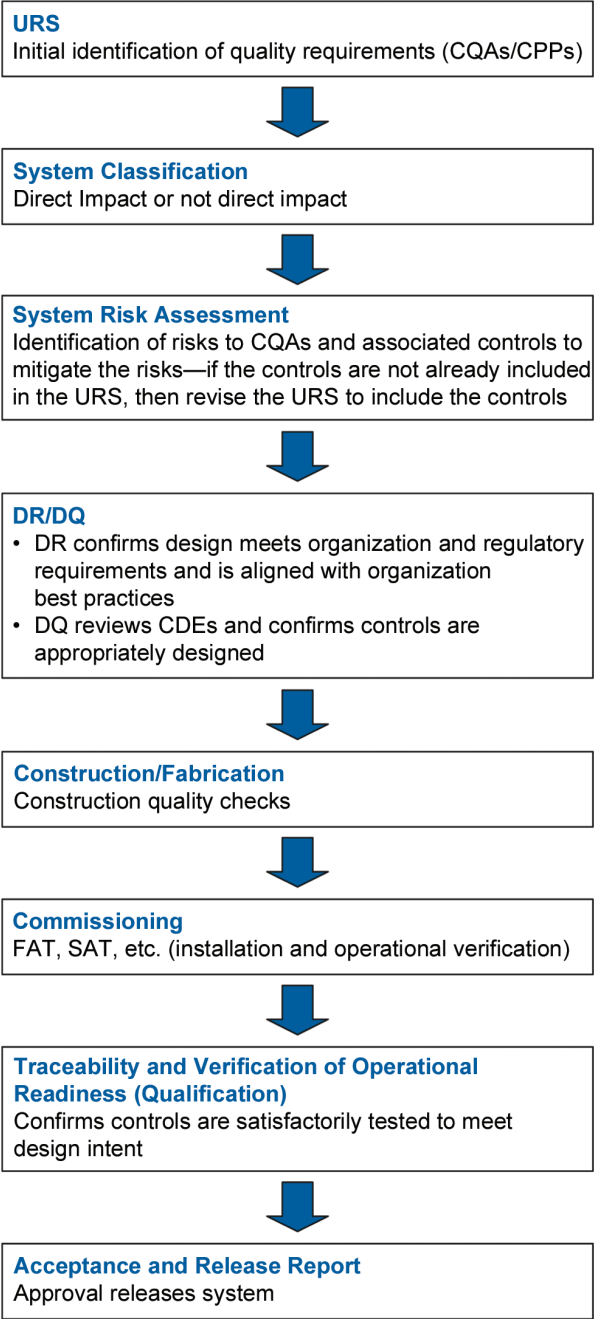
5.1.2 Process Overview

A diagram showing the process workflow is shown in Figure 5.2.

Notes:

- The construction/fabrication stage is not required for a standard off-the-shelf or catalog system, such as a freezer.
- The diagram is simplified and does not include ECM.

Figure 5.2: Process Workflow [1]



Improvements suggested in the *ISPE Baseline Guide: C&Q (Second Edition)* [1] have been incorporated into the process workflow descriptions in this Guide.

For a project the system boundaries should be defined, so that the scope of work for commissioning and qualification can be defined, together with the relationship of associated systems, such as independent monitoring systems, smoke detection, and fire protection systems.

5.1.2.1 User Requirement Specifications

The user must define their requirements (see Chapter 3). Typically, this requires collaboration of the user and the Quality Unit to produce a concise document that contains the requirements but does not repeat specification or contractual details.

The requirements may be defined in a URS – see the example in Appendix 1. The URS should be supported by relevant company specifications and standards.

For simpler systems a purchase order or catalog cut sheet may be used provided the quality critical aspects are defined with the associated acceptance criteria. When using a purchase order or catalog cut sheet, compliance with company standards/specifications should be confirmed or deviations accepted.

5.1.2.2 System Classification

Using the same terminology as the *ISPE Baseline Guide: C&Q (Second Edition)* [1], systems are classified as:

- Direct impact systems – are commissioned and qualified
- Not direct impact systems – are commissioned

The process descriptions in this CTC Guide are direct impact systems. How to determine the classification of a system is discussed in the *ISPE Baseline Guide: C&Q (Second Edition)* [1].

The same principles can be applied to not direct impact systems up to and including commissioning, except for Design Qualification (DQ). Usually a commissioning summary report is produced closing out the activities, confirming that the installation and operation meets the specifications, and closing out any engineering changes.

With an experienced project team, systems are often informally classified in order to help define the documentation strategy, and the classification formalized in a commissioning and qualification strategy and/or plan.

5.1.2.3 System Risk Assessment

For walk-in cold rooms, freezer systems, and warehouses, it is worth developing the SRA. Note that the independent monitoring system is typically qualified separately because it is not the same system as the CTC system. For simpler systems, such as off-the-shelf or catalog refrigerators and freezers, it may be appropriate to use a standardized approach.

The timing for the initial risk assessment can be defined by the project team. The assessment is used as an engineering tool until the final revision is completed based on the Issue for Construction (IFC) design. Then the assessment and summary report becomes a GMP document, and managed as such.

Companies may have their own risk assessment approach. The SRA process described below is the approach proposed in the *ISPE Baseline Guide: C&Q (Second Edition)* [1].

- “A small team develops an initial baseline draft that includes development of the process description and

- *identification of design and procedural controls*
- *A broader team reviews this initial baseline draft, making any necessary revisions to ensure the information is correct and no risks or controls have been omitted, and assigns the system risk profile*
- *As a general guide, the initial session by the small team and the review session by the broader team should not exceed more than a few hours each if the preparation work is complete and teams with the correct level of expertise are used.*

The initial draft can be completed at the time the project deems appropriate and is a Good Engineering Practice (GEP) document that can be used to assist design development. It becomes a GMP document when it is revised to reflect the final approved design.”

“Developing the Baseline Draft

The small team that develops the initial baseline draft should consist of the following:

- *Experienced facilitator to guide the process*
- *Quality Unit representative*
- *Process/system SME (may be one SME or could additionally include a system vendor representative)”*

“Performing the System Risk Assessment

After the initial baseline is completed, a broader team reviews the document and determines the risk profile. This review team generally consists of representatives from a broader group with different experiences to provide a balanced perspective; this team could include:

- *Operations (Production)*
- *Maintenance*
- *Engineering*
- *Process development*
- *Vendor representatives*

Note: *The time taken for the review generally increases with the number of people in the team, thus maintaining a balance between experience and the number of team members is important. Prior to determining the risk profile, the team identifies the sources of the risks (potential failure modes) along with design aspects and/or procedural controls that would mitigate or eliminate the risk. Risk profiles are assigned per definitions shown in Table [5.1].”*

Table 5.1: System Risk Level Definitions [1]

Risk Level	Definition
Low	Considered acceptable with no actions needed. The likelihood of the risk occurring with the defined controls operating is considered low, and the detection is robust.
Medium	Normally considered unacceptable and requires mitigation through design (CAs/CDEs) and/or procedural controls. However, the SME reviewers may consider a medium risk as acceptable for the specific process/system and decide to accept the risk.
High	Considered unacceptable and requires mitigation through design and/or procedural controls. In certain situations, the SME reviewers may determine that these controls are not appropriate, e.g., implementation costs outweigh the incurred costs if the risk occurred, operational challenges, or other reasons. The SMEs may recommend acceptance of the system with the high risk; in this case, the assessment needs to be accepted by high-level management.

Note: The risk level is defined considering the overall process; thus, if the process is changed, the System Risk Assessments may need to be reviewed/updated. This aspect should be addressed through change management.”

The *ISPE Baseline Guide: C&Q (Second Edition)* [1] adds:

“Determination of the risk profile should also be based on the SME’s experience with the process/system. While defining the risk of the failure or hazard occurring, the SME should consider that the controls listed are in place and operating as intended.”

The procedural and design controls (CAs and/or CDEs) defined during the SRA are considered part of the system requirements and should be included in the URS. If a URS is not being used, they can be added to the C&Q plan, or, if a traceability matrix is used as a tool to manage the C&Q execution process, the CDEs could be added to the traceability matrix only.

Use of the System Risk Assessment

The strategy using the SRA should be defined and incorporated into the relevant document, for example, change control or C&Q plan.

There are many ways that the SRA can be used, such as determining the lowest risk system as part of bid analysis, or as a tool to assist during design development, to ensure that the final design has an acceptable risk profile.

For standard systems there may be no value in completing a draft early in a project, apart from early use of available resources.

A typical process for a complex system is:

- Complete the initial SRA

As the design progresses the SRA can be reviewed and revised to suit the design, to ensure that it has an acceptable risk profile, and the design review process started.

Once the design is complete:

- Complete the design review/DQ

Completing the SRA ensures that the team understands how the system operates knowing the associated risks and risk controls (CDEs). The SRA ensures that the proposed system design has a suitable risk profile.

Completion of the alarm table using the data from the assessment helps define the system critical alarms/instruments. Note that the critical alarms typically come from the independent monitoring system.

A model risk assessment is described in Appendix 5.

5.1.2.4 Design Review/Design Qualification

Design Review

Based on the documents provided (for example, the URS and company standards/specifications), the design authority/supplier proposes a design or an off-the-shelf or catalog system that meets the requirements. The proposed design is reviewed by company SMEs to:

- Ensure that the user requirements and company design standards and specifications are understood and incorporated into the design documents
- Determine compliance to quality, business, health, safety and environmental, national and local codes, operations, maintenance, project, and technical requirements
- Incorporate lessons learned from previous projects into the proposed design to ensure that it is robust and will perform reliably, and that the performance can be documented
- Confirm appropriate integration with available utilities
- Ensure that CAs are met
- Ensure all stakeholders have the opportunity to review the design
- Ensure that preferred suppliers have been specified if applicable
- Ensure that the design proposed is cost effective, i.e., that design options, lifecycle costs, and risks have been identified and addressed

The process is typically iterative, with the reviews conducted as the design is developed from concept to detailed design. It is easier and more cost effective to change a design before it is constructed than during or after construction, and therefore, there is less impact on capital cost and schedule. As the design develops, input from all interested parties should be considered to avoid later changes.

For systems that may affect product quality, it is a recommended practice to split design reviews into two categories:

1. Engineering or technical reviews
2. Quality (GMP) focused reviews (DQ)

It is recommended that the review considers both aspects to improve understanding of the design for all parties.

To facilitate the process companies may develop and utilize knowledge capture systems such as:

- Design review checklists
- Design guides that define the preferred way of designing a given system type

These approaches should ensure that organizational experience and preferences are both captured and considered. They should be written so that they do not preclude novel concepts.

The review method used should be defined and agreed upon. The method and participants may vary depending on the system type. For complex systems such as a warehouse and the associated HVAC system, the use of a multidisciplinary team to ensure that key viewpoints are considered is recommended. For simpler systems such as an off-the-shelf or catalog freezer, a single review with the appropriate SMEs may be sufficient.

Actionable items from the design review should be implemented through drawing and specification changes. The means of tracking these should be defined, such as by an action items log or assigning the relevant SME with responsibility for tracking and confirming that the items have been addressed. This should be a formal process for a direct impact system.

Note: During the design review phase documents relevant to the areas surrounding the chamber and to the utilities serving it (e.g., hot water, chilled water, glycol (heat transfer fluids), electricity, floor loading limits based on fully loaded CTC with fork lift, the impact on HVAC and air balancing for the surrounding area) could be included in the review in order to properly understand the interaction between the CTC and the rest of the facility. This will help to ensure that the overall design is robust. From the commissioning perspective, the design review also allows the commissioning team to become familiar with the project and to verify that the design has adequate provision for the tests required, as defined in the equipment specification and URS.

Design Qualification

For direct impact systems, at a specific point in the design process (usually at the end of detailed design) DQ is performed to verify that the project design to date is fit for use. As with design review, DQ should focus on two main areas:

- GEP
 - Does the design satisfy user preferences as expressed in the user requirements?
 - Can the design be constructed, commissioned, operated, and maintained?
- GMP
 - Does the design meet the quality CAs/CDEs as expressed in the user requirements?
 - Is the design aligned with the SRA?
 - Does the design meet regulatory expectations?

This scope allows the conclusion that the design is fit for its intended use. This may be a very simple statement for an off-the-shelf or catalog type system, confirming that it addresses the quality CAs/CDEs. The GEP aspects are typically addressed through the design review process.

It should be noted that regulatory guidances expect a design review/DQ. ICH Q7 [25] suggests a design review and the EU Annex 15 [26] and PIC/S Annex 15 [27] state:

“Design qualification (DQ)

The next element in the qualification of equipment, facilities, utilities, or systems is DQ where the compliance of the design with GMP should be demonstrated and documented. The requirements of the user requirements specification should be verified during the design qualification.”

5.1.2.5 Construction Documents and Construction Support

After the detailed design is finished, the design team should complete bid documentation, resolve construction questions, and perform on-site construction reviews. Engineers also may be involved in activities related to the receipt and installation of equipment and systems to verify that they were delivered and installed in a manner consistent with the design. This approach of integrating installation verification with construction is efficient, although it requires robust ECM to ensure that at the end of construction the documentation accurately reflects the installation.

System Risk Assessment Completion

Once the design is finalized and has been accepted (i.e., completed design review/DQ), the SRA can be finalized, as the design controls are defined. The risk assessment and summary report are typically approved by the system owner or SME representative and the Quality Unit.

Engineering Change Management

It is critical to ensure that:

- Any proposed changes to the approved (design review complete, DQ approved) design are acceptable to the design authority (as they are usually responsible for code compliance)
- The documentation at the completion of commissioning accurately represents the installation, and operation of the system.

In order to do this, an ECM system is required to ensure all proposed changes are considered and accepted where appropriate. It should cover the project from the approval of the design until qualification is complete, when the site change control process takes over:

- Site ECM for commissioned systems
- Site quality change control system for qualified systems

5.1.3 Commissioning

Note: This section considers the CTC. For guidance on commissioning and qualifying the independent monitoring system, refer to the appropriate ISPE GAMP guides [28].

The science and risk-based approach to commissioning and qualification uses the engineering documents to provide the documented evidence that the system is installed and operates to meet the design specification (i.e., is commissioned). The format and naming of these varies depending on company practice. The terminology used in this Guide is:

Installation Verification: Documentation of installation verification includes a record of the major components, confirmation that the system has been installed per manufacturer's recommendations, calibration has been conducted as required by the specifications, and confirmation that the specified utilities have been correctly connected.

Operational Verification: Documentation of operational verification provides evidence that the system operating temperature ranges (Critical Quality Attributes (CQAs)) meet the specifications during simulated operations.

As well as the temperature mapping and testing described below, the normal functions of the system automation should be verified, for example:

- Power failure recovery

- Refrigeration system failure operates the standby refrigeration system
- Engineering monitoring system alarm remote notification systems operate

Note that the operational verification of the independent monitoring system and the associated functions are covered in the documentation associated with that system.

5.1.3.1 Challenging the Area with Operational Activities

The area being mapped should be challenged with specific known operational activities during the thermal mapping process. These activities should simulate normal processes and functions that occur during routine operations, such as opening interior doors, opening exterior doors (receiving dock doors, shipping dock doors), running equipment (e.g., thermoformers, stretch wrappers, forklifts, conveyors), where there is a potential impact to the conditions. By performing these functions and operating equipment during the execution of the mapping studies, cause and effect relationships can be identified for the thermal profile.

- Quality-related alarms should be tested and captured in the calibration program, to ensure that the alarm function is tested on a periodic basis
- Perform open-door testing where applicable (simulating operations)

Open-Door Recovery Test

An example of a typical operational activity is to examine recovery time for the internal chamber temperature after opening the access door and simulating loading/unloading. It is recommended to determine the time that the chamber door may remain fully open before the monitoring sensors show that the conditions are no longer within the acceptable range. This test can also help to determine how quickly the temperatures recover and return to the normal operating temperature range. For an open-door recovery test for a new CTC, it is recommended to:

- Use a minimum (simulated) load with minimum thermal mass or no load in the storage unit (i.e., no thermal mass stored is the worst case for this test, as adding thermal mass slows the rate of temperature rise)
- Locate temperature sensors as recommended for thermal mapping (with special attention to the sensors located in proximity to the storage locations nearest the door being tested)

It is recommended to set the data acquisition rate to < 1 minute intervals on the mapping system for this test, because of uncertainty in the time that the test may take. Generally, there are no specific acceptance criteria for this test unless a recovery time is specified in the unit manufacturer's specifications. An open-door recovery test should be performed as an integral part of the mapping study to enable acquisition of the mapping data showing the potential impact of the loading/unloading activities.

Note: For larger walk-in CTCs fitted with vinyl curtains or an airlock, the units may be able to maintain the internal conditions (temperatures) indefinitely, as the impact of passing through the curtains is short term and has a negligible impact on the temperature. Vinyl curtains should be well-fitted, rest on the unit walls, and sit just clear of the floor.

- Temperature mapping should be conducted on an empty chamber for no less than 24 hours
- Where there are duty/standby systems, both should be tested
- Temperature study should be conducted on a "loaded" chamber for no less than 24 hours
- Mapping sensors should be located as defined in the test strategy document
- Test records should include a diagram showing the mapping sensor placement

- Perform power loss temperature rise/recovery testing where applicable (an engineering test)

Power Outage/Thermal Degradation Test (Recovery from Power Failure or Power Interruption)

Another type of test that may be helpful from a business continuity, disaster recovery, and operational perspective for a refrigerated unit is a simulated power outage test. This test measures the amount of time the CTC remained within the desired temperature range once electrical power is turned off.

Note: Although it is expected the CTC will have redundant and backup power supplies, the results from this test provide data to define how quickly materials need to be moved to another location if the backup power systems failed, or if maintenance requires shutting off all power supplies. Doors to the storage chamber should remain closed during this test.

For a power outage/rate of temperature rise (and subsequent recovery) test for a refrigerated CTC, it is recommended to use a minimum (simulated) load with the minimum thermal mass or no load in the CTC.

There is another test that is recommended as a commissioning activity before the storage unit is put into use, if possible, with the mapping sensors in place, and the data acquisition rate to ≤ 1 minute intervals for this test, because of uncertainty in the time that the test may take.

Once power is restored to the CTC, the amount of time required for the unit to return to the required temperature can be of value for operational considerations for recovery from a disaster, or after a maintenance activity (i.e., elapsed time for recovery of temperature sensor(s) from the time when power is restored until the last temperature sensor(s) returned to the acceptable range). If recorded from an ambient temperature condition, this time can be a useful overall reference of the system's cooling capacity.

Generally, there are no specific acceptance criteria for this test unless a recovery time is specified in the unit manufacturer's specifications.

- Cover system defrost where applicable – it is typical to cover multiple defrosts, usually three to ensure the worst case is included (typically with clean coils so that there is a maximum heat loss to the surrounding space; where there are duty/standby systems both should be tested)
- There should be a record of all adjustable set points

Note: The level of liquid nitrogen or CO₂ is maintained at its set point (liquid nitrogen or CO₂ freezers)

The tests may be in different document or sections of a single commissioning document, and as part of Factory Acceptance Testing (FAT) or the Site Acceptance Testing (SAT).

Note: The test record should include a list of the settings used for any adjustable set points.

These tests will commission the system, as shown in Figure 5.2. The additional activities required to qualify the system are also shown.

5.1.3.2 Preoperational Safety Checks

This phase is intended to eliminate as many risks as possible regarding the safety of personnel during commissioning and operation.

A section of the commissioning report (or site acceptance test) should describe and document the appropriate procedures.

Procedures such as the “lock out/tag out” can determine that utilities, e.g., chilled water, gas, steam, and electrical power, are provided to the system once the following are available and documented:

- Conformity of the electrical system included in the CTC system and correct settings, e.g., earthing (grounding), overloads/fuse ratings
- Evidence of pressure test of utilities lines
- Testing/certification of safety devices, e.g., pressure release valves, emergency lighting
- Door release mechanisms
- Freezer heated air vents
- Presence of pressure/temperature safety release devices (with right settings)
- Emergency escape routes
- Required personal protective equipment

5.1.3.3 Safety during Commissioning

There may be a separate project safety plan for a project or a job safety analysis covering all phases of commissioning that should be prepared by the personnel who will execute the site activities. The analysis should describe the potential risks, their probability, and should identify protection devices. Operators or technical personnel should be informed about the outcome of this analysis when receiving permission to execute commissioning activities. Information about scheduled activities and related risks should be provided to other personnel that may be in the area at the same time.

The job safety analysis should be reviewed by the personnel responsible for safety. An example job hazard analysis template is provided in Appendix 7.

When judged appropriate, the testing area or room should have signs informing personnel of risks or forbidding access.

Additionally, the project typically has a waste handling strategy or plan. [29]

5.1.4 Documentation Standards

The documentation practices used for engineering documentation (preoperational documentation) should follow a specified standard, which typically is not as detailed as the requirements for a GMP record (such as a batch record).

The *ISPE Baseline Guide: C&Q (Second Edition)* [1] states:

“Testing and verification documentation are usually written in the local language. Summary documents are generally in English to facilitate regulatory review.

The use of forms/templates is encouraged to promote consistency. Forms/templates can be approved once when created and then utilized multiple times without pre-approval if the relevant content is not changed. Where this approach is used, there should be a procedure to ensure that the form/template used for data entry is current at the time of use.

When forms/templates are used, the SME should review the form to ensure that it includes all of the required testing for the system being verified; if additional tests are required, the form/template should be supplemented with additional testing.

Test scripts may be approved individually and assembled into a test plan/report. If test data sheets are used to supplement test scripts, they should be approved in accordance with the requirements defined in the test plan.

The test procedures should be clear and unambiguous such that the results are repeatable and reproducible, as explained below:

- *Repeatable: The same result would be achieved if the same procedure is repeated by the same trained and qualified personnel.*
- *Reproducible: The test procedures should be performed the same way each time and the same result would be achieved by different trained and qualified personnel.*

The test document should indicate the order that the cases are to be executed. Acceptance criteria needs to be established and it should be clear that the tests demonstrate that the acceptance criteria have been met.”

Documentation should be completed following the ALCOA+ principles so that the test and inspection records are:

- Attributable
- Legible
- Contemporaneous
- Original
- Accurate
- Complete
- Consistent
- Enduring
- Available

5.1.5 Discrepancy Resolution Process

Commissioning test failures that can be corrected through adjusting the equipment may be addressed in the commissioning test document, with an explanation and a retest record added for approval by the assigned SME.

Commissioning test failures that require system modifications should be noted as discrepancies and addressed using ECM. Impacted documentation should be revised to reflect the installed system.

5.1.6 System Risk Assessment

The SRA records the system CQAs, process parameters, and the associated CDEs (design risk controls) and procedural risk controls for hazards that could impact them. The scope of the Installation Verification (IV)/Operational Verification (OV) is determined by GEP. The SRA defines the CDEs and related acceptance criteria that are tested during commissioning.

Note that for simple systems an SRA may not be required.

5.2 Qualification

The traditional approach delivers the following C&Q documents (note the list only includes documents that would be impacted by the change in strategy).

By Engineering:

- Commissioning test, inspection and set up documents (documented evidence that the installation and operation meets the specifications)
- Commissioning summary report

By Validation:

- Installation Qualification (IQ) – documented evidence that the installation meets the specifications
- Operational Qualification (OQ) – documented evidence that the system operation meets the specifications
- Performance Qualification (PQ) where applicable
- Qualification Summary Report

Adopting a science and risk-based approach provides benefits not only in terms of reducing the number of documents, but also results in a risk assessment that provides an understanding of the quality risks associated with the system.

By Engineering:

- IV – documented evidence that the installation meets the specifications
- OV – documented evidence that the system operation meets the specifications

By Validation:

- SRA
- Commissioning and Qualification Summary Report

The commissioning and qualification summary report includes:

- Verification that installation and operation meets specifications
- Verification that the CDEs (design controls) identified in the SRA are installed and operate in accordance with the specifications by cross-referencing the relevant commissioning test documents
- Verification that the procedural risk controls described in the SRA are implemented in the operational SOPs
- Verification that the system is incorporated into the maintenance and calibration system
- Verification that staff are trained on the operational, maintenance, and calibration SOPs
- Verification that a spare parts list and associated consumables are available as appropriate

5.3 Handover and Release

Handover/release usually represents turnover of the system to the end user, with the handover of operation and maintenance. C&Q activities are usually completed. (The timing of the handover may be defined differently on a case-by-case basis according to specific contract stipulations).

There should be a documented record confirming that the functionality and operation of the system is acceptable and commissioning can be considered complete. The *ISPE Baseline Guide: C&Q (Second Edition)* [1] suggests that handover/release of the system to the end user is through the approval of the C&Q summary report.

A punch list (snag list) is typically used to manage completion of the system to meet the specifications. Major items on this list would prohibit handover; however, other items that can be resolved following handover are generally accepted. These items should be included on an action list to facilitate tracking to completion after handover. Typical categories used for a punch list include:

- Items that affect performance – to be addressed prior to handover
- Items that affect safe operation of the equipment – to be addressed prior to handover
- Items that block next activities/verifications – to be addressed prior to handover
- Minor items that do not impact operation – to be addressed after handover
- Desirable changes/improvements – to be addressed after system handover/release through the change control system

Following handover, responsibility for operation and maintenance of the system is transferred to the end user and the maintenance team. Appropriate operation and maintenance procedures should be established.

5.4 Calibration Strategy

The calibration strategy needs to be defined to ensure the requirements are met before testing is completed in operational verification.

There are operational benefits to be derived from a considered calibration strategy, for example:

- Many of the instruments on the refrigeration system can only be calibrated when the system is emptied for major maintenance; therefore, they are calibrated following the manufacturer's recommendations.
- The control sensor can be verified at a single point, as the control functions are usually based on whether the system is above or below the operating set point. If variable set points are used, for example in an incubator, then the sensor calibration needs to cover the working range.
- Calibration of the monitoring sensors should encompass the operating range and ambient conditions. For example, if the storage condition is 2°C–8°C, the calibration low point should be lower, for example below 0°C, and the high point must be above ambient room temperature, for example above 25°C. This is necessary to ensure that any excursions from the storage condition can be properly investigated. Excursions below the storage condition may present a different risk to those excursions above the storage condition. For example, freezing the load may represent a catastrophic risk to the product. The calibration strategy needs to account for this; otherwise temperature excursions may not be adequately investigated. This system provides the quality record of the storage conditions.
- Ensures that quality-related alarms are tested on a periodic basis; every 2 years is suggested.

5.5 Test Strategy

5.5.1 Outcomes

Testing is intended to:

- Confirm that the storage space remains within the defined limits with respect to temperature uniformity and stability over time across usable storage space
- Determine locations representing operating temperature extremes to establish the strategy for monitoring and alarming the space used for storage and/or processing

Note: It is assumed that temperature is quality critical. If there are other parameters, such as humidity or light, they should be considered using the same approach.

The operational temperature limits should be defined, together with any allowable excursions (acceptable limits, time periods).

Note: Some companies apply MKT to the acceptable operation of long-term storage areas such as warehouses. MKT is not used for refrigerators, cold rooms, or stability chambers.

There are many variable factors to consider when planning temperature mapping for a system. It is recommended that they are considered and the approach to be used put into a test strategy document for the Quality Unit to approve.

5.5.2 Risk Assessment

Prior to developing the test strategy, evaluate potential sources of excursions by considering the following factors:

Note: A company may use Computational Fluid Dynamics (CFD) modeling to evaluate the impact of these factors but the results still need some level of confirmation through actual testing.

- Purpose of the facility: storage, process cooling, transportation preconditioning
- Size: speed and magnitude of changes are impacted by the footprint as well as the height (Note: Pay attention to potential stratification with heights over 4.6 m.)
- Layout/configuration: wall and obstructions, fixed racks, temporary racks, pallets on floors, duct supply and return locations, door location
- Geographic location of facility: to define ambient conditions and seasonal variation
- Facility layout: physical location within a facility, location of windows, skylights, doors, traffic and access frequency
- Operating characteristics of the HVAC/refrigeration system: defrost cycles, redundant systems, number of units supporting storage space and how they interact with each other.

For example, a cold room may have duty standby systems that both operate in the event of a high temperature. This could impact temperature distribution compared to a simpler design that disables the duty system and starts the standby system in the event of a duty system failure or high temperature.

Additionally, there should be a clear definition of the acceptance criteria/expected results. Typically, this is provided in the URS, change record, C&Q plan, or equivalent document.

Temperature mapping is used to confirm that the system is operating to meet specifications, that is, the temperature range within the unit is acceptable.

It is a standard practice in the pharmaceutical industry to provide permanent monitoring of critical parameters using an independent system. Advantages of this practice include:

- Calibration of the monitoring sensors can be carried out without impacting the system operation
- The independent system becomes the Quality “system of record”
- The data can be used to provide continuous verification of acceptable system performance

Mapping results may be used to define the number and location of the permanent monitoring sensors for a warehouse, cold room, or chamber.

Note: While MKT may be used for the temperature monitoring for a controlled room temperature warehouse, the Guide authors recommend using this only for abnormal situations (i.e., do not use during testing to justify poor performance). Commissioning should be used to confirm that the room maintains the normal operating range.

A company can use multiple sensors in the area to provide continuous verification of the CTC performance. This can be used as justification for not conducting periodic mapping. (See Chapter 7.)

5.5.3 When to Perform Temperature Mapping

Temperature mapping should start as early as possible during commissioning. This allows the early detection and resolution of any performance issues. It also allows potential operational constraints to be identified and discussed with end users, to determine if the constraints are acceptable or require resolution.

Where there are systems that support operation of a CTC such as the refrigeration system for a cold room, HVAC system for a warehouse, or a condenser water cooling system for a refrigeration system, these systems should be identified and commissioned or in a known (and monitored) state prior to temperature mapping, to help ensure complete and representative testing, and facilitate investigation in the event of performance problems. For example:

“It can be acceptable to temperature map an area served by an HVAC system where a chilled water system which supports HVAC system operation has not been fully balanced, if:

- *the chilled water system is known to provide an adequate supply of chilled water at a constant temperature*

Note: the Building Management System (BMS)/Building Automation System (BAS) could be used to provide supporting data that:

- *the chilled water supply temperature remains within the operating limits*
- *the HVAC system cooling control valve does not open completely – indicating that an adequate supply of chilled water has been available for the duration of the test” [30]*

5.5.3.1 Prior to Temperature Mapping

Before temperature mapping begins, the commissioning plan/temperature mapping strategy should be approved by the system owner or SME representative and the Quality Unit.

The level of detail, number of personnel, or number of suppliers involved in temperature mapping can vary depending on the type of project. For example, commissioning of an air-conditioned warehouse in a new facility can involve several groups (e.g., construction manager, system installation contractors, and commissioning contractors). For a custom-built refrigerated storage room purchased through a design/build qualify contract, the requirements and responsibilities may lie with one company.

Roles and responsibilities should be clearly defined. *ISPE Baseline Guide: C&Q (Second Edition)* Section 6.4 contains an example document approval matrix [1].

Temperature mapping equipment requirements should be defined and the system calibrated.

Documents that should be available from the supplier/construction company (as applicable), include:

- Control system functional description
- Control system I/O list
- Calibration certificates as applicable
- Control system drawings (e.g., control panel layout, wiring diagram)
- Equipment and instrument lists
- Instrument and equipment data sheets
- Materials receiving and inspection reports/material certifications
- Equipment receipt and/or installation verification records
- Cable tests (continuity, insulation)
- Grounding (earth) tests
- Motor insulation “megger” and direction of rotation tests
- Fuse and breaker ratings/overload settings
- Control loop wiring checks
- Piping leak (hydrostatic) and evacuation tests
- Pipework cleaning/flushing records
- Record of cleaning for ductwork
- Ductwork and Air Handling Unit (AHU) leakage tests
- Operation and maintenance manuals
- Spare parts list
- Consumable parts list
- List of fluids/lubricants

- As-built drawings/specifications
- Specifications for materials of construction

The test requirements will depend on the type of system, and how critical performance is to the product and/or business. Project specifications should define documentation and test requirements.

5.5.3.2 Factors Influencing the Mapping Strategy

The mapping strategy should consider:

- Any internal heat gains, e.g., equipment, personnel, and how the strategy will address worst-case conditions
- How minimum and maximum load conditions will be addressed
- Seasonal variations
- How operational aspects will be addressed, e.g., any cooling load, loading/unloading, defrost cycles

When defining mapping locations, the potential storage locations should be considered:

- Do not use the floor as a location for a fixed load; it needs to be possible to clean the area
- Adequate space should be provided for cleaning walk-in cold rooms, freezers, and warehouses, but a mobile load may be stored in the space
- Storage locations for stand-alone units should follow manufacturers' recommendations to ensure adequate space inside for air movement
- The storage locations and extreme points of the stored load, e.g., top and bottom
- The source and direction of the conditioning airstream (a diagram may be helpful)
- Proximity to heating/cooling sources (motors, lights etc.)
- Adjacency to doors, windows, skylights, and the impact of their use
- Proximity to refrigeration coils and the impact of defrost
- Operational requirements: how long doors will be open, how many people, traffic patterns, equipment
- Operation of an aspirating smoke detection system

For units used for conditioning of the load, the unit should have the capacity to condition components within the desired time and temperature per the user requirements. Mapping sensors should be in the load to monitor conditions. This type of process equipment is not within the scope of this Guide.

Defrost cycles should be considered when outlining test procedures. Defrost temperature ranges and alarms should be defined according to user requirements or chamber profile. Keep in mind when mapping that the impact on the environment of a defrost cycle of a clean coil is likely to be worse than that for a coil with ice buildup.

Mapping system sample rate should be at least once every 5 minutes and may need to be more frequent if changes in temperature are anticipated due to cycling of defrost, cooling, or open-door tests (1 minute or less intervals).

Load

The mapping is used to confirm that the CTC will perform satisfactorily under all load conditions. In order to confirm this it is usually tested empty (with no or minimum load), and with a simulation to represent a full load.

The load can impact temperature distribution in two ways:

- Impacting the airflow – cartons will change air distribution in the chamber; for this reason the test load typically simulates the largest package that may be stored in the CTC
- Adding thermal mass – depending on the airflow the thermal mass may act as an insulator between the conditioning airflow and the perimeter walls of the CTC

It may not be necessary to fully load the CTC to simulate a fully loaded test condition. If the design is symmetrical then a partial load can be justified. Similarly, the use of a carton representing the full size of the load may be adequate to demonstrate the impact on the air distribution of a full load, with simulated thermal mass included where required (typically in the form of water bottles). The test strategy should include the rationale for the load simulation that will be used.

It is worth noting that for racking, the use of backstops ensures that the gap provided for air circulation, cleaning, and pest control is not compromised. For smaller off-the-shelf or catalog systems, the manufacturer will recommend the clearances needed when loading the unit. This information should be included in operational SOPs to ensure that the CTC is used correctly.

5.5.3.3 Selection of Mapping Equipment

Selection of the equipment to be used for mapping should consider the required accuracy and repeatability related to the acceptance criteria, for example:

A unit may have acceptance criteria of 2°C to 8°C, (standard storage conditions per USP). If the mapping system has an accuracy of $\pm 0.5^\circ\text{C}$ these provide results (allowing for rounding of the results) to meet the acceptance criteria of 2°C to 8°C.

The user requirements may have a tighter design specification, e.g., expected results of 3°C to 7°C.

Some low-cost mapping data loggers have accuracy specifications of $\pm 2^\circ\text{C}$. Use of these devices would not be appropriate for this application unless the acceptance criteria or expected results were revised to allow for the potential error.

Type of Probes

Temperature sensors with a rapid response time are typically used for temperature mapping, to help provide accurate information on system performance:

- If humidity has been defined as a CQA
- If there are no significant sources of humidity in the CTC it may be assumed that absolute humidity will equalize within the CTC and use of a small number of sensors may be all that is needed to provide the required data, for example, one at a low level and one at a high level (in the case of stratification). If the humidity is significantly higher outside the conditioned area, additional probes should be considered for the nearest storage locations.
- If there are significant sources of moisture, then the humidity could vary significantly within the conditioned space, and a greater number of humidity mapping sensors are needed. The number and location can be developed using similar concepts to those used for the temperature mapping.

Where RH is the criterion, the levels should be checked based on temperature variations using the measured RH at the local temperature from the mapping sensors to determine the RH in the rest of the space. Using duplex sensors (where available) can simplify the data analysis.

Users should also consider the probes ease of use and calibration, which impact measurement uncertainty and must be incorporated in mapping and monitoring/alarming practices.

Use of a radio data logger system can provide advantages compared to traditional thermocouple-based systems. The thermocouple system requires connection to the measuring and recording instrument, calibration, installation in the correct location, and being careful to route the cables so they are not damaged during testing. After testing the thermocouple must be removed and recalibrated. The radio data logger is calibrated annually, and the unit can be put in the location and data seen on the receiver as soon as it is turned on.

Buffered probes may be used for monitoring if the performance relationship is known compared to the characteristics of the stored product; however:

- Buffered probes should not be used as controlling probes; the buffer provides a response delay usually of a short duration to allow for temperature upsets due to defrost, door openings, etc.
- Buffered probes should not be used for mapping when the goal is to see the actual room (air) temperature

Note: Most situations call for monitoring in the air. Confirm the proper application of the monitoring practice with the Quality Unit.

Mapping sensor locations should be documented to allow a test to be repeated if necessary. The proposed load test strategy should be reviewed and approved by the Quality Unit and the system owner.

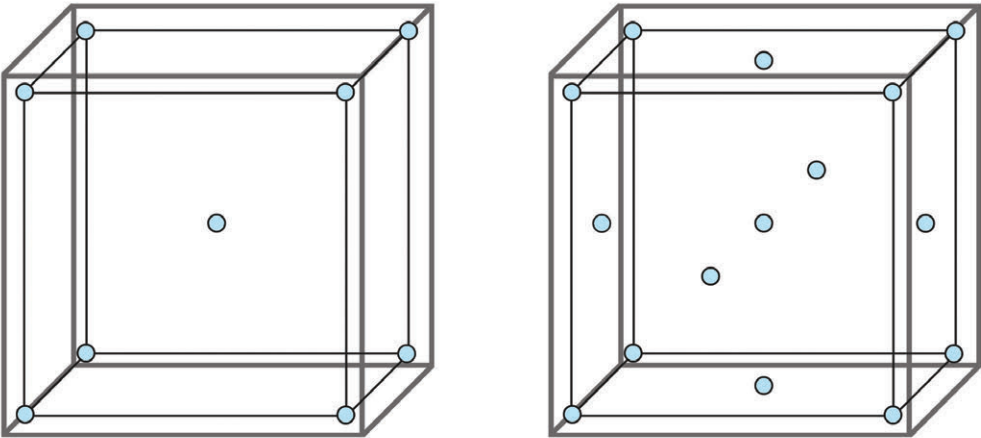
5.5.3.4 *Determining the Number and Locations for the Mapping Sensors*

The diagram on the left side in Figure 5.3 shows the suggested minimum number of mapping sensor locations to be used for a system with a chamber volume up to approximately 2 m³.

The diagram on the right side in Figure 5.3 shows the minimum number of mapping sensor locations suggested for a system with a chamber volume of up to 20 m³.

More information is provided in Table 5.2.

Figure 5.3: Example Sensor Locations



Note: The inner box represents the working area (where product is stored), the circles represent the sensor locations.

There are some strategies that require a sensor every 3 meters (12 feet). This Guide suggests that the location and number of sensors to be used should be established based on a scientific rationale: if there is no reason for the temperature to change then there is no reason for an additional sensor. For example, a long wall with equal conditions on the inside and outside will be the same temperature along its length, so there is no value adding multiple sensors. Consider:

- Where product will be stored
- The potential major influences on storage locations (heating/cooling)
- Obstructions that may impact conditioning airflow

Table 5.2: Recommended Placement of Mapping Sensors

Location	Rationale
Adjacent to temperature sensor used for control of the chamber	Provides data that can be used to assess the mapping data compared to the temperature the system controller records
Storage rack corners including geometric center of rack	Provides data for the extreme storage locations on the rack
Storage locations adjacent to ingresses/ egresses (door)	A potential area for temperature extremes
Storage locations near air supply and return outlets	A potential area for temperature extremes
Storage locations opposite or under evaporators	To monitor potential location for lowest temperatures, and the impact of defrost where applicable
Storage locations near air distributors (i.e., fans, blowers, etc.)	A potential area for temperature extremes
Shelving	A potential area for temperature extremes if the shelving has the potential to create dead air pockets leading to temperature deviations.
Storage locations adjacent to physical or structural obstructions, and in CTC corners	A potential area for temperature extremes if there are pockets of dead air
Bottom and top of any potential stored product	Mapping the height of component storage ensures that the worst-case locations are identified.
Storage locations adjacent to lights	Ensures that the light does not create warm spots when turned on that may impact component temperature

There should be a scientifically based rationale developed for the number and location of the sensors; however, there are some basic requirements:

- For a system with external walls (e.g., a warehouse) or where the system performance can be influenced by external conditions (e.g., where the refrigeration system uses an air-cooled condenser), data should be available on the external conditions. This data may originate from another calibrated reference source, e.g., the BMS/BAS (not the mapping system). This is to show the relationship of the conditions at the time of testing with the design temperature range, allowing the designer to evaluate the data.

For a conditioned room or standard off-the-shelf (catalog) system within a facility/room, there should be data on the facility/room conditions at the time of testing outside the store, to ensure that any open-door tests are representative of normal operating conditions. The data could be provided by an independent calibrated system (such as a BMS/BAS) or a sensor included in the mapping system.

When defining the number of sensors and their locations, factors to consider include:

- The mapping sensor locations should be increased as necessary to monitor areas where changes in the conditions are likely to be found during use, e.g., conditions for product located near a conditioned air discharge.
- Where there are multiple air conditioning units or HVAC outlets which operate consistently, the area treated by each unit may be treated as a zone, with the mapping sensor layout developed to suit those zones.
- An area may be laid out so that it can be divided into zones that will perform similarly, e.g., there may be end zones, and internal zones. Temperature mapping a representative zone may provide adequate data to represent performance of the whole area.

A rationale that describes the location and number of sensors to be used should be developed and included in the test strategy. This should be reviewed and approved by the system owner or SME delegate and the Quality Unit. An example is provided in Appendix 2.

An SRA example is found in Appendix 5.

6 System Monitoring

It is considered a good practice to use an independent system to monitor the critical parameters in CTCs. The location of monitoring system probes can be informed by the results obtained during mapping. The monitoring system used, and the location of the probes, will typically be based on a monitoring program or policy at the facility.

6.1 Monitoring Systems

The system used depends on the company and may include one of these three main types of monitoring system:

- Stand-alone devices, e.g., thermometers, chart recorders, data loggers
- Control systems, e.g., BMS/BAS systems, Supervisory Control And Data Acquisition (SCADA) systems
- Monitoring systems, e.g., Environmental Monitoring Systems (EMS)/Continuous Monitoring Systems (CMS)

The basic benefit of a monitoring system is to provide an independent temperature record of a CTC to document that heating/cooling systems are operating as intended and have maintained temperatures inside the CTC within specification. Additional benefits may include:

- Alarming and notification (alerts users of CTC emergencies such as equipment failure or open doors)
- Duplex sensors (independent sensors for control and monitoring provide redundancy and reduce risk)
- Automation (reduces human effort, and introduces the possibility of real-time data visibility for meaningful alarms)

6.1.1 *Stand-Alone Devices*

The most basic form of monitoring involves stand-alone devices such as maximum and minimum thermometers, chart recorders, and data loggers. These devices must be periodically checked by an operator, the results manually collected and/or downloaded, and evaluated against specifications. These monitoring solutions are simple, but involve significant labor costs and bring significant potential for human error.

6.1.2 *Control Systems*

Control systems may include BMS, BAS, and SCADA. Control systems leverage the existing sensors and/or network in a controlled temperature space, and utilize these resources for the purpose of monitoring. These systems may control the space being monitored and may not represent a truly independent system. In addition, control systems are inherently complex and customized, and therefore may require increased effort to qualify compared to other monitoring solutions.

6.1.3 *Independent Monitoring Systems*

Independent monitoring systems with a redundant network of temperature sensors have become commonplace. Often the sensors are data loggers connected to a central server via an Ethernet connection, but may also use wireless or radio frequency connections. These systems are typically referred to as EMS or CMS. These systems are optimized for monitoring, reporting, and alarming, rather than control. Typically, this makes them simpler to implement, and easier to qualify. In addition, as these systems are not used for control, they allow calibration of the monitoring probes to be performed without impacting the operation of the control system.

6.2 Temperature Sensors

There are two main types of temperature sensors in use for monitoring CTCs:

1. Thermocouples
2. RTDs and thermistors

6.2.1 Thermocouples

Thermocouples are an inexpensive sensor often found in monitoring systems. Type J, K, and T thermocouples are common in monitoring and mapping applications. The accuracy of a thermocouple is affected by its type and the operating temperature. Thermocouples provide fast response times but cables and joints can be prone to damage, and excessive manipulation will cause measurement drift. While thermocouple accuracy is limited, they may be an effective choice for measuring extreme temperatures, such as cryogenic storage applications. As with any sensor selection, the accuracy and stability of the thermocouple should be evaluated, based on the intended use.

See IEC 60584 [31] for information on the maximum permitted error for thermocouples.

6.2.2 Resistance Temperature Detectors and Thermistors

RTDs and thermistors are available to various grades and manufactured in a variety of housings. The operating principle considers the change in resistance of the sensor as the temperature changes.

An RTD is constructed from pure metal (typically platinum, but may also be nickel or copper) with the resistance of the device increasing with the temperature. A thermistor is made using a composite material with the resistance generally decreasing as temperature increases (Negative Temperature Coefficient (NTC)). There are Positive Temperature Coefficient (PTC) thermistors, but they are less common and typically less sensitive.

Overall, RTDs and thermistors are high quality sensors that are considered reliable and accurate and may last more than 10 years while maintaining the specified accuracy. Both sensor types are suitable for monitoring applications, though there are important differences. RTDs are typically larger with slower response times, but are generally interchangeable sensors that meet established standards. In comparison, thermistors are smaller sensors with faster response times, however, thermistor electronic characteristics vary by manufacturer and therefore must be integrated into a device for proper operation and accuracy. The selection of an RTD or thermistor should be evaluated based on the intended use.

Accuracy

The accuracy of an RTD is defined by its class according to IEC 60751 [32]; it is affected by the operating temperature. It is worth noting that overall loop accuracy may change when an RTD is connected to a device or transmitter.

- A class A device will be within $\pm 0.15^{\circ}\text{C}$ at 0°C
- A class B device will be within $\pm 0.3^{\circ}\text{C}$ at 0°C

There are no international standards for thermistors, but they typically are similar in accuracy to a class B RTD. When integrated into a device with appropriate electronics, thermistors can be more accurate than a typical RTD.

The accuracy, stability, and calibration method of the RTD or thermistor should be evaluated based on the intended use.

Note: Temperature is the primary parameter of concern when monitoring CTCs. However, RH may be a relevant parameter in CTCs such as stability chambers, incubators, and warehouses. Where RH is a critical factor, humidity may be mapped and monitored following the general practices outlined for temperature. Humidity sensors could be located in the same locations – or fewer locations used with the impact of the temperature considered for other locations – as absolute humidity will be very similar throughout the space. Combination sensors are available that provide both temperature and RH measurements (see Section 5.5.3.3).

6.3 Number Required and Location of Monitoring Probes

Mapping versus Monitoring Probes

Mapping probes are located in the storage locations to confirm that all locations remain in the defined limits during operation.

Monitoring probes are the independent means of verifying that the system remains operating within the qualified range.

There should be a strategy to support the number and location of monitoring probes. The at-risk (worst-case) locations should be identified during mapping of the CTC. Some examples are:

Off-the-shelf Refrigerator or Freezer: A single probe in the warmest location is often used for these systems. This location is based on a simple risk assessment (see Appendix 5) showing that the most likely failure mode is high not low temperature; therefore this strategy has the greatest possibility of detecting any abnormal conditions.

Walk-in Cold Room: Probes in the coldest and warmest locations are often used in these rooms to ensure that operation of the room remains in the qualified range and to detect significant changes in the room, e.g., cooling system function/capacity, room operations, room thermal performance. This approach supports the periodic review assessment strategy.

Controlled Room Temperature (CRT) Warehouse: The strategy varies because of a few factors, but typically includes measuring hot and cold locations. This may mean more than two probes are required to cover seasonal temperature changes, taking into consideration:

- Number of warehouse external walls, orientation versus local wind direction
- Design of HVAC systems; one system means a common supply air condition throughout the space; multiple zones may mean different conditions in each conditioned zone
- Segregation from loading/unloading docks
- Skylights/openings that may impact local conditions

6.3.1 Considerations for Permanent Sensor Placement

Once mapping is complete, the test data should be reviewed to determine the number of permanent monitoring sensors and where they should be located. Monitoring sensors should be firmly mounted to prevent accidental or inadvertent sensor movement to other locations. Some CTCs may have monitoring sensors already installed by the manufacturer – the position and suitability of such sensors should be evaluated against the mapping data.

For small units (< 2 m³) it may be acceptable to use one monitoring sensor as a single sensor will provide robust detection of any operational or performance issues. This sensor should be in a location that is representative of the worst-case location. This is typically the hottest location for a CTC operating below room temperature. It should be noted that it may not be convenient or reasonable to monitor the worst-case location, in which case a similar representative location may be considered.

For larger systems, or where there are multiple cooling systems, more sensors may be justified to ensure that an adequate record is available for the positions where stored product will experience the maximum and minimum temperatures (and/or maximum or minimum humidity, where applicable) during normal operation.

The optimum location can be determined by reviewing the monitoring data considering periods or tests that represent normal use, for example:

- Operation in working hours
- Operation outside working hours
- Loading
- Unloading

The mapping results and the data analysis can be interpreted and summarized in a report recommending the optimum location(s) for approval by the Quality Unit and the area/system owner.

Where testing has demonstrated that equipment failure can have a significant impact on local conditions (e.g., where conditioning is provided by a number of independent HVAC units), the data can be used to recommend actions to be taken in the case of such an event, for example, putting a defined rack out of use.

Factors affecting the placement of permanent sensors include:

- The results of the CTC mapping should determine the number of permanent monitoring sensors and their locations
- Risk-based worst-case storage locations
- Coverage based upon single and/or dual temperature control unit placement
- Volume of space to be controlled
- Density of materials storage and potential obstructions
- Loaded and empty space air circulation temperature results derived from OQ testing
- Mass of materials that enter the CTC vault at higher-than-operating temperature that adds to the heat load of the space. This decreases the ability for the cooling units to bring those products quickly to long-term storage temperature. This issue is inclusive of shipping gels, which frequently arrive at the location of the CRT and will add significantly to the heat load. Materials like this should be cooled in a separate area if possible before being transferred to CTCs used to store temperature-sensitive products.
- Ability of the circulating fans or diffusers to adequately move cooling air
- High and low storage locations where warmer air rises
- Any independent heat sources in the CTC such as lighting, or supporting production equipment, where applicable
- Open-door temperature or storage locations near CTC ingresses and egresses
- Frequency of personnel and lift truck ingress/egress operation during working hours

- Operations outside working hours
- Minimization of nuisance alarms
- Air conditioning, or lack thereof, of the CTC loading area where preconditioning of the air can limit humidity and consequent frost on evaporator coils
- Although there is no formula for the number of sensors in any particular CTC, the distribution must be satisfactory to evaluate the uniformity of temperature within the CTC. The risk areas, doors, airflow, volume load, total volume dimension (size), 3-dimensionality of the space, internal and external heat sources, and obstructions should be taken into account.
- Sensor(s) should not interfere with normal product movement or storage or be subject to damage from those movements
- Sensor(s) should be positioned in multiple geometric planes to accurately convey the temperature of the CTC

Data used to determine the number and location of the monitoring sensors may be interpreted in several ways. Two example interpretations are provided in Appendices 2 and 3.

It is normal to experience short-term temperature excursions due to typical operations such as door opening or personnel and truck movement. There are two approaches normally used to minimize nuisance alarms from these events:

- Putting the sensor into a container of thermal damping materials or equivalent (i.e., buffering). Sensor buffering can provide a simulation of the thermal response time experienced by the stored product due to the packaging acting as an insulator, significantly damping out any short-term temperature excursion on the temperature of the stored product.
- Putting time delays into the alarm system so that an alarm is activated only if the temperature has been outside the alarm set point for more than a defined time. Delays can be used to filter out nuisance alarms from short-term excursions caused by known recurring events such as frequent door opening.

Where either approach is used it should be documented, and include a supporting rationale, as both approaches can reduce the sensitivity of monitoring and alarming.

6.3.2 Thermal Damping Materials for Temperature Monitoring Probes

Monitoring probes may have alarm delay times or be placed in thermal damping materials (typically fluids such glycol) to avoid nuisance alarms, allowing for short-term temperature variances due to operational activities such as loading and unloading chambers.

Probes in Air: Time delays should be justified as having no product impact. The justification should consider the impact of the sample frequency of the monitoring system on the time delay set point. If a product has an acceptable excursion limit of 30 minutes and the monitoring system has a sample frequency of 15 minutes, then every alarm is potentially an Out of Specification (OOS) situation.

Probes in Thermal Damping Material: Probes in damping material allow for short-term temperature variance. The dampening material must simulate the storage of product in their primary containers. The dampening material should react to CTC temperature changes quicker than the primary container/packaging. Justification for the use of thermal damping materials should be documented.

Justification should consider:

- Container size and material of construction
- Material selection (e.g., glycol, glycerin, aluminum block, glass beads etc.)
- Location of probe within the container or material
- Primary containers and secondary packaging
 - The thermal dampening materials should always have a thermal mass less than that of the product being stored
 - If a variety of containers with commensurate mass are to be stored in the same CTC, then a conservative approach should be taken when designing the probe medium and its own container

Note: If liquids are used, probes should be specified to be suitable for immersion in liquid, with procedures to confirm the liquid level is as specified, with the probe location in the liquid correct, e.g., centrally located in the container. If the fluid is likely to evaporate or spill, procedures should ensure that the liquid level is maintained, and prevent product contamination.

Buffering is not usual for room temperature, and could affect MKT.

6.4 Determining Sampling Frequency

Monitoring systems typically have a configurable sampling frequency, and may utilize different sampling frequencies for different CTCs. It is a good practice to define the sampling frequency for each CTC in the relevant monitoring policy, procedure, or specification. The sample frequency selected for a given CTC is dependent on the behavior of the CTC, the value or sensitivity of the product stored, and the allowable duration of temperature excursions. CTCs with longer allowable excursions can typically tolerate longer sampling frequencies. Typical sampling frequencies range from 5–15 min. Sampling more frequently than every 5 minutes provides little gain and will quickly use data storage space. Sampling less frequently than every 15 minutes loses data resolution and is not recommended.

6.5 Determining Alarm Set Points and Alarm Delay Settings

It is considered good practice to define the alarm strategy for the CTC. A frequently used approach is provided here as an example.

6.5.1 Engineering Alarms

These are all the alarms generated from the temperature conditioning system and the associated control system. These alarms include system fault, system high or low pressure, high or low temperature, and temperature sensor fault (open or closed circuit). These alarms require an engineering action.

Temperature alarm set points are defined on the basis of the system “as commissioned” performance, typically set just outside the 3 standard deviation range to avoid nuisance alarms, but ensure indication of potential problems as early as possible.

Note: If the data from an operational period (at least 24 hours to encompass normal operations) is analyzed then the mean and standard deviation can be calculated. For a normal distribution of data, 99.7% of the results lie within ± 3 standard deviations of the mean. Therefore, if these set points are used to define normal operating alarms there will be a very small number of false alarms, but only a very small chance that a change in operational performance remains undetected.

6.5.2 Quality Alarms

These are the alarms from the independent monitoring system, usually configured based on the defined temperature storage limits.

For example, a unit with the operational range 2°C to 8°C may have alarm set points of 3°C and 7°C as alert limits, with action alarm set points of 2°C and 8°C.

Note: The exact alarm limits chosen depends on procedures in use at the site. For instance, some facilities may choose to set alarm limits that account for the measurement uncertainty of the monitoring sensors, as presented in Appendix 3.

6.5.3 Alarm Delays

Alarm delays are a feature of many automated monitoring systems. The function of an alarm delay is to filter out short excursions caused by normal use, such as the temperature changes caused by opening the chamber door for a short period. If an open-door test is performed, the results can be used to help determine appropriate alarm delay settings.

Probes mounted in air and probes in thermal damping material require different delays. For a probe mounted in air, the delay time should be sufficient to allow the air temperature to recover, avoiding nuisance alarms, that is, alarms not indicative of abnormal system operation. For probes mounted in a thermal damping material, the same concepts apply. However, the delay time, if obtained through the thermal mass of the sensor “system,” takes time to respond to air temperature changes. These arrangements of a sensor in a container of thermal material (used to avoid risk of freezing or evaporation) are designed to represent the worst-case stored material. This is usually the smallest container in the minimum packaging stored in that CTC. Where this approach is used, there should be a document providing the rationale for the system design, and demonstrate that use of delays and/or thermal damping are suitable for the specific products stored in the CTC.

Adopting a “standard” delay time for each class of CTC reduces the risk of setting an incorrect delay time that would exist if each unit has individual settings, while still providing a reliable indication if there are operational issues with the system, either due to the conditioning system, or because of an operator inadvertently leaving the door open.

The units usually have a wide range of operational loads, but what is appropriate for one load may not be for another. Lower temperature units in particular, take time to recover from being accessed, particularly if they have very little in them, that is, no stored thermal mass. The systems require time to recover physically. (The cold air in the unit is denser than the surrounding ambient air hence it will flow from the chamber to be replaced by warmer ambient air, which will take time to cool to the operating temperature of the chamber. For warm units the reverse effect is observed.)

The impact of the time to recover and the overall ability of these standard systems to maintain temperatures within limits must be considered when determining alarm delays, with the combined goals of protecting product, minimizing nuisance alarms, and simplifying administration.

7 Periodic Evaluation and Frequency of Thermal Mapping Studies

CTCs should be periodically assessed to determine the need for revalidation (remapping). The control systems, equipment, monitoring, and procedures for CTCs should be part of this periodic evaluation to ensure the unit still operates in a verified/qualified manner.

7.1 Risk Assessment

It is recommended that a risk assessment be used to determine the frequency of periodic review and the need for repeating the mapping of the CTC. This assessment should be aligned with a site's validation master plan and be based upon various risk factors including, but not limited to:

- Local regulations/regulatory expectations for maintaining a validated state and periodically repeating the validation studies
- Regulatory inspection observations and/or internal audit observations
- History of performance in maintaining the desired temperature range
- Critical equipment repairs/replacements
- Temperature deviations, i.e., alerts, alarms
- Standard Operating Procedure (SOP) changes
- Change controls to the area, to the temperature-controlling system, to the temperature monitoring system

Note: The impact of any changes to the system, for example, to the configuration of the building, doors, racking system, de-stratification fans, etc., is assessed through the change control system.

Periodic mapping exercises may be required based on the results of a risk assessment exercise or whenever significant modifications are made to the facility or the temperature-controlling equipment. However alternative approaches may be employed where supported by a scientifically sound risk assessment. The risk assessment for the requirement and frequency of remapping should include the following considerations, but is not limited to:

- Risk of failure to maintain uniform temperature (e.g., high risk from a refrigerated storage facility directly exposed to extreme temperature fluctuations versus a low risk from an ambient warehouse enclosed within a facility controlled to moderate temperature fluctuations)
- The storage of high value/criticality products and/or temperature-sensitive products may warrant more frequent reverification/requalification of the CTC for protection of investment and maintaining availability of product for critical quality and patient-focused considerations.

Typical industry practice for periodic assessments varies; however, these should always be performed in alignment with the relevant risk assessment and with additional consideration given to critical product storage. The rationale for this frequency should be documented for the facility and/or equipment.

The *ISPE Baseline Guide: C&Q (Second Edition)* provides industry guidance for the periodic review for CTCs. To evaluate system performance, Appendix 12 discusses periodic review assessment of monitoring sensor data by appropriate SME(s). [1]

7.2 Re-evaluation Requirements

The re-evaluation should include, but is not necessarily limited to, a review of:

- Changes made to the system, subsystem, or supporting system
- Changes made to the storage location (i.e., rack/bin configuration)
- Changes to material/product stored in chamber, e.g., high risk versus low risk
- Calibration history
- Equipment use logs
- Deviation investigations
- Maintenance and repair history
- Temperature mapping studies
- Operational procedures
- Existing verification/qualification documentation
- Ambient temperature profiles

Re-evaluation requirements should consider a data-driven evaluation of the chamber's performance. Several strategies exist to support this including:

- Periodic remapping of the chamber
- The use of data generated by a sufficient number of monitoring probes to demonstrate confidence in continued performance including the option of continuous temperature mapping
- Periodic review of alarms to detect chamber operational performance

These approaches, with examples, are detailed below.

7.3 Periodic Remapping

Remapping data for the CTC is evaluated against the original commissioning or qualification mapping data.

A risk-based approach may be developed to apply varying degrees of rigor depending on the material stored, and the system operating ranges (units operating with a close temperature range can be considered higher risk than those with a wider operating range).

For reverification/requalification, chamber mapping may be performed with an "as-found" material-loaded unit, utilizing sensor locations from the initial chamber mapping, if possible.

Mapping exercises should demonstrate that there has been no degradation of temperature control.

7.4 Periodic Evaluation of Process Monitoring Data

Data collected and summarized from CTCs equipped with multiple permanently installed monitoring probes may be an acceptable substitute for requalification temperature mapping studies, provided the probes have a demonstrated history of successful operation that includes calibration and maintenance.

If this approach is to be adopted, the number and location of the sensors needs to be determined based on the ability to monitor the impact of the conditioning system performance and the system construction.

The storage locations are defined, and any revisions are addressed through change control, allowing reassessment of the number/location of the monitoring probes.

7.5 Examples of Approaches for Different Equipment

The current practices vary from organization to organization. Typical approaches can be as follows:

- Assessing the system performance on a periodic basis to ensure that it is operating as commissioned

This can be through review of performance data and the system maintenance history, or through a maintenance routine that is specified to verify performance/check potential sources of problems. The method used will depend on the type of system, for example:

- For a CRT storage area (warehouse with a conventional HVAC system):
 - The building envelope will be inspected to ensure that there is no visible degradation of the materials of construction, or of seals/curtains that would impact ingress of unconditioned air
 - The system alarm history for the period since the last periodic maintenance (temperature monitoring system, HVAC system temperature/humidity, and airflow volume sensors as applicable) will be reviewed to see if there are any negative performance trends indicated, observations will be noted in the maintenance record
 - The HVAC system will be visually checked to ensure that the system balance remains as commissioned; through verification that balancing dampers remain in their as commissioned/ marked settings
 - Heating/cooling coils are checked and cleaned if necessary
 - The control system records are checked to confirm that control valves or fan/motor are not operating at 100% (indicating that system performance limits are being reached or that there are utility system issues), unless the control algorithms require that, e.g., dehumidification.
- For walk-in refrigerators/freezers:
 - The system envelope will be inspected to ensure that there is no visible degradation of the materials of construction, door or window seals, door curtains, or air leakage paths, as indicated by moisture or ice buildup, or thermal image scan
 - The system alarm history (temperature monitoring system, refrigeration system, and general alarms) will be reviewed for the period since the last periodic maintenance to see if there are any negative performance trends indicated, observations will be noted in the maintenance record
 - The system will be checked to ensure that the refrigeration system set point is being maintained, e.g., compressor suction pressure

- The maintenance history will be checked to ensure that:
 - > Coils have been cleaned, if needed and applicable
 - > Fan operation has been checked
 - > Operation of the unit cooler control valves and thermal expansion valves has been checked
 - > Defrost settings and operation of the defrost system have been verified, if applicable
- Check safety interlock functions
- For standard stand-alone refrigerators and freezers:
 - The system envelope will be inspected to ensure that there is no degradation of the materials of construction, door seals, or air leakage paths, as indicated by moisture or ice buildup
 - The system alarm history (temperature monitoring system, refrigeration system, and general alarms) will be reviewed for the period since the last periodic maintenance to see if there are any negative performance trends indicated, observations will be noted in the maintenance record
 - The system will be checked to ensure that the refrigeration system set point is being maintained, e.g., compressor suction pressure
 - The maintenance history will be checked to ensure that:
 - > Coils have been cleaned, if needed and applicable
 - > Fan operation has been checked
 - > Operation of the unit cooler control valves and thermal expansion valves has been checked
 - > Defrost settings and operation of the defrost system have been verified, if applicable

System performance as commissioned/qualified can be evaluated against the current data from the monitoring system to ensure that the system performance remains unchanged.

Three examples of an approach that may be used to provide this assurance are shown in Figures 7.1, 7.2, and 7.3.

Figure 7.1: Cold Room Layout Example Showing Mapping Sensor Locations and Location of Test Load

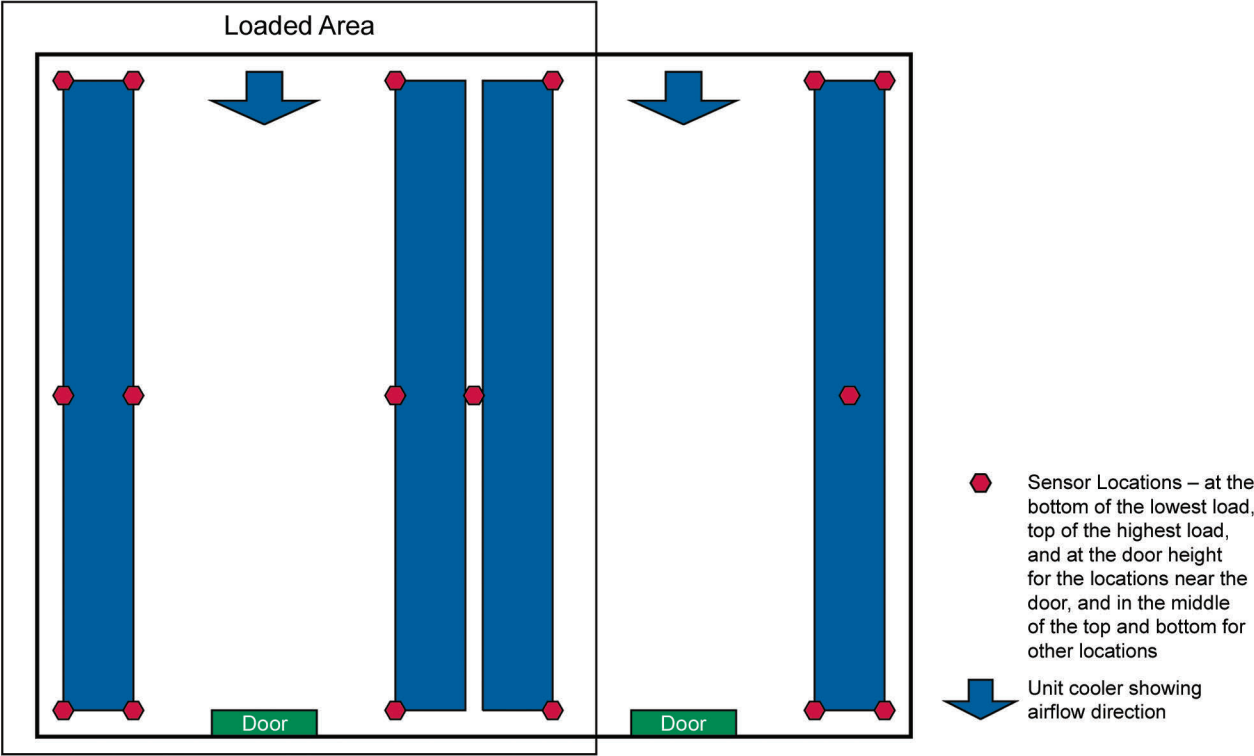


Figure 7.1 shows a plan of the unit with the sensor locations used for the mapping of the unit. The design is symmetrical about a vertical axis, so load testing was carried out by placing a simulated load in the areas shown inside the blue box marked "loaded area".

The load included thermal mass, as the mass of the load may impact the load temperatures in locations adjacent to the outside walls.

Figure 7.2 shows the locations proposed for a monitoring system to provide continuous verification that the system is performing to meet the specifications, to be confirmed from the initial mapping. Note that the impact of reduced performance from a unit cooler or significant degradation of the cold room structure would be detected by the sensors in the locations shown.

Figure 7.2: Proposed Sensor Locations to Provide Continuous Evidence of System Operation to Meet Requirements

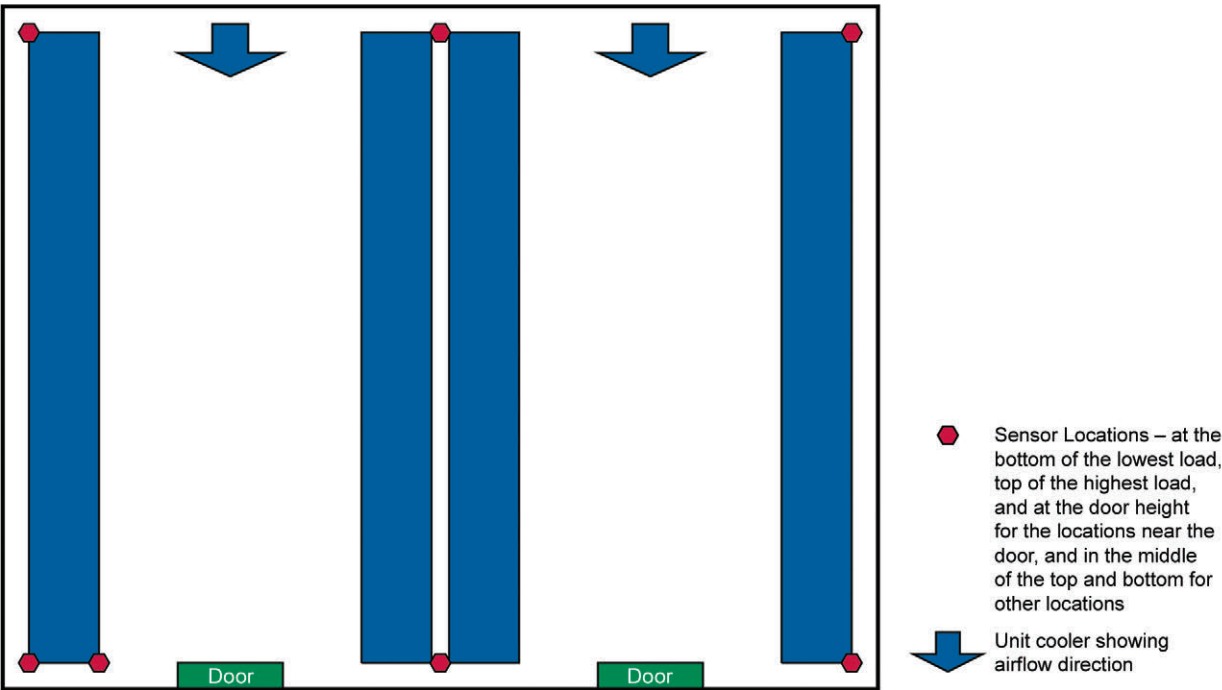


Figure 7.3: Cold Room Layout Example Showing Mapping Sensor Locations and Location of Test Load

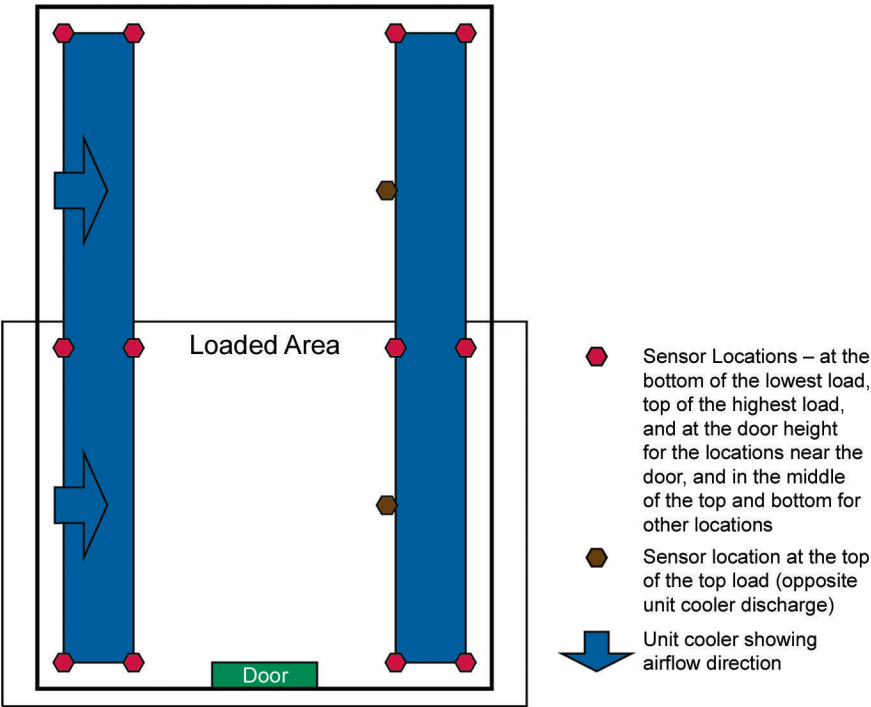
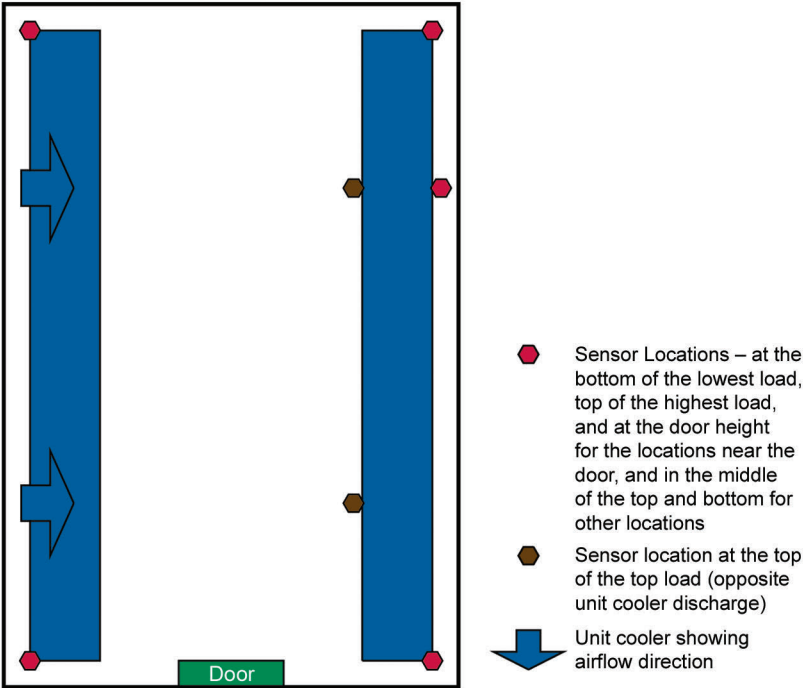


Figure 7.3 shows a plan of the unit with the sensor locations used for the mapping of the unit. Load testing was carried out by placing a simulated load in the areas shown inside the blue box marked “loaded area.”

The load did not include thermal mass, as the mass of the load may not impact the temperatures adjacent to the outside walls, but an empty box with external dimensions matching the size of the largest stored volume will be used to provide a worst-case disruption of airflow.

Figure 7.4 shows the locations proposed for a monitoring system to provide continuous verification that the system is performing to meet the specifications, to be confirmed from the initial mapping. Note that the impact of reduced performance from a unit cooler or significant degradation of the cold room structure would be detected by the sensors in the locations shown.

Figure 7.4: Proposed Sensor Locations to Provide Continuous Evidence of System Operation to Meet Requirements



8 Maintenance

This chapter discusses factors that should be considered in a maintenance program. For further information on maintenance programs see the *ISPE Good Practice Guide: Maintenance (Second Edition)* [33].

It is recommended to completely read and understand the operation and maintenance manual before operating the equipment. The supplier of the equipment should provide general recommendations for the maintenance and calibration of the equipment as well as the frequency of maintenance, in regard to the specific application. These recommendations may not recognize the high value of the stored goods in this application. The potential impact of the maintenance program should be assessed.

8.1 Frequency

Regardless of the maintenance strategy adopted, the frequency of cleaning and maintenance should be considered in a maintenance program. The supplier of the equipment should provide general recommendations for the frequency; however, the equipment criticality or classification may also be considered to determine frequency.

8.2 General Cleaning and Maintenance

The approach to cleaning and frequency of cleaning should likewise be considered. Dry cleaning is preferred in order to minimize risk of contamination.

Regular inspection is recommended to ensure that there is no unseen microbial/fungicidal growth. Any growth detected should be addressed.

Regularly check and ensure that there is no frost buildup in the chamber. It is recommended to defrost at least once per year or more frequently as needed.

Where applicable, filters should be checked, cleaned, or replaced as necessary.

Check, and clear off as necessary, the air circulation pathways in the chamber.

Periodically check the power supply to the temperature-control equipment to determine if it is within the acceptable limits.

8.3 Cleaning of the Coils

The coils (unit cooler and air-cooled condenser) should be cleaned regularly and checked for any damage to ensure that the performance levels are maintained. Dirty coils increase energy consumption by preventing effective heat dissipation.

To clean coils, gently vacuum, brush, or wipe with a wet paper towel. Damaged or distorted fins should be straightened with a suitable comb or coil brush. This inspection should also ensure that there is no corrosion of the fins. If the unit has delicate metal parallel fins on the coil, it is important to vacuum, brush, or wipe in the direction of the fins. Be careful not to bend metal fins.

Regular inspection should also confirm that the defrost cycle is effective (where applicable), especially if the cycle is adjusted to account for seasonal variation. Adjusting the defrost cycle can have implications on the validated state of the system as longer defrost cycles can increase the chamber temperature and drive the chamber out of control limits needed for the storage conditions of the product. The adjustments should be part of OQ and PQ of the CTC, to ensure that it remains in a validated state.

Where cleaning or sanitizing agents are used, compatibility with materials of construction should be assessed.

8.4 Functional Testing of the Unit Coolers/Defrost Elements

There is usually no feedback from the solenoid valve on the unit coolers, so the only way of checking their correct operation is to carry out, periodically, a functional check to verify the performance of each unit cooler. This activity should also include the standby refrigeration system for those unit coolers equipped with one.

8.5 Check the Condensate Pipework

The condensate pipework should be checked to ensure that it is not damaged and that it has the correct slope and flows freely. Trace heating or insulation also should be checked.

The condensate drip tray and drain, if present on the system design, should be cleaned periodically.

Periodic verification during operation is also recommended; this should ensure that the pipe supports are adequate and have not loosened.

8.6 Leak Testing

It is considered a good practice to periodically check the pipe runs, as well as the equipment for leaks, using one of the electronic test devices available. This may be a mandatory requirement, depending on region.

- The system refrigerant fluid level and pressure against the system specification should be periodically checked and documented. Early detection may help avoid system failure.
- It may be necessary to keep a record of the refrigerant used for the system. Environmentally friendly refrigerants are recommended.
- Equipment rooms may be fitted with leakage detectors/alarms; these should be calibrated and checked regularly.
- End users should be aware of the local regulatory requirements.

8.7 Heating Elements

Some CTCs like stability chambers, incubators, and heating chambers (ovens) are fitted with heating elements. These should be periodically checked for:

- Continuity, as faulty or broken elements may lead the chamber to underperform as well as pose safety risks, e.g., fire, explosions
- Scaling, as in the case of autoclaves where the heating elements are in contact with water. Descaling should be done as necessary.

- Any loose connections for the elements, and tightening as necessary
- Functionality of safety devices for the elements such as a safety thermostat

8.8 Door Seal

Periodically inspect the door seal to make sure it is tight and free of damage, and that there is no microbial growth present. A good seal is important for keeping warm or cold air outside of the unit for both the benefit of energy efficiency and unit performance. Microbial growth on the seal is usually a result of moisture buildup due to a worn seal. Replace any damaged seals. Keep gaskets free of frost and ice, if applicable, by brushing frost off the gasket frequently.

8.9 Thermal Scanning

The panels, doors, door seals, and sealing of utility penetrations can degrade over time (foamed panels may be better in this respect than panels filled with mineral wool). A thermal scan of the installation will allow any degradation to be seen, which is as a deviation from the baseline scan taken during initial OQ, and then an appropriate action is taken. As a minimum, the penetrations should be visually inspected to ensure that they are sound with no leakage/ingress of condensation. For conduit penetrations, the internal sealing also should be checked to ensure that condensation is not leaking inside the conduit.

8.10 Air Circulation Fan

Some equipment may be equipped with a circulation fan to effectively control temperature in the chamber. In such cases, condition-based monitoring of the motor and blower may be employed, by checking the electrical current drawn or bearings of the motor, so as to timely replace/repair the fan assembly as necessary. Fan bearings should be periodically lubricated. Fan belts, where applicable, should be free of wear or damage. Nonmetallic fan blades start to deteriorate over time, especially on the blades and fixings. Periodic visual inspections are recommended.

8.11 System Alarms

The entire alarm system should be periodically tested by direct or simulated temperature change against the specified alarm set points and proper functionality of audible and visual alarms at local and/or remote alarm stations (e.g., emails, phone calls, or text-based alerts) should be confirmed and documented.

Regardless of the root cause, product losses from cold storage-related failures can be prevented with the proper design and operation of the alarm system, and proper end user response to an alarm event. A risk-based evaluation may be adopted for establishing the end user response. It is recommended that the equipment supplier's recommendations be followed for setting up a backup CTC, for example, requiring a 30 minute runtime before loading, in the case of the CTC failing. Periodic testing of the alarm system should help to ensure that it will work when needed.

The system maintenance procedures can be used to provide confirmation that the system performance remains satisfactory, supporting a reduced requirement for remapping where appropriate. The data from the monitoring system should be reviewed periodically, to look for any negative performance trends.

8.12 Operational Considerations

The following are other areas to consider with regards to operation and maintenance.

Dehumidification

Dehumidification should be considered for a cold room or freezer. The life of the equipment can be extended where fewer defrost cycles are required (reduced expansion/contraction). Dehumidification can also reduce the risk of ice forming on a freezer floor, which presents a safety hazard. For large distribution facilities it may be worth considering the layout, and providing access to a walk-in freezer through a cold room, in order to reduce moisture in the freezer.

Equipment Durability

When specifying new equipment or planning a new facility, it is worthwhile defining the system design life, that is, the expected operating lifespan without significant maintenance. High quality refrigeration systems can have an expected life of 12 to 15 years. With regards to durability and redundancy, it may be worth considering a configuration that utilizes optimal energy consumption and efficiency while meeting redundancy and backup requirements. Standby systems are usually installed for an installation holding material of a significant commercial value.

8.13 Suggested Maintenance Procedures

Some suggested maintenance procedures for different system types are:

CRT Storage Area (Warehouse)

1. The building envelope should be inspected to ensure that there is no visible degradation of the materials of construction, or of seals/curtains that would impact ingress of unconditioned air.
2. The system alarm history for the period since the last system maintenance (temperature monitoring system, HVAC system temperature/humidity, and airflow volume sensors, as applicable) should be reviewed to see if there are any negative performance trends indicated, and observations should be noted in the maintenance record. Periodic performance trends review, e.g., monthly, should also include an alarm history review.
3. The HVAC system should be visually checked to ensure that the system balance remains as commissioned through verification that balancing dampers remain in their as commissioned/ marked settings.
4. Heating/cooling coils are checked and cleaned, if necessary.
5. The control system records should be checked to confirm that control valves are not operating at 100% (indicating that system performance limits are being reached, or that there are utility system issues), unless the control algorithms require that this is what the control valves should do, e.g., for dehumidification.
6. Functionality checks should be done for emergency components such as exits, door locks, and lighting.

Walk-in Refrigerators/Freezers

1. The system envelope should be inspected to ensure that there is no visible degradation of the materials of construction, door or window seals, door curtains, or air leakage paths, as indicated by moisture or ice buildup.
2. The system alarm history (temperature monitoring system and refrigeration system general alarms) should be reviewed for the period since the last maintenance to see if any negative performance trends are indicated; observations should be noted in the maintenance record. Periodic performance trends review, e.g., monthly, should also include an alarm history review.

3. The system should be checked to ensure that the refrigeration system set point is being maintained, e.g., compressor suction pressure.
4. Functionality checks should be done for emergency components such as exits, door locks, and lighting.
5. The maintenance history should be checked to ensure that:
 - Coils have been cleaned, if needed and applicable
 - Fan operation has been checked
 - Operation of the unit cooler control valves and thermal expansion valves has been checked
 - Defrost settings and operation of the defrost system have been verified, if applicable

Stand-Alone Refrigerators and Freezers

1. The system envelope should be inspected to ensure that there is no degradation of the materials of construction, door seals, or air leakage paths, as indicated by moisture or ice buildup.
2. The system alarm history (temperature monitoring system and refrigeration system general alarms) should be reviewed for the period since the last maintenance to see if any negative performance trends are indicated; observations should be noted in the maintenance record. Periodic performance trends review, e.g., monthly, should also include an alarm history review.
3. The system should be checked to ensure that the refrigeration system set point is being maintained, e.g., compressor suction pressure.
4. The maintenance history should be checked to ensure that:
 - Coils have been cleaned, if needed and applicable
 - Fan operation has been checked
 - Operation of the unit cooler control valves and thermal expansion valves has been checked
 - Defrost settings and operation of the defrost system have been verified, if applicable

Incubators and Ovens, Heating Chambers

1. The system envelope should be inspected to ensure that there is no degradation of the materials of construction, door seals, or air leakage paths, as indicated by moisture buildup.
2. The system alarm history (temperature monitoring system and heating control system general alarms) should be reviewed for the period since the last maintenance to see if any negative performance trends are indicated; observations should be noted in the maintenance record. Periodic performance trends review, e.g., monthly, should also include an alarm history review.
3. The system should be periodically checked to ensure that the heating system set point is being maintained, e.g., electrical power supplied.

4. The maintenance history should be checked to ensure that:

- Heating elements have been cleaned, where applicable
- Heating elements are not broken
- All electrical connections are appropriately tightened
- Electrical and temperature controls are functional
- Fan operation, where applicable, has been checked
- Operation of the safety thermostat has been checked
- The chamber breather or temperature–pressure relief valve has been checked to be open/functional
- The fire suppression system, where applicable, is functional
- Door and/or closure integrity is maintained and aligned to prevent ingress and egress of air

Where a system has multiple monitoring locations, e.g., the control sensor, and an independent temperature monitor located to monitor the hot/cold locations in the unit, preventive maintenance is not required but is considered best practice to confirm satisfactory unit operation. Independent monitoring may provide the justification to extend the interval between preventive maintenance activities. The review of current system performance data against the as qualified data by an SME can be used to confirm consistent system performance.

This review should assess the thermal performance of the unit insulation and the conditioning system, allowing a suitably experienced SME to determine if there has been a significant change in performance that may require maintenance activities to improve or if the system is still performing acceptably.

9 Decommissioning

Decommissioning is a general term for a process to remove something from active status. It may be a precursor to putting facilities/equipment into storage, repurposing, or demolition/disposal. [34]

It is important that a plan to decommission equipment be well designed and documented. This can be done in the strategy document, procedure, or business plan documents. Robust planning ensures control of any situation regarding the need to decommission a single asset or a whole facility.

Aspects to consider in relation to decommissioning include:

Reasons to Decommission

- Equipment is nearing the end of the manufacturer's suggested lifecycle. (Manufacturers know best the materials and parts used for their products and are the best resource for understanding the lifecycle of their equipment. This activity can be planned and scheduled when the equipment is purchased.)
- Multiple failures and/or repairs
- Problems discovered during scheduled preventive maintenance activities

Considerations Prior to Decommissioning

- Testing: A final qualification test is recommended to document that the CTC was decommissioned while in a qualified state of operation.
- Ensure a replacement storage area for samples is available. New equipment may need to be purchased and implemented as a replacement for the decommissioned equipment.
- Proper documentation is as important when decommissioning a piece of equipment as it was during the initial qualification.

Considerations for Responding to a Test Failure

- Document the failure using an approved process such as a procedure and/or deviation protocol. Be sure to include sample movement information.
- If the equipment is unable to be repaired or it is not cost effective to repair, then document the reason and justification in the deviation.
 - Physically tag the equipment "Do Not Use" until removed from the area.
 - External monitoring temperature data can be used to document the condition of the CTC up to, during, and after the failure. Document all samples that were transferred to determine if the samples are in a safe-to-use condition.

More information is contained in the *ISPE Good Practice Guide: Decommissioning of Pharmaceutical Equipment and Facilities* [34].

10 Packaged Transport Systems

10.1 Aircraft Transport

At the time of publication, the International Air Transportation Association (IATA) is building a network of certified pharmaceutical trade lanes that meet consistent standards and ensure product integrity to help the air cargo supply chain better serve the pharmaceutical market. This program is called the Center of Excellence for Independent Validators in Pharmaceutical Logistics (CEIV Pharma). See the IATA Guidance for Vaccine and Pharmaceutical Logistics and Distribution for more information. [35]

For temperature-controlled aircraft transport, there are two types of Controlled Temperature Shipping Solutions (CTSS): active or passive.

10.1.1 Active Controlled Temperature Shipping Solution

Active CTSS have active temperature control, which adjusts the temperature inside the container to maintain the specified temperature range. Active containers have electrical cooling and heating systems. They are typically powered via internal batteries/rechargeable batteries or an external electrical source. It should be noted that when in use (i.e., in an aircraft) the units operate on battery power.

10.1.2 Passive Controlled Temperature Shipping Solution

Passive CTSS, on the other hand, are insulated boxes or containers without any active temperature control. Passive systems maintain a temperature-controlled environment inside an insulated enclosure. They have no active mechanisms or moving parts and require no human interaction in order to function when in use. They rely upon intelligent design and the presence of suitable preconditioned coolant/insulants such as chilled or frozen gel packs, phase change materials (such as dry ice or other specialist materials) in order to perform.

Passive CTSS typically comprise a cardboard outer insulated container with polystyrene, polyurethane, or vacuum-insulated panels. Alongside the thermal media, they act in concert to resist changes in product temperature driven by adverse external temperatures.

These systems require preparation of components under controlled conditions and the system can be packed and sealed prior to dispatch. The company may need to provide in-transit handling instructions to address known possible scenarios, such as delayed in transit or storage conditions for in-between flights. For example, the CTSS may have the dry ice topped up to extend for the duration, or a unit using a special phase change material may be stored in a cold room to help recharge the material.

There can be risks during transport particularly during any transfers *en route* where a pallet may sit on the tarmac waiting for a connecting flight. These can be managed by using a suitable specialist freight company, or the end user can use risk mitigation techniques, such as covering the top and bottom of the load with thermal blankets.

10.1.3 Unit Load Devices

Unit Load Devices (ULDs) are specially designed cargo pallets and containers used to carry air cargo on wide-body aircraft and specific narrow-body aircraft. They allow a large quantity of cargo to be bundled into a single unit, and are powered to either cool or heat the air within the container.

There are many different types of ULDs, but only a few are designed for shipping temperature-sensitive product by aircraft. When operated within the design limits and properly loaded, these containers enable precise end-to-end temperature control throughout every stage of the cold chain transportation cycle to ensure that the products inside are not affected by variable conditions experienced during the journey.

Temperature-controlled containers come in a range of sizes and technical specifications. The key features for ensuring consistent temperature control in the most commonly available standard temperature-sensitive airline containers are [36]:

- Insulated container shell constructed in an all-composite material
- Electric heating and compressor cooling system
- Rechargeable batteries that can be charged at standard AC power connection points
- Air circulation system inside the container to maintain a low temperature gradient within the entire cargo space
- Air circulated by the heating and cooling unit is close to the set temperature
- Control unit to operate the container
- Validated recording solutions for temperature and battery information
- Ability to secure unit for tamper evidence
- Accessible from all sides by forklifts, even when fully loaded, eliminating the need for roller beds to move the containers in the warehouse and to and from trucking docks

These systems are typically leased, with the specialist company providing supervision of their units during transit. Services can include checking the ULD, recharging it if there is a significant layover, etc.

10.1.4 Qualification Approaches for Active Shipping Containers

There are two possible approaches:

- Unit qualification
- Supplier assessment

10.1.4.1 Unit Qualification

It is desirable for the supplier to temperature map the individual unit to confirm its satisfactory operation and to qualify the service supplier, ensuring that their maintenance and calibration quality systems are robust, as this type of equipment is usually provided by a service supplier. The pharmaceutical or biopharmaceutical organization should specify use of mapped units.

10.1.4.2 Supplier Assessment

Qualification of the manufacturer/supplier of the units: the design and factory testing, and maintenance and calibration program should be reviewed to ensure they are robust.

Ensure that quality agreements with airlines/freight forwarders are complete so that expected handling/storage, etc., is adequate for the intended purpose.

Usually the pharmaceutical or biopharmaceutical organization includes a temperature logger with the shipment, to provide independent assurance that the specified conditions were maintained during shipping.

The method for loading and unloading the unit should be qualified and procedurally defined. Usually the unit arrives on site at temperature. Loading may take place in CRT or cooled conditions.

Typically loading procedures include a pre-loading check that the ULD is visually clean and odor free, has no visible damage, has the correct temperature setting, and an adequately charged battery. The load is then put into the ULD (being careful not to block any air vents) and secured, the doors closed, and the security seal added.

When the CTSS is unloaded the shipment should be placed into quarantine until the information from the data logger confirms that the shipment was within the defined limits for the duration of the shipment. If the shipment was not within the defined limits during shipment, a nonconformance or deviation should be raised and the implications of the excursion are investigated.

Note: With some of these systems it is possible to obtain data during shipping from the ULD.

10.2 Trailers and Ocean Reefers

10.2.1 Temperature-Controlled Trailers

Temperature-controlled over-the-road trailers can be used to move significant amounts of temperature-sensitive pharmaceutical product. An approach offered in North America and Europe takes into account the fact that trucks in most jurisdictions are manufactured to specifications that allow the end users to leverage accepted, voluntary industry standards and focus on the Critical Process Parameters (CPPs) that affect the operational performance of every load carried.

Trailers have a different set of environmental and usage profiles compared to *in situ* custom-built CTCs that are owned and controlled by each pharmaceutical or biopharmaceutical organization, for example:

- Loading and unloading operations
- Defrost cycles
- Loading patterns

Moisture, as a degradation element, is much more prevalent in trailers due to the operational profile, that is, continuous running for a several days, followed by periods of downtime when the trailer is empty. The operational stresses present a challenge to the integrity of joints in the insulation covering. The primary causes for moisture ingress in trailers that may not occur in cold chambers are:

- Running the cooling system while loading
- Running the evaporator fans while loading (even if the unit is not cooling but is preconditioned)
- Loading from a non-conditioned dock area (or dock area with higher temperature and/or humidity than the desired trailer set point) or from a non-sealed dock

A best practice is to precondition the trailer (so that the thermal mass has been cooled), then load with the refrigeration system turned off.

Rigorous inspection of each trailer should be performed prior to loading to account for potential damage by moisture along with potential damage to the sidewalls by forklifts. An approach that takes these differences into account while leveraging the construction standards is presented in this section of this Guide.

Note that the standards require routine recertification of the trailers, including thermal performance.

These considerations apply to trailers used for long distance transport, as well as those used for local delivery purposes. Transportation purchase orders should specify product (load) temperature requirements.

Distribution of temperature-sensitive pharmaceutical or biopharmaceutical product via a temperature-controlled trailer is a unit operation that requires qualification. The mapping of every trailer is not considered a substitute for adopting a defined methodology that identifies suitable equipment and applies a control strategy to the day-to-day operations in order to maintain suitable temperature ranges for pharmaceuticals.

Manufacturers who build temperature-controlled vehicles for the food and pharmaceutical/ biopharmaceutical industry adhere to voluntary standards that allow the end user to take a “family” approach (i.e., several units of the same make and model) to qualification. The focus of activities should be on the key operational and maintenance elements. This can help to establish high confidence in this distribution unit operation.

Trailer Design

Trailers are designed to maintain product temperature, although they may have a specially intended use such as rapidly cooling material to a desired temperature, known as blast freezing. Trailers specified to maintain product temperature procedurally should specify equilibrating the trailer to the appropriate temperature before loading or have a process defined through studies. As the heat or cold penetrates the walls, roof, and floor of the trailer, circulating airflows capture the environmental heat or cold. This environmental heat or cold is then conditioned to the desired temperature in the HVAC unit(s) of the trailer.

Note: Fresh fruits and vegetables produce heat from respiration; this adds load and an allowance for normal degradation of the insulation is a design consideration factored into the sizing of the HVAC unit. Pharmaceutical and biopharmaceutical products do not give off respiration heat; therefore, the HVAC units have additional reserve cooling capacity when the trailers are used for hauling these products.

Trailers used for freezing only are often only provided with means of cooling; therefore special consideration should be taken when such a trailer is used in winter, or when environmental conditions are below the storage temperature specification. This is uncommon in North America/Europe, but there are many locations where this may be a concern.

Trailers used for cooling material from an unspecified temperature to a specified temperature should have appropriate conditioning studies performed. Design considerations for rapid cooling of material should include cooling BTU, airflow velocity, and air change rates. These characteristics impact the design and quantity of HVAC units.

Depending on the geographical area, manufactures build and test trailers to meet the standards of one of two organizations:

1. In North America, the governing body is the American Trucking Association (ATA) Technology & Maintenance Council (TMC) [37]
2. In Europe, the standards are part of the Agreement on the International Carriage of Perishable Foodstuffs and on the Special Equipment to be Used for such Carriage (ATP) [38]

TMC Classifications

TMC is a voluntary code from the US that classifies controlled temperature vehicles as either C35, C65, F, or DF [37]. See Table 10.1.

In North America, manufacturers can voluntarily certify the capability to one of the performance criteria shown in Table 10.1. Most trailers sold in North America are built to correspond to the F classification.

A stamped metal plate providing details of the certification of these trailers should be requested when the order is placed with the trailer manufacturer. Trailer manufactures may issue a plate with the certification post-manufacturing if the vehicle identification number is provided. A quality agreement with the carrier should confirm that the trailer, delivered as new, was either F- or DF-rated, regardless of the presence or absence of a stamped metal plate.

Table 10.1: TMC Controlled Temperature Vehicle Classifications [37]

Used with permission from ATA's Technology & Maintenance Council (TMC), <http://tmc.trucking.org> <<https://tmc.trucking.org>>

Vehicle Designation	Use	Loading and Transportation Temperature
C65	Transport of chilled commodities	≥ 65°F (≥ 18.3°C)
C35	Transport of chilled commodities	≥ 35°F (≥ 1.7°C)
F	Transport of frozen commodities	≥ 0°F (≥ -17.8°C)
DF	Transport of deep frozen (not quick frozen) commodities	≥ -20°F (≥ -28.9°C)

ATP Classifications

ATP defines a number of categories:

- Insulated equipment
- Refrigerated equipment ("passive" type systems)
- Mechanically refrigerated equipment
- Heated equipment
- Mechanically refrigerated and heated equipment

Each category has subcategories with specifications. See ATP for more information [38].

Both the TMC and the ATP evaluate the insulation and HVAC unit in order to certify a unit.

10.2.2 Ocean Reefers (Air-Conditioned Containers)

As with trailers, manufacturers who build temperature-controlled ocean reefers adhere to voluntary standards (in this case ISO 1496-2 [39]) that allow the end user to take a "family" approach (i.e., several units of the same make and model) to qualification. The focus of activities should be on the key operational and maintenance elements. This can help to establish a high confidence for this distribution unit operation.

Also called intermodal containers, the most common lengths in use are 20 ft (6 m), 40 ft (12 m), and 53 ft (16 m). They operate using heating, cooling or heating, and cooling systems at a standard range of cargo temperatures: -70°C through +30°C.

The same concepts described in Section 10.2.1 apply to reefers. Because the duration of a shipment may be longer by ocean reefers, they are often specified with features such as:

- Controlled CO₂ level
- Redundant refrigeration systems (i.e., independent duty standby systems)
- Built-in, backup power generation

The standard design supplies the conditioned air through tee slots in the base, with the load acting as an integral part of the design to ensure airflow to the end of the reefer. When the reefer is not fully loaded, it is usual practice to add dunnage or material so that the airflow goes to the end of the reefer, ensuring that the reefer is fully conditioned.

The typical system design maintains the return air temperature at the system set point.

Modern refrigeration systems monitor the temperature difference across the cooling coil, and initiates a defrost cycle when the difference is too small, showing excessive ice buildup.

The performance testing requirements of ISO 1496-2 [39] do not specify the maximum ambient temperature at which the system must operate. (It should be remembered that the net gain is a combination of radiation, conduction, and convection.) ISO requires the cooling capacity to be 125% of the heat load that passes through the walls at an ambient temperature of 38°C and a temperature inside the container of -18°C.

The requirement is for the average temperature in the reefer to meet the specifications. The temperature monitoring points are 100 mm from the side wall.

10.2.3 Insulation Degradation

The degradation of insulation is the same for trailers and reefers. The container floor, roof, walls, and doors are insulated with polyurethane. Polyurethane is a general term used to refer to a family of two part plastics. A blowing agent typically CO₂ with other gases, provides the foamed characteristic of the material along with a very high insulation value. Over time the blowing agent gas migrates out of the substrate and it is replaced with air and water vapor. This transition occurs slowly over a period of years and is normal. The temperature-control unit is sized accordingly to ensure a useful life of 15 years with timely maintenance to protect the core insulation of the trailer/ocean reefer.

Note: Testing performed on a reefer includes ISO 1496-2 Test No. 15b) – Functional test of a thermal container at high ambient temperatures while being cooled by a mechanical refrigeration unit (MRU) [39].

10.2.4 Qualification Approach

The qualification approach is similar for trailers and ocean reefers.

The user requirement for this unit operation is straightforward: the product must be maintained at its designated storage condition and arrive at the destination in acceptable physical condition.

As the industry standards ensure that there is a limited trailer-to-trailer/reefer-to-reefer performance difference for each classification, the qualification approach falls to the end user company to consider the impact of loading patterns, inspectional criteria, maintenance, as well as the specification of an appropriate trailer/ocean reefer classification considering the load temperature requirements and external conditions for a specific route. Companies may use different approaches, conducting a “characterization” test to confirm satisfactory performance, or qualification of a particular type of unit for a given lane. The qualification exercise typically considers the loading and unloading and the load pattern to ensure that proper airflow can be achieved under conditions of heat and cold to maintain the product within the acceptable temperature range. Even temperature distribution is dependent on airflow.

Loading practices that block airflow impede the addition or removal of heat and may result in product temperature that may not be controlled within the required range.

Loading/unloading environments (e.g., in a sheltered garage or in an exposed sunny car park) need to be considered as part of the intended qualification approach as extreme surrounding environmental temperatures impact the time taken for the trailer/ocean reefer to return to their target temperatures.

Consideration needs to be made for operational conditions compared to the test conditions used for the certification tests. The loading method should be documented. An alternative is to use thermal chambers in which the loaded trailer/ocean reefer can be placed to gather both hot and cold conditions.

Note: Where trailers or reefers are used for a new route with significantly different temperatures from those used during certification, remapping or monitoring of shipments may be required to confirm the satisfactory operation of the units.

The data from the standards can be used to support use of different categories of system where appropriate. For example, as mentioned in Section 10.2.1, the F-rated trailer is the most common. A successful study on an F-rated trailer implies that a DF-rated trailer would also be suitable because of the additional insulation and conditioning capacity.

Where a specialist unit is to be used that is not built and tested to recognized standards, commissioning and qualification needs to be carried out in a similar manner as a static CTC. Usually it is not feasible to test at the operational extremes of temperature, so a similar strategy may be used as is used for static CTCs: the hottest and coldest locations are determined through mapping and then monitored. The data from these will show if the heat load exceeds the unit capabilities. If the system provides heating and cooling, then these worst-case locations may differ depending on whether the system is heating or cooling.

Regular reviews of independent monitoring data from the load monitoring systems can help identify any potential performance trends.

10.2.5 Loading Patterns

Loading patterns are the key determinant in ensuring uniform product temperature within the trailer/ocean reefer and should be based upon qualification testing to support best temperature distribution. The product should be loaded on pallets that allow free airflow over the length of the trailer/ocean reefer and the product should not be in contact with the walls. Double stacking refrigerated goods should be carefully considered, as airflow distribution in the rear of the trailer/ocean reefer may not be sufficient to maintain uniform temperature under extreme conditions. Use of air bags can provide support to the load as well as an improved degree of insulation from the sidewall, but they impact airflow. Loads should be suitably braced to prevent movement during transit.

For reefers: The load should be pushed as far forward (toward the machine end) of the reefer as possible, covering whatever floor is not covered with plywood or other material to ensure good air distribution and conditioning of the empty part of the reefer. If this is not done the rear section of the reefer may not be kept at temperature – as a result when power is turned off to unload the reefer may warm up faster. In addition, “chimneys” created by openings between pallets/boxes should be avoided to ensure the air flows back to the back of the reefer.

Inspection Criteria

The quality agreement between the carrier and the company should define the shared responsibilities of inspection.

A pre-use inspection checklist in the procedures should include the following items as they have a direct impact on the suitability and function of the trailer/ocean reefer:

- Check for physical damage
- Confirm temperature set point
- Lack of alarms
- Confirm air distribution chute is intact (if applicable)
- Air supply, air returns, and floor drains are clear of debris
- Check repairs to interior and exterior surfaces

- Confirm door seals are intact
- Note general cleanliness of the unit and conditioning equipment (visual inspection) with no odor present

Points to consider when using an ocean reefer:

- It may take some time to precondition the reefer
- The load should ideally be placed centrally along the axis of the reefer
- Dunnage should be used to ensure airflow to the end of the reefer
- The location of any temperature monitors should be defined to ensure they are measuring product not air temperatures
- Reefers suffer from lack of power during loading/unloading/transshipping. Often thermal blankets are used to mitigate the risk of temperature excursions.

Intermodal containers have unique usage patterns, including power interruptions during loading and unloading operations from a truck to the ship at the origin, then loading to the truck from the ship at the destination and during any transfers *en route*. Various controls are used to minimize these. Some of the better managed ports monitor time without power, and have target maximum power off times they try to maintain; others provide premium services to manage power off durations to within defined maximum times. The end user can use risk mitigation techniques such as using thermal blankets.

11 Appendix 1 – Example User Requirements Specification

This appendix presents sample requirements in each are as an example URS for a 2°C-8°C walk-in cold room in a distribution center.

1. Purpose and Scope

The purpose of this document is to define the user requirements for the direct impact walk-in cold room. The scope of this document applies to the new cold room at the distribution center.

Revise this document as necessary throughout the system or equipment lifecycle. Use it as the basis to assess changes to the system.

2. Definitions and Abbreviations

Term	Definition

Abbreviation	Definition

3. References

Document Identifier	Title
<SOP-yyyy>	<Creation and Management of a User Requirements Specification>

4. System Description

The walk-in cold room provides storage for finished drug products at 2°C to 8°C. The cold room is a stand-alone unit located in the CRT space of the distribution center. It is constructed from pre-insulated panels, mounted to the floor, cooled using two (duty standby) cooling systems. The internal racking is fixed.

5. User Requirements

The cold room user requirements are defined in the tables below. Each table contains user requirements for a functional area or discipline such as Process, Design and Construction, Operations and Maintenance, etc.

The tables are structured as follows:

- ID Number
- Requirements

- Type: a drop-down menu allowing the author to select Quality, Business, or Other as the type of attribute
- Source: to record the source of the requirement

Cold Room User Requirements Specification

A cold room is provided to store packaged finished product or labeled product using a storage temperature of 2°C to 8°C.

6. Process Requirements – Capacity

ID No.	Requirement	Type	Source
6.1	The cold room must have a minimum storage capacity of 300–1200 mm × 800 mm euro pallets, each with a maximum loaded height of 1100 mm and weight of 250 kg.	Business	Functional Requirement/ Design Specification
6.2	There must be a minimum clearance of 150 mm underneath stored material for cleaning.	Quality	Quality Standard
6.3	The location of the cold room racks must provide sufficient space (457 mm) between the rack and the wall to allow access for cleaning and maintenance.	Quality	Quality Standard
6.4	The cold room must be equipped with a racking system suitable for the storage of plastic pallets including back stops.	Business	Functional Requirement/ Design Specification/ Unit Manual
6.5	Cold room design needs to be sized based on the use of electric stacker trucks within the cold room. Space between the racking should be a minimum of 3000 mm.	Business	Functional Requirement/ Design Specification/ Unit Manual
6.6	Each storage location on the rack must to have the facility for a label to designate the location and classification: • Quarantine • Rejected	Quality	Functional Requirement/ SOP
6.7	The cold room should have one entry door. The door to the cold room shall be access controlled.	Business	Functional Requirement/ Design Specification/ Unit Manual

7. Process Requirements – Product Physical Properties

ID No.	Requirement	Type	Source
7.1	Material loaded into the cold room does not need to be 2°C–8°C.	Other	Process/Product Requirements

8. Process Requirements Critical Quality Attributes and Critical Process Parameters

ID No.	Requirement	Type	Source
8.1	The cold room must capable of maintaining all storage locations within the room at 2°C–8°C.	Quality	Product Requirements/ SOP
8.2	The cold room shall be fitted with alarms to detect excursions.	Quality	Product Requirements/ SOP

9. Automation and Records

ID No.	Requirement	Type	Source
9.1	Each cooling system may have an independent control system with a fault alarm (if feasible).	Business	Functional Requirement/ Design Specification/ Unit Manual
9.2	The cold room must have the capability for installing two independent monitoring sensors – the locations for these to be determined during temperature mapping.	Quality	Design Specification/ SOP
9.3	Failure of the primary cooling system should automatically initiate the standby cooling system.	Business	Unit Manual
9.4	Increased temperature above the expected operating range will initiate the standby cooling system.	Business	Unit Manual
9.5	There must be an engineering alarm if a refrigeration plant fails, or develops a fault.	Business	Unit Manual

10. Design and Construction

ID No.	Requirement	Type	Source
10.1	The equipment finishes must be of hygienic design, smooth, cleanable, impervious, and resistant to cleaning/sterilizing agents.	Quality	Functional Requirement/ Design Specification/Unit Manual
10.2	The lighting capacity needs to be based on a luminance minimum of 100 lux at floor level.	Other	Local Regulations
10.3	Equipped with sufficient possibilities to manually open the doors in case of a power loss or emergency.	Other	Unit Manual
10.4	Equipped with fire suppression and fire detection systems that are tested and monitored.	Other	Unit Manual
10.5	Noise levels generated by the equipment shall be less than 75 dBA.	Other	Local Government Requirements
10.6	The door must be adequate for the equipment used to load the storage racks.	Business	Unit Manual
10.7	Door gaskets should be designed to permit ease of maintenance/replacement.	Business	Unit Manual
10.8	The door should capable of being fitted with a device to minimize the impact of door opening for loading/unloading.	Business	Product Requirements/ SOP

11. Utilities

ID No.	Requirement	Type	Source
11.1	Electrical power is available 240/380-415 v ½ phase 50 Hz.	Business	Unit Manual
11.2	Drainage is available.	Business	Facilities Drawings
11.3	Pressurized water is available for fire protection.	Business	Facilities Drawings

12. Operations and Maintenance

ID No.	Requirement	Type	Source
12.1	The system must have a design lifecycle that meets company standard.	Business	Strategy Document
12.2	The equipment must be designed for ease of maintenance with access provisions for internal and external equipment and meet company safety standards.	Business	Unit Manual/Safety Standard

13. Miscellaneous

Not applicable.

12 Appendix 2 – Sample Document: Number and Location of Mapping Sensors

The number of sensors to use in a mapping project, and where to place them in the area to be mapped, are two of the most common questions when mapping. Guidance on the number and placement of mapping sensors for a given unit size or type is provided in Chapter 6. It is recommended that the decisions and rationale supporting sensor placement be documented for each mapped area or unit, as recommended in Chapter 6. This rationale could be a separate document, or a section within a validation plan or mapping protocol document.

Many commercially available CTC units have a volume of less than 2 m³, and can be mapped with the generic sensor placement provided in Figure 5.3. Units in this size class following the recommended sensor placement may not benefit from a documented rationale unless it is necessary to account for an unusual use case.

Similarly, CTC units or areas with volume between 2 m³ and 20 m³ can be mapped with the generic sensor placement provided in Figure 5.3. Because CTCs in this size class may be custom or site-built, they may benefit from a documented rationale.

Finally, controlled temperature warehouses (with volumes over 20 m³) have structures and configurations that are unique. Therefore warehouses are most likely to benefit from a documented rationale for sensor placement.

This chapter provides a sample rationale document supporting sensor placement in a CRT warehouse, defining the number and location of temperature mapping sensors to be used and the load test rationale. This document may be adapted for use in other situations.

12.1 Introduction

This document is intended to provide a record of the rationale used to select the sensor locations for the temperature mapping of Warehouse ABC, and the associated load test strategy.

The mapping will be used to determine the number and location for the permanent monitoring sensors, based on the product storage locations that experience maximum and minimum temperature conditions during simulated normal use.

It should be noted that the product stored has no critical humidity requirements; therefore, humidity will be neither mapped nor monitored.

12.2 Approach

The following points were considered:

- External heat gains and losses
- Internal heat gains and losses

External Heat Gains and Losses

The major heat gains and losses to the warehouse are due to the influences of the external environment, with the most significant being through the north wall and ceiling.²

² In this example, the north wall and ceiling are the only exterior walls. All other walls are adjacent to additional controlled spaces.

All other influences will be consistent; areas within the warehouse which are adjacent to the temperature-controlled storage room, will be air conditioned. During normal use the doors will be opened for loading and unloading, influencing the internal conditions.

Vinyl curtains are placed at each entrance, with inflatable seals to seal onto the truck sides for loading and unloading.

Internal Heat Gains and Losses

- Lighting: the lighting is from low energy fluorescent lamps; these are left on during the working day
- Product: the product stored in the warehouse is supplied at the storage temperature, so will not put a significant heating or cooling load onto the space
- Equipment: the equipment used comprises a stacking forklift truck; this is a relatively small intermittent load
- Personnel: on average there are approximately two people working in the area – the heat gains are, therefore, small and transient
- Air conditioning is by a single system with outlets equally spaced in the aisles as shown on the attached drawing.³

12.3 Sensor Placement

The areas considered during the mapping are those areas where product will be stored.

The sensor locations proposed have been generated using the French Standard FD X15-140 Measurement of air moisture – Climatic and thermostatic chambers – Characterization and verification [22] for guidance.

The chamber has been divided into zones, with each zone supplied by an air conditioning vent.

The end zones are different to the center zones in that they have the end wall/door influences. However, the symmetrical nature of the design of the central zones has allowed the number of mapping sensors proposed to be reduced.

A temperature sensor should be placed on the lowest and highest product storage points, and at each corner of the room on the racking/product nearest the perimeter.

One temperature sensor will be located in the center of these locations at mid-level.

³ A drawing could be attached for clarity.

12.4 Monitoring System Sensor Placement

The mapping will be used to define the number and location of the permanent locations that will be used for mounting the system monitoring sensors.

Load Testing

The warehouse will be mapped using two “load” scenarios. In both cases there will be a minimum amount of thermal mass so that the impact of the change is seen quickly:

- “Empty”
 - The warehouse will be mapped empty, so that there are no constraints on airflow, with the lowest average airflow velocity in the room.
- “Full”
 - The unit is considered as a number of zones that operate independently; the front (i.e., nearest the door) zone and a half will be filled with empty boxes simulating the largest stored product case. This scenario will give minimum thermal mass, but maximum interruption of airflow, allowing the effect on the temperature distribution to be seen in the filled zones.

Note: the area selected was based on the fact that the warehouse is symmetrical.

Summer/Winter Conditions

The mapping sensor layout is uniform when the data is analyzed the following will be considered:

- The North wall of the warehouse and the roof are exposed to external conditions, potentially influencing the temperature at high level and along that wall
- Local weather data from the Meteorological Office provided weather data showing the 90% and 95% summer/winter conditions to be:
 - Summer x
 - Winter y

The design is specified to maintain internal conditions in all conditions up to and including the 95% limits.

The initial mapping is to be in summer, the data obtained from this may be used to recommend a reduced number of sensors to be maintained through to the winter season. The mapping will be considered complete when seasonal variation has been experienced that exceeds the 90% design conditions.

13 Appendix 3 – Example Use of Measurement Uncertainty

Note: It is the organization's responsibility use a risk-based approach on how measurement uncertainty is addressed. Measurement uncertainty is usually described in the risk assessment or procedure. Some companies justify it through engineering methods while others acknowledge that it exists but it is not built into their process. Another approach is to define measurement uncertainty and then apply laboratory rounding rules. What makes addressing measurement uncertainty a success is the company's robust or proven strategy or procedure to base their practice on. This appendix is one example.

This example describes the use of measurement uncertainty to determine the monitoring sensor locations from the mapping data. For further information on measurement uncertainty see the NSAI – National Metrology Laboratory (NML) [40].⁴

The maximum temperature and minimum temperature should be found during the mapping activity. This data can be analyzed to determine the locations representing temperature extremes.

- Where monitoring sensors can be located at the maximum and minimum temperature spots, alarm limits can be determined based on Critical Operational Data (COD)/Measurement Uncertainty (MU) of sensor and calibration standards
 - Where monitoring sensors cannot be located at the maximum and minimum temperature spots, an offset should be added to the COD/MU to account for the offset between the maximum and minimum temperature spots and the monitoring sensor(s)
1. For alarm sensor(s) installed at the locations of maximum and minimum temperature as determined in mapping, the COD/MU for each device should be included when establishing upper and lower limits for the alarm range
 2. For alarm sensors permanently mounted at a location other than the maximum and minimum temperature locations as determined in mapping, the offset between the maximum and minimum temperature spots and the monitoring sensor(s) should be determined. When establishing upper and lower limits for the alarm range, any offsets due to the sensor location should be included in addition to the COD/MU for each device

In addition, the measurement uncertainty of both types of sensors (i.e., the sensors used during temperature mapping and the permanently installed monitoring sensors) should be considered in establishing the alarm limits. Measurement uncertainty calculation and application:

For the low limit: $DL \geq \text{Limit} + (GB - U_{sr})$

For the high limit: $DL \leq \text{Limit} - (GB - U_{sr})$

Where:

DL = Decision Limit

Limit = PAR (Proven Acceptable Range), Action or Safe Operating Limits

Guard Band (GB) = $k * u_{system}$

k = coverage factor which must be ≥ 2.00

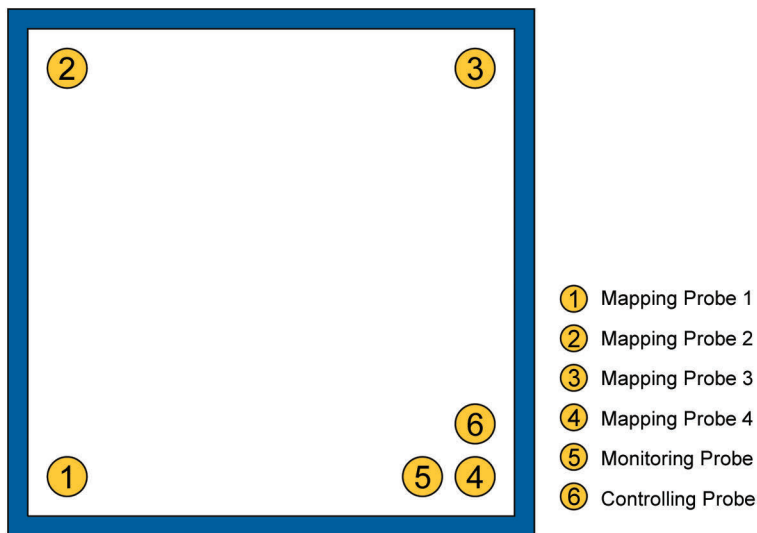
u_{system} = calculated measurement uncertainty

U_{sr} = Required Uncertainty of Limits (usually = 0 to indicate no allowable excursion from the alarm value)

⁴ Dubhaltach Mac Lochlainn of the National Standards Authority of Ireland (NSAI) provides further guidance on the use of measurement uncertainty. See the ISPE website for further information.

Examples

Figure 13.1



Given: $u_{T1} = u_{T2} = u_{T3} = u_{T4} = u_{T5} = u_{T6} = 0.125^{\circ}\text{C}$

U_{sr} for each sensor = 0°C (no data available to allow for temperatures above or below the Proven Acceptable Range (PAR))

$k = 2$ (corresponds to confidence $\geq 97.7\%$ for single sided alarm; 2σ)

PAR = 2°C to 8°C

$GB = u_s * k = 0.125^{\circ}\text{C} * 2 = 0.25^{\circ}\text{C}$

Example A

If all sensors are within $GB = 0.25^{\circ}\text{C}$ of each other, the differences in temperature can be attributed to the measurement uncertainty of the sensors and the location of the monitoring sensor can be at any one of the locations of the mapping sensors. The alarm limits are:

$$DL_H = 8 - (GB - U_{sr}) = 7.75^{\circ}\text{C}$$

$$DL_L = 2 + (GB - U_{sr}) = 2.25^{\circ}\text{C}$$

Example B

If sensor 1 shows highest temperature at 7.0°C and sensor 4 shows lowest temperature of 6.5°C at that same time, the following applies:

- If you leave the monitoring sensor 5 in the existing location (same as sensor 4), include 0.5°C offset and measurement uncertainty for sensors 1 and 4 in calculation (i) for the high alarm limit. The low alarm limit calculation (ii) remains the same as in Example A:

$$i. \quad DL_H = 8 - (0.5 + 2\sqrt{(u_{T1}^2 + u_{T4}^2)} - U_{sr}) = 8 - (0.5 + 2(0.177)) - 0 = 7.15^{\circ}\text{C}$$

$$ii. \quad DL_L = 2 + (GB - U_{sr}) = 2.25^{\circ}\text{C}$$

- b. If you relocate the monitoring sensor 5 to the location of the high temperature (position 1), include 0.5°C offset and measurement uncertainty for sensors 1 and 4 in calculation (ii) for the low alarm limit. The high alarm limit calculation (i) remains the same as in Example A:

i. $DL_H = 8 - (GB - U_{sr}) = 7.75^\circ\text{C}$

ii. $DL_L = 2 + (0.5 + 2\sqrt{(u_{T1}^2 + u_{T4}^2)} - U_{sr}) = 2.85^\circ\text{C}$

Example C

If sensor 1 shows highest temperature at 7.0°C and sensor 2 shows lowest temperature of 6.0°C and sensor 4 shows value of 6.5°C at the same time as the sensor 1's high and sensor 2's low, the following applies:

- a. If you leave the monitoring sensor 5 in the existing location, include 0.5°C offset and measurement uncertainty for sensors 1 and 4 in calculation (i) for the high alarm limit and the 0.5°C offset and measurement uncertainty for sensors 2 and 4 in calculation (ii) for the low alarm limit:

i. $DL_H = 8 - (0.5 + 2\sqrt{(u_{T1}^2 + u_{T4}^2)} - U_{sr}) = 7.15^\circ\text{C}$

ii. $DL_L = 2 + (0.5 + 2\sqrt{(u_{T2}^2 + u_{T4}^2)} - U_{sr}) = 2.85^\circ\text{C}$

- b. If you relocate the monitoring sensor to the location of the high temperature, include 1.0°C offset (difference between the high and low temperature locations) and measurement uncertainty for sensors 1 and 2 in calculation (ii) for the low alarm limit. The high alarm limit calculation (i) remains the same as in Example A:

i. $DL_H = 8 - (GB - U_{sr}) = 7.75^\circ\text{C}$

ii. $DL_L = 2 + (1.0 + 2\sqrt{(u_{T2}^2 + u_{T1}^2)} - U_{sr}) = 3.35^\circ\text{C}$

- c. If you relocate the monitoring sensor 5 to the location of the low temperature (position 2), include 1.0°C offset (difference between the high and low temperature locations) and measurement uncertainty for sensors 1 and 2 in calculation (i) for the high alarm limit. The low alarm limit calculation (ii) remains the same as in Example A:

i. $DL_H = 8 - (1.0 + 2\sqrt{(u_{T1}^2 + u_{T2}^2)} - U_{sr}) = 6.65^\circ\text{C}$

ii. $DL_L = 2 + (GB - U_{sr}) = 2.25^\circ\text{C}$

- d. If you place monitoring sensors at each of the high and low temperature locations, then there is no offset and only the single device measurement uncertainty is included in the calculation. The monitoring sensor in the high temperature location will only alarm for high temperature excursions (i) and the monitoring sensor in the low temperature location will only alarm for the low temperature excursions (ii):

i. $DL_H = 8 - (GB - U_{sr}) = 7.75^\circ\text{C}$

ii. $DL_L = 2 + (GB - U_{sr}) = 2.25^\circ\text{C}$

Example Protocol

1. Define system
 - a. Description: -20°C freezer, 20 ft × 20 ft × 10 ft (6 m x 6 m x 3 m) with redundant compressor/evaporator systems located in CRT warehouse with single door opening to warehouse and one wall exposed to external temperatures
 - b. Process Parameters (Acceptance Criteria): -17°C to -3°C. *U_{sr} is considered "0" because the product will be considered unacceptable if the temperature goes outside of these parameters as defined in stability studies.*
2. Establish rationale for sensor locations. Using the guidance in Section 6.3 of this Guide, identify sensor locations and document them as follows (this example shows an excessive number of sensors for the space only to demonstrate various rationales):
 - a. Sensors 1 to 9: Basic layout: high and low at each corner plus one in middle of room
 - b. Sensors 10 to 25: high, middle and low on fixed racks due to restriction of air flow and temperature stratification
 - c. Sensor 26: external door to unconditioned space with potential impact on room condition
 - d. Sensors 27, 28, and 29: external wall due to temperature extremes greater than 10°C from room set point
 - e. Sensor 30: potential exposure to heat from defrost cycles of the system
 - f. Sensor 31: monitoring external conditions (for information only; not subject to acceptance criteria)
 - g. Sensor 32: control and/or monitoring sensor location
3. Define Equipment
 - a. Model
 - b. Description
 - c. Calibration information
 - i. Identify measurement uncertainty of sensors and data collector (from Calibration Report)
 - Accuracy = $\pm 0.125^{\circ}\text{C}$
 - ii. Calculate GB for system
 - $\text{GB} = k * u_s = 2 * 0.125 = 0.25^{\circ}\text{C}$
 - d. Define Acceptance Criteria
 - i. For the low limit: $\text{DL} \geq \text{Limit} + (\text{GB} - \text{U}_{sr}) = (-17) + 0.25 = -16.75^{\circ}\text{C}$
 - ii. For the high limit: $\text{DL} \leq \text{Limit} - (\text{GB} - \text{U}_{sr}) = (-3) - 0.25 = -3.25^{\circ}\text{C}$

- e. Setup parameters
 - i. Sample rate: once every 5 minutes
 - ii. Test time: at least 24 hours (include time to capture defrost cycles and switchover of redundant systems)
- 4. Place equipment in space
- 5. Verify that values are being recorded
- 6. Record start time
- 7. Record end time
- 8. Evaluate data for temperature uniformity for each sensor
 - a. Minimum reading with time and sensor number
 - b. Maximum reading with time and sensor number
 - c. If minimum and maximum readings are within the Acceptance Criteria for the room, then the uniformity is acceptable for this system
 - d. Compare minimums and maximums to the monitoring device values and determine if relocation of the monitoring sensor is required or an offset in the alarm strategy is required or other action is necessary
 - e. Mean kinetic temperature (for CRT warehouse with areas allowing specified temperature excursions over specified periods of time only)

14 Appendix 4 – Sample Monitoring Sensor Strategy Report

Note: This is a sample document presenting a recommendation for monitoring sensor locations based on thermometric data from the commissioning monitoring sensor. The amount and location of the sensors should be determined based on the CTC storage boundaries, racking configurations, and storage material requirements. See Section 6.3 for more information. The sensors were located in thermal fluid as this is the preference of the system owner who provided this example.

14.1 Introduction

The freezer was mapped using 36 independent sensors connected to a temperature mapping system during commissioning, with the mapping sensors located as shown in Figure 14.1. A series of tests were performed to ensure that the system met performance specifications during simulated use.

Internal CTC temperatures will be monitored by the BMS, with an independent chart recorder used as a backup system. The mapping data obtained during the commissioning tests was used to determine the BMS and chart recorder sensor locations.

14.2 Product Temperature

The monitored room air temperatures are intended to be kept between -15°C and -25°C. The product is placed in the room at temperatures within this range; so it is not considered necessary to separately monitor the product temperature.

Temperature monitoring sensors will be located in a small container of glycerin, to simulate the thermal response of the product packaging.

Note: in this example, the site had data confirming that this was equivalent to the material to be stored in the unit.

Monitoring System

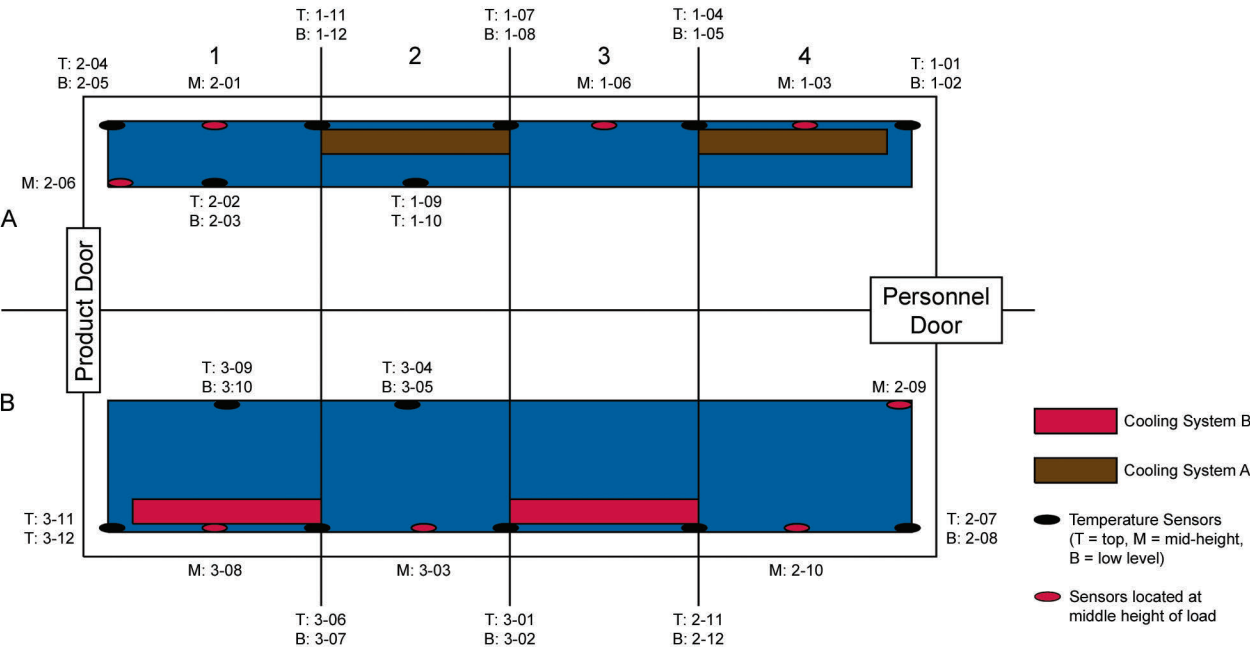
The BMS and chart recorder sensor elements will be placed in the room and monitor worst-case air temperature based on the conclusion contained in this report. There are six temperature sensors for use as freezer temperature monitors.

For a CTC with a broad operating range (> 6°C), a failure would most likely be caused by significant mechanical failure that is highly detectable. [1]

Approach

The following test results were used as the basis for determining the BMS and chart recorder sensor locations, so that they represent the worst-case conditions (minimum and maximum temperatures) during simulated normal use of the freezer. The sensors are intended to be placed in a location that will see the most rapid changes in the air temperature, in order to achieve the earliest possible alarm in case of an out of limit operation.

Figure 14.1: Mapping Sensor Locations



Commissioning Test 1 – Baseline Readings

This test was used to confirm the proper operation of the refrigeration systems by monitoring system operational parameters. The freezer operation was monitored for a minimum of 24 hours, in steady state conditions with no door opening.

Commissioning Test 2 – Open-Door Test Empty

This test was used to determine the effect on the freezer temperature while the freezer sliding door was left open. The test was conducted with the system furthest from the door (Cooling System A) operating, to provide worst-case conditions i.e., minimum forced-air mixing adjacent to the door). The door was opened for a one-hour period with the internal vinyl curtains closed; the test period was adequate to indicate the conditions that the system would maintain if the door was left open.

Commissioning Test 3 – Open-Door Test Loaded

This test was used to demonstrate the system performance loaded, using empty boxes to simulate the load (maximum volume, minimum thermal mass).

The test was conducted with the system furthest from the door (Cooling System A) operating, to provide worst-case conditions i.e., minimum forced-air mixing adjacent to the door). For each load, an open-door test was conducted. The door was opened for a one-hour period with the internal vinyl curtains closed; the test period was adequate to indicate the conditions the system would maintain if the door was left open.

Note: The intent of the open-door test is to understand the impact on the room conditions; the data can be used to define operational constraints in the operations SOP, for example: “Do not keep the door open for more than 5 minutes.”

Table 14.1: Test Results

1. Baseline Readings, Empty	Temperature (°C)	Location
Minimum average temperature		
Maximum average temperature		
Minimum temperature		
Maximum temperature		
2. Open-Door Empty	Temperature (°C)	Location
Minimum average temperature		
Maximum average temperature		
Minimum temperature		
Maximum temperature		
3. Open-Door Loaded	Temperature (°C)	Location
Minimum average temperature		
Maximum average temperature		
Minimum temperature		
Maximum temperature		

Considerations

The locations recommended for the BMS sensors are those generally representing the minimum and maximum temperatures during the tests, considering the following:

- The sensor should ideally be accessible for ease of calibration
- The sensor should provide a reading reasonably representative of the minimum and maximum temperatures product will be exposed to during normal operation of the freezer
- The highest risks of temperature excursions are during door opening. During operation, normal traffic into the room is performed with the opened door through the curtains of the cold room
- The sensors should be distributed, as much as possible, to try to represent the whole freezer
- The central area of the freezer is considered to be more stable in terms of temperature than the perimeter zones, as there are no doors and the external environmental influences are significantly less
- The accuracy of the mapping sensors used is $\pm 0.5^{\circ}\text{C}$

Minimum Temperature Locations

The location experiencing the minimum average temperature and minimum temperature during test 1 was location x. The minimum value recorded during testing was $x^{\circ}\text{C}$, recorded at sensor x during test 1 and at location x during test 2.

Maximum Temperature Locations

The maximum temperature locations are influenced by the defrosting of the unit coolers and the door opening. This creates a heat gain as the dense cooled air leaves the freezer and warm air enters the room to replace it, continuing to enter after the door is closed through the vents to ensure that the freezer is not under pressure or vacuum.

The location showing the maximum temperature during test 1 was x. The maximum average temperature was x°C. During test 2 the maximum average temperature of x°C was obtained at location x with the maximum of x°C obtained at location x.

14.3 Conclusion

Based on the data the following locations were considered for the BMS sensors.

The locations selected for the temperature monitoring sensors are:

- The locations for the **minimum** temperature are: _____
- The locations for the **maximum** temperature are: _____

Note: The following graphs showing the mapping data for each test were attached to the example:

- Data from all mapping sensors
- Data the sensors selected to provide the monitoring data

These showed that the monitoring locations selected provided a reasonable record of the worst-case locations (i.e., hottest and coldest).

15 Appendix 5 – Model System Risk Assessment for a Walk-in Cold Room/Freezer

15.1 Introduction

This is an example of a model SRA for a cold room developed based on a typical unit. The concept of a model is that this version can be used within the company as a template to develop a specific SRA, thus saving time and helping to achieve a level of consistency.

The SRA is focused on the quality critical requirements. It does not consider business or safety requirements; these aspects are addressed through the engineering process.

Notes:

- The unit in this example is fitted with a vinyl curtain inside the door. This allows unrestricted access, but when no one is passing through the curtain, it prevents the dense cold air from leaving the room to be replaced by room air. This saves energy and helps maintain the conditions in the room.
- One of the risk controls used is the mapping conducted as part of the commissioning (operational verification) of the system. This should be carried out following the test strategy developed by engineering and approved by the Quality Unit. See Chapter 5 for details.
- In many cases humidity is not a critical parameter for the materials being stored in a cold room, and it is therefore, not controlled – as is the case for this example. Some companies dehumidify cold rooms to reduce the need to defrost the cooling coils and to reduce the risk of microbial/sporicidal growth [41]. Dehumidification in freezers also helps reduce the formation of ice.

15.2 The Model System Risk Assessment

15.2.1 Instructions

Review the system description and compare with the system to be qualified – editing the description and risk assessment to suit the system to be qualified.

Notes: The risk assessment process scope is defined in the title; system “feeds” are considered qualified; system product contact materials are considered to be verified through the commissioning process, thus they are not in the scope of the assessment; the maintenance and calibration quality system is considered robust, i.e., calibration failure is not considered as a risk.

15.2.2 System Description

For this example, the internal dimensions of the cold room are 5 m wide by 10 m long, and 3 m high.

The room is constructed of 100 mm pre-insulated panels making up a complete structure, comprising the walls, floor and ceiling with a single manually operated door; a vinyl strip curtain is mounted inside the door to reduce the loss of cold air/ingress of outside air when the door is used.

The room is fitted with racks on both sides to store the product, with product being manually loaded/unloaded into the room.

The room has two independent cooling systems, each system with a unit coolers mounted at each end of the room, blowing conditioned air down the length of the room.

15.2.3 Control System Description

The cooling systems operate in duty standby mode automatically changing the duty system on a daily basis. The standby system will automatically start if there is an alarm from the duty system, or when a high temperature is detected by the control system.

The units are set so that the duty unit has continuously operating evaporator fans to maintain airflow with the standby unit fans only operating when the unit is required.

System defrost is a defined cycle using electric heaters in the coils set up on a user defined time base.

The refrigeration systems are powered from the utility electrical system with standby generator backup, the control and monitoring systems are powered by an Uninterruptible Power Supply (UPS).

The system will provide the following alarms to the BMS, and locally to a lamp/sounder:

- Refrigeration System Failure (details of the specific alarm can be found on the refrigeration control system)
- Room Low temperature
- Room High Temperature

Note: These are engineering alarms. Quality alarms come from the independent monitoring system.

Table 15.1: System Risk Assessment Example

Ref	Process Description	CQA	CPP	Impact on CQA	How CQA can be Impacted	Recipe Parameter (if applicable)	Is there an associated alarm?	Design Control	Procedural Controls	Comment	Residual Risk Determination
1	Cold room operates continuously with duty cooling system cycling – duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Quality (Product Temperature)	Control Room Temperature between 2°C–8°C	Direct	Product temperature can be OOS, if controlled temperature is out of range due to failure of refrigeration system/ defrost.	N/A	Yes (Independent temperature monitoring probe)	The system has duty standby systems – the standby system will become active if the duty system fails. If the controlled temperature exceeds high limits, standby system will be activated until the temperature comes back into range.	SOP defines action if independent monitoring system alarms.	Testing will confirm system changeover functionality is as specified. Alarm time delay is to be determined during commissioning activities.	Low
2	Cold room operates continuously with duty cooling system cycling – duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Quality (Product Temperature)	Control Room Temperature between 2°C–8°C	Direct	Product temperature can be OOS, if controlled temperature is out of range due to door opened too long.	N/A	Yes (Independent temperature monitoring probe)	A vinyl curtain is installed inside the cold room doorway to minimize the impact of an open door.	SOP will require the cold room door to be closed after use and defines action if independent monitoring system alarms.	Open-door testing will determine if there is a limit for open-door test, and alarm delay times will be applied to independent monitoring system alarms.	Low

Note: See **Process Description** for Operations Sequence/Process Flow.

Table 15.1: System Risk Assessment Example (continued)

Ref	Process Description	CQA	CPP	Impact on CQA	How CQA can be Impacted	Recipe Parameter (if applicable)	Is there an associated alarm?	Design Control	Procedural Controls	Comment	Residual Risk Determination
3	Cold room operates continuously with duty cooling system cycling - duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Quality (Product Temperature)	Room Temperature Distribution (airflow)	Direct	Product temperature distribution can be OOS due to evaporator unit fan failure causing inadequate airflow.	N/A	Yes (fan failure alarm on controller and alarm from independent temperature monitoring probe)	The system has duty standby systems – the standby system will become active if the duty system fails or the controlled temperature exceeds high limits.	SOP defines action if independent monitoring system alarms.	Cold room mapping will determine the location(s) for the independent monitoring probes(s) covering the worst-case locations (hottest/ coldest).	Low
4	Cold room operates continuously with duty cooling system cycling – duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Quality (Product Temperature)	Room Temperature Distribution (airflow)	Direct	Product temperature distribution can be OOS due to stored product interfering with airflow.	N/A	Yes (Independent temperature monitoring probe)	The design defines the possible storage locations through the shelving positions.	SOP defines acceptable product storage locations to avoid interference with airflow.	Cold room mapping will encompass the operating range of the cold room minimum and maximum stored product through a defined testing strategy agreed to by the Quality Unit.	Low

Note: See **Process Description** for Operations Sequence/Process Flow.

Table 15.1: System Risk Assessment Example (continued)

Ref	Process Description	CQA	CPP	Impact on CQA	How CQA can be Impacted	Recipe Parameter (if applicable)	Is there an associated alarm?	Design Control	Procedural Controls	Comment	Residual Risk Determination
5	Cold room operates continuously with duty cooling system cycling - duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Purity (Product)	N/A	N/A	High humidity in cooler would provide an environment conducive to mold growth.	N/A	No	N/A	SOP defines required preventive maintenance. Annual PM will require visual inspection and cleaning of the cold room as needed.	N/A	Low
6	Cold room operates continuously with duty cooling system cycling – duty system has continuously running fans, control of cooling fluid is by thermostat, system used time based defrost; unit used as required – loaded and unloaded – material entering is within the storage temperature range	Purity (Product)	N/A	N/A	Condensate could drip onto stored materials during defrost.	N/A	No	Insulated condensate pans under all evaporators. Insulated condensate drain lines from the drain pans out of the cold room.	There will be a planned maintenance routine for regular visual inspection to ensure integrity of insulation and ensure there is no dripping of condensate.	Insulation installation is verified during commissioning. Design Control is captured in the company engineering standard.	Low

Note: See **Process Description** for Operations Sequence/Process Flow.

Table 15.2: Alarm Classification

Ref	Operations Sequence/ Process Flow	Alarm Type	System Quality Attribute	System Process Parameter	Alarm	Comment
1	See Process Description	Quality (Product Temperature)	Quality (Product Temperature)	Control Room Temperature between 2°C–8°C	Yes (Independent temperature monitoring probe)	The independent system is the quality system of record
2	See Process Description	Quality (Product Temperature)	Quality (Product Temperature)	Control Room Temperature between 2°C–8°C	Yes (Independent temperature monitoring probe)	The independent system is the quality system of record
3	See Process Description	Quality (Product Temperature)	Quality (Product Temperature)	Room Temperature Distribution (airflow)	Yes (fan failure alarm on controller and alarm from independent temperature monitoring probe)	The independent system is the quality system of record
4	See Process Description	Quality (Product Temperature)	Quality (Product Temperature)	Room Temperature Distribution (airflow)	Yes (Independent temperature monitoring probe)	The independent system is the quality system of record

Note:

- **Quality Critical:** Alarm that is triggered from a parameter that has direct impact on the system quality attributes.
- **Process Critical:** Alarm that is triggered from a parameter that has indirect impact on the system quality attributes.

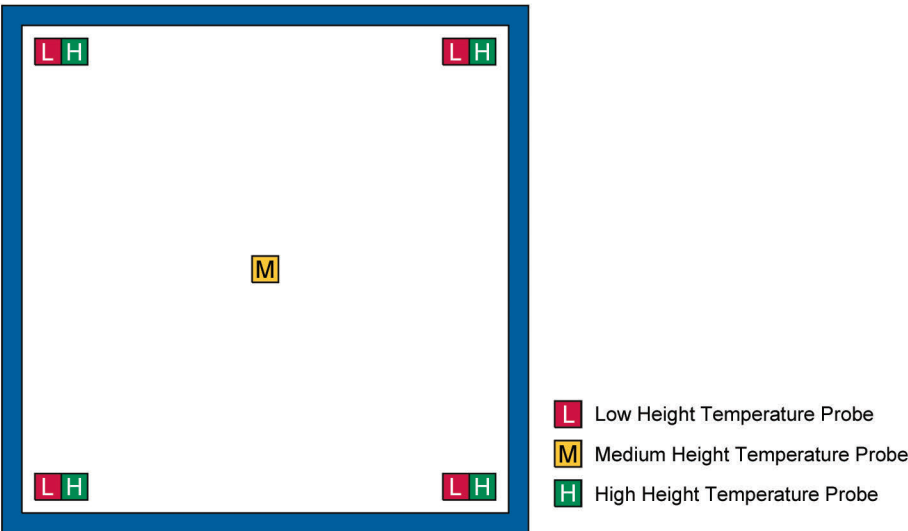
16 Appendix 6 – Example Testing Strategies

16.1 Example 1: Mapping Strategy for a Walk-in Conditioned Space

This Appendix presents **an example** of developing the mapping strategy for a 20 ft × 20 ft × 10 ft (6 m × 6 m × 3 m) space for storage of up to ten double stacked pallets of materials at a controlled room temperature.

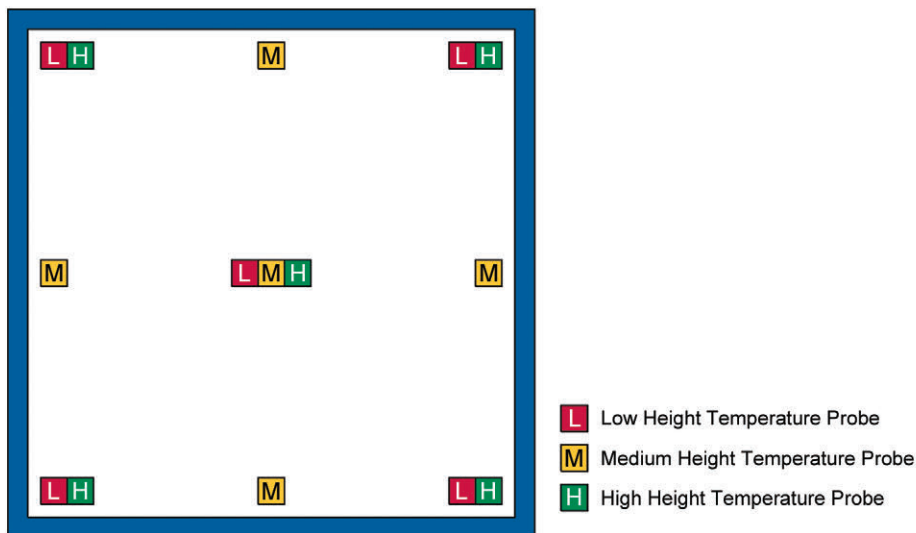
1. The purpose of the facility should be identified. This example facility is to be used for storage of up to ten double stacked pallets of materials at a controlled room temperature. There are no process steps (e.g., rolling heated material into the space) that could cause variable heat loads or impact the temperature of other materials located nearby. Additional mapping probes to detect potential impact on the spaces temperature profile are, therefore, not needed.
2. The size of the facility is given as 20 ft × 20 ft × 10 ft (6 m × 6 m × 3 m). In general, the three-dimensional space should be divided into at least three planes:
 - Upper level of space
 - Middle level of space
 - Lower level of space
3. Using the concepts described for spaces up to 70 cubic ft (2 m³), probes should be located on the upper and lower levels at the four corners of the space and a single probe in the center of the box (see Figure 16.1).

Figure 16.1: Mapping Locations for a CRT Space up to 70 cubic feet (2 cubic meters)



4. For spaces up to 700 cubic ft (20 m³), add a probe in the center of each face of the spaces walls, ceiling and floor (see Figure 16.2).

Figure 16.2: Mapping Locations for a CRT Space up to 700 cubic feet (20 cubic meters)

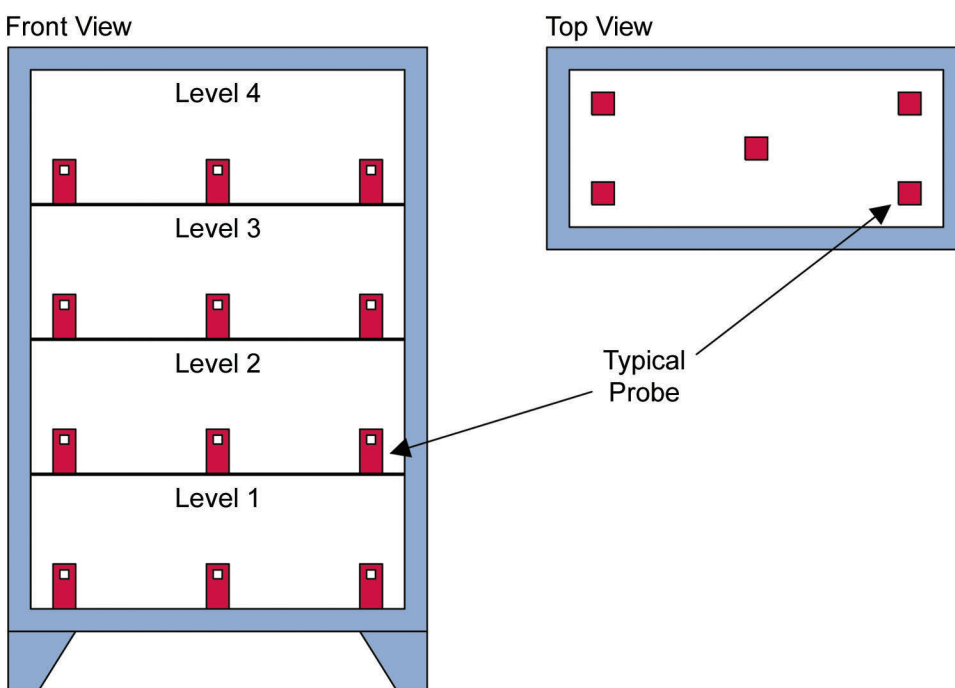


5. For spaces greater than 700 cubic ft (20 m³), the configuration given in Figure 16.2 should be used and supplementary probes should be added, as appropriate, to demonstrate uniformity of temperature and conformance with product requirements. There is no definitive standard or regulatory guidance to define the number of thermocouples or temperature recording devices used in mapping a space. A rule of thumb based on review of existing facilities is to allow a distance no greater than 30 ft (9 m) between both the horizontal and vertical direction of sensors. For this example, the volume is 4000 cubic ft (113 m³), so the configuration in Figure 16.2 is appropriate as a starting point. Vertical and horizontal distances are not greater than 30 ft (9 m) so additional probes are not suggested beyond the probes identified in Figure 16.2.
6. The air supply for the space is a ceiling diffuser with four returns located in each of the corners. There are no racks, shelves, or other obstructions to airflow within the space. The pallets have the potential to be stacked to a height of 6 ft (1.8 m), so will not create significant obstruction to airflow. Material will not be stored near the ceiling along the walls, so there is no concern for "hotspots" in these locations that would not already be detected by the probes mounted at the middle plane along each of the walls. No additional probes would be suggested beyond the probes identified in Figure 16.2.
7. The room is inside a general purpose building that is conditioned throughout the year to 20°C (68°F) so there is no concern for external temperature extremes compared to the controlled environment that could penetrate the walls/ceilings/floors, nor create temperature control issues when the door is opened to move material or people. No additional probes would be suggested (such as near windows, doors, walls, etc.) beyond the probes identified in Figure 16.2.
8. There are lights in the facility; however, there is no product to be stored near the fixtures that might be impacted by heat generated during operation. There are no motors, compressors, computers, or other heat generating equipment located in the space that might impact temperatures in the immediate vicinity. There are no materials being stored in the facility that have the potential for exothermic/endothermic reactions that might impact temperatures in the immediate vicinity. No additional probes would be suggested beyond the probes identified in Figure 16.2.

9. The HVAC system consists of a dedicated single air unit which is a recirculation system with a supply fan, cooling and heating coils. It is located remotely from the space. There is no redundancy which might prescribe the need to test conditions for each unit or during switchover. There are no defrost cycles that would need to be identified and included in the testing cycle, and with no defrost cycles, or significantly colder/hotter temperatures of supply air, there is no need for additional probes to define potential impact on product stored in the immediate path of supply air. Failure of the air system could have an impact on environmental conditions in the space. This potential impact should be identified and accounted for in the design of the system. Appropriate testing would then be performed to demonstrate capability during commissioning. For this system, the facility is located inside an environment already controlled to conditions within the limits of the facility; therefore, failure of the air system would not impact the temperature of the space. If there were a potential for failure, the testing strategy would include testing under failure conditions to demonstrate how long the facility can maintain conditions appropriate for storage.

16.2 Example 2: Mapping Strategy for a Refrigerator

Figure 16.3: Mapping Locations for a Refrigerator

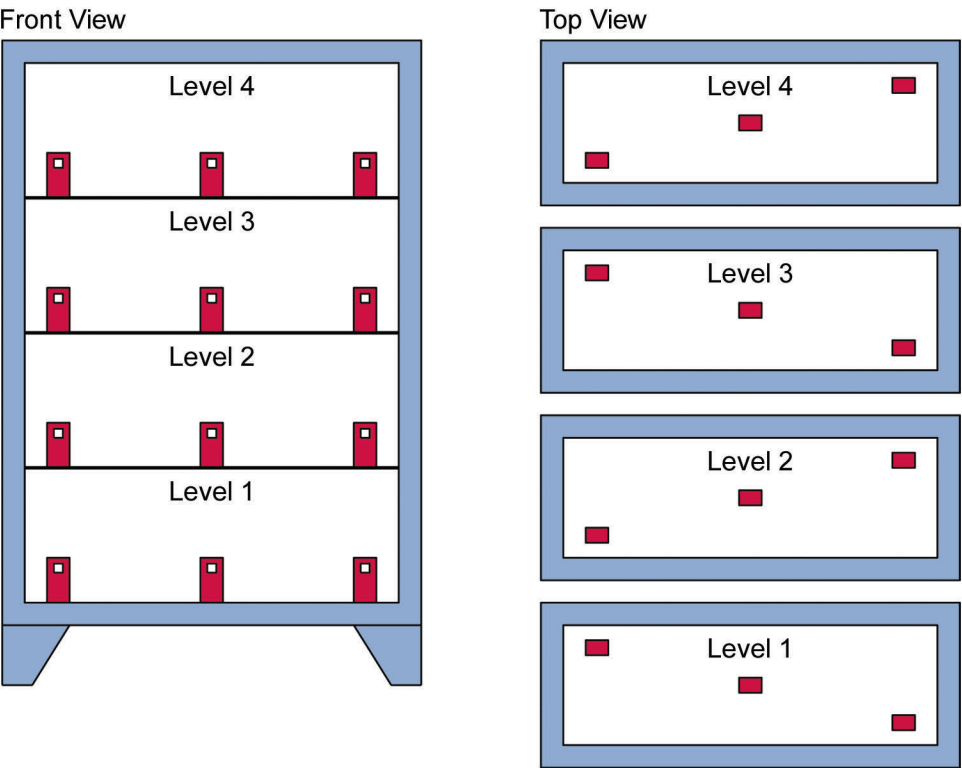


A mapping strategy for standard unit refrigerators and walk-in units follows the same development steps as described in Example 1 for a walk-in unit. Due to the potential for airflow blockage from the shelves and/or material stored on the shelves, each level should be considered a mapping zone with a complement of probes assigned to each level.

The strategy should consider the unit and potential to relocate intermediate shelves, ideally defining the lower and upper limits, with the test load designed so that shelving may be moved as required during operation. One concept that has been successfully used is to define the limiting parameters for the load on a self-adhesive label attached to the unit door e.g., allow 3 in (75 mm) clearance on the side front and rear of the unit, lowest shelf position 4 in (100 mm) from the bottom of the unit, minimum top clearance 4 in (100 mm), maximum room temperature load 20 lb (9 Kg).

Figure 16.3 shows a typical layout using five probes per shelf. If shelves allow circulation, the intermediate shelf probes may be replaced with a probe in the center of the refrigerator. Another option is to place three probes per shelf, alternating corners (front and back) on each level Figure 16.4. An assessment of the potential circulation will dictate the density option, five per shelf for the poorest circulation (solid shelves densely packed), three per shelf for intermediate circulation (wire shelves packed very full), and probes in four corners and middle of refrigerator for good circulation (wire shelves with lots of space for air circulation between contents).

Figure 16.4: Alternative Mapping Locations for a Refrigerator

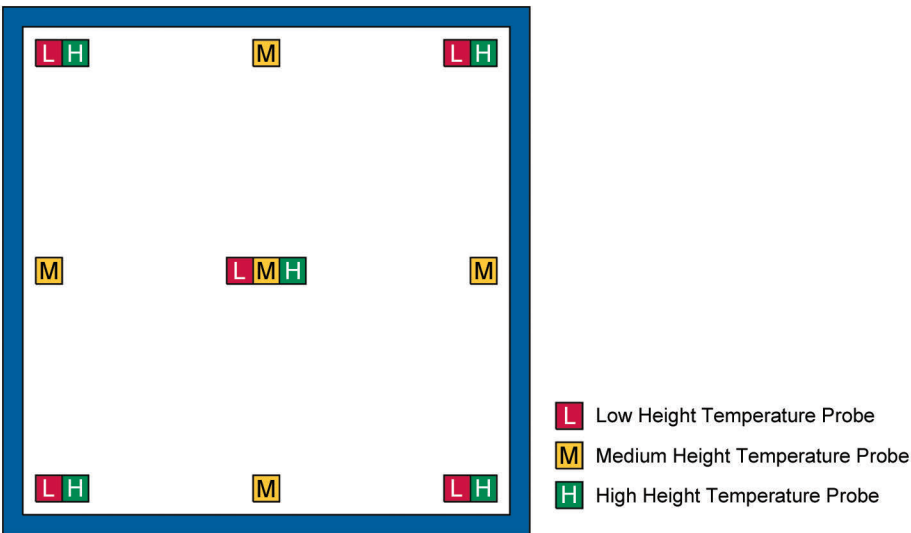


Temperatures may be monitored in the load or a suitable buffer to record the load temperature rather than the air temperature if there is concern for short temporary fluctuations outside of the acceptable ranges due to defrost cycles, door openings, or other predictable occurrences. The air temperature is affected more by repeatedly opening and closing the door for smaller systems, due to the temperature differences between the internal space and the surrounding environment. The volume of stored material can act as a “heat sink” to maintain the temperature of the space. These loads can be either actual product or simulated product with similar thermal properties. If using simulated product, what is being used and how it is comparable (e.g., 20 ml of water is being used instead of insulin because the thermal properties of water are identical to insulin and will behave in the same manner as insulin stored in this facility and the smallest container stored in the facility is 20 ml) should be documented. The internal air temperature distribution should be mapped in the empty and full state to demonstrate operating capability throughout all states of content. Full state is defined as the maximum amount anticipated being stored in the unit (this should follow manufacturer’s recommendations and be aligned with the requirements in the URS) provided the rationale and control over operational conditions is documented. Afterwards temperature distribution should be checked under conditions of normal use.

16.3 Example 3: Warehouse Mapping Layout and Rationale

The development of a mapping strategy for a warehouse should follow the steps identified above. For example, start with the basic room configuration identified in Figure 16.5. Temperature probes are positioned so that there is not more than 30 ft (9 m) vertical separation or 30 ft (9 m) of horizontal separation.

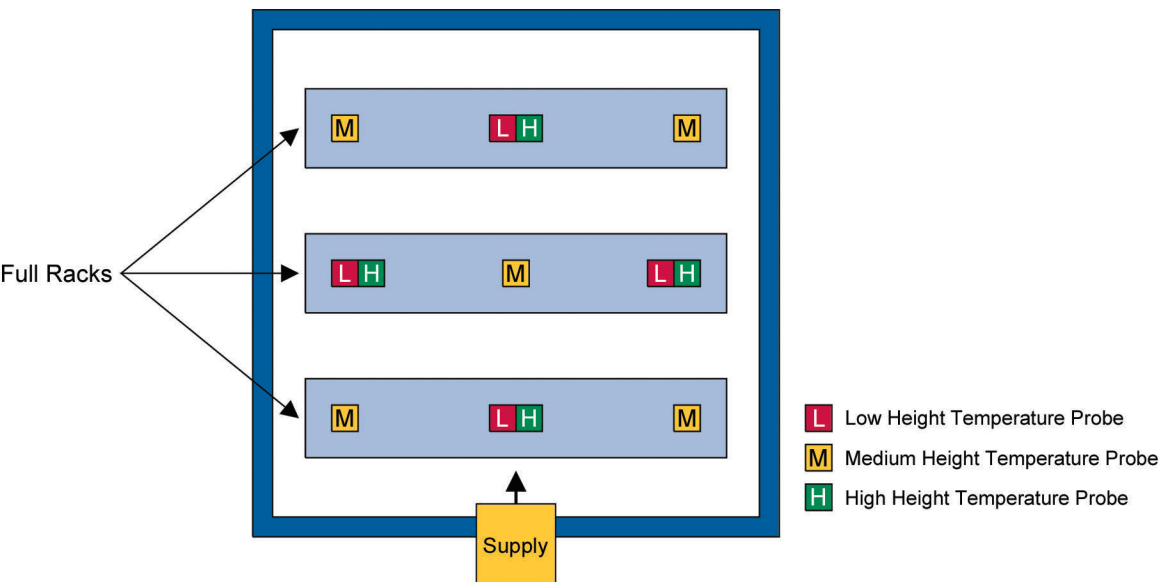
Figure 16.5: Mapping Locations for a Warehouse (basic configuration)



Identify airflow restrictions based on the location of the air supply and location/orientation of full pallet racks, drums, tanks etc. Additional temperature probes should then be added in areas of restricted airflow. In the case of full racks and/or shelves, the restricted airflow may impact the temperature profile so additional probes should be added based on expected airflow pattern.

Figure 16.6 shows how the locations for additional monitoring probes could be added if the area conditioning air supply is from one direction as indicated in the drawings.

Figure 16.6: Additional Mapping Locations for a Warehouse (where HVAC is from one direction)



Mapping of warehouses requires both empty and full state (again, full is defined as the maximum amount anticipated being stored in the facility). In the case of warehouses, the mass of the material is not as significant a factor as the volume, in terms of altering airflow patterns. The full state may be met by simulating the rack loads with empty storage containers (cartons, boxes, pallet boxes, etc.) to obstruct the airflow patterns, as they would when holding actual product. However, if the heating/cooling is provided centrally, the racking nearest the walls may be considered as a layer of insulation; therefore, thermal mass would influence performance. This should be considered when developing the strategy and defining the test strategy. It may be achieved by loading a representative section of racking with a simulated load incorporating thermal mass.

Mapping of controlled room temperature warehouses has the same requirements.

With this mapping strategy outlined, the probe locations should be reviewed again to verify and document the locations critical to the facility capability.

16.3.1 Warehouse Load Testing

When defining how the unit will be tested, the following should be considered:

- Is the area used to condition the stored material, or does it arrive within the specified conditions?
- What is the worst-case condition:
 - An empty store with the maximum specified load arriving at the most extreme conditions permitted?
 - A store with the maximum load; this may be considered in terms of the impact of the thermal mass or load volume impeding airflow?

Note: it is typically considered that testing to represent the extreme conditions, empty and fully loaded, provides adequate assurance of operation in partially loaded conditions. Where a system is divided into zones, the load testing may be justified on a zone basis.

- What will be the worst-case conditions in use:
 - Doors held open for the maximum specified time in extreme environmental conditions?

The area/system owner may also want to use this opportunity to test system performance with simulated cooling equipment component failures, to provide data that may be useful to defend continued use of the system during operation with an equipment failure.

For more examples showing suggestions for mapping temperature sensor layout and supporting rationale for some common cold room layouts, see Appendix 4.

16.3.2 Summer/Winter Testing

An organization may use a risk assessment to determine if seasonal testing is required. Risk mitigation factors to consider include:

- A robust software program was used to calculate the HVAC design/predict area performance
- The commissioning practices used for the HVAC system confirm that the components perform to meet the design specification (e.g., fan volumes with clean and simulated dirty filters, heating and cooling coil capacities)
- The number and location of the monitoring sensors are adequate to provide evidence if the environment exceeds the specified conditions ("continuous mapping" in effect)

- Impact of external temperature extremes (when conditions are expected to exceed limits of normal operation by $\pm 6^{\circ}\text{C}$ for greater than 3 hours). Unless the warehouse is located in a temperate geographic location and not subject to extreme temperatures, the mapping exercise should be performed both during summer and winter in order to assess the worst-case scenarios, as extremes of temperature may adversely affect the temperature distribution within the warehouse storage area due to insulation capabilities, equipment performance, and other factors
- If initial mapping is performed under conditions of extreme external temperature, then one supplemental mapping should be performed within the first year to test under the opposite season i.e., one with heating and one with cooling. If the initial mapping cannot be performed during one of these extreme conditions, this could be allowed for in the monitoring strategy e.g., by provision of additional sensors, or identification of the hot/cold spots during the heating/cooling seasons. The full state could also be tested in the subsequent seasonal remapping, as:
 - Seasonal testing is required for installations that could be impacted by seasonal extremes
 - A warehouse is not typically filled to capacity in the first six months

Alternatively, organizations may perform mapping of warehouses over a full year. The external conditions are compared to the specified design conditions, and a judgment made to determine if the results adequately represent all seasons (likely seasonal extremes). These results are used to help define locations for the permanent monitoring locations and to ensure that locations consider the impact of seasonal variations on the locations likely to experience minimum/maximum temperatures. These uses should be taken into consideration during the comparison of the results.

For controlled temperature rooms, e.g., cold rooms, it may be considered adequate to map the system once, on the basis that the surrounding environment is controlled and the monitoring system provides adequate assurance that the internal conditions are being maintained, any impact from the outside temperatures on the performance of the heating/cooling system would be visible from the monitoring system, as the temperature profiles in the conditioned space would not be affected.

The rationale for the load and seasonal testing should be defined in a document that is reviewed and approved by the Quality Unit and the area/system owner.

A sample document is included in Appendix 2.

It is considered good practice to review the initial monitoring data and compare the results with the defined requirements of the URS, to ensure that the requirements have been met.

Where user requirements have not been met, the designer/supplier should be asked to review the data and make recommendations for corrective action, to ensure that the system complies with the design specification. Depending on the contracts in place these modifications may be at the design authority's or equipment supplier's expense, and should be managed through ECM.

16.4 Example 4: Test Document/Protocol

- A. Define the system to be tested and confirm the calibration of the instrumentation to be used. The response time⁵ for the instrumentation IEST-RP-CC006.3 [42] recommends a 63.2% time constant of no more than 15 seconds for areas with stricter environmental control specifications. This correlates to 75 seconds to achieve 99% (five time constants). This response time should be taken into account when matching temperature sensors with the analytical procedure. The system used should be selected keeping in mind the temperature accuracy required of the facility probes, qualification requirements for the mapping instrumentation, data collection, output and retention capabilities, and compliance with the appropriate electronic record requirements (21 CFR Part 11 [43]/ Eudralex Annex 11 [44]).
- B. Define the facility operations (static versus dynamic activities to simulate operational states) and control systems (set points, operational parameters)
- C. Define acceptance criteria/expected results as appropriate
- D. Probe type and locations Note the strategy defined the rationale for locations
- E. Define the load test and sequence of loading/unloading
- F. Define how test issues will be resolved
- G. Define pre-test requirements (commissioning, calibration, signage, training, etc.)
- H. Define test steps, mapping under various operational states such as defrost cycles, switch between redundant systems, door openings, people activity, power loss, etc.
- I. Record test start and stop times
- J. Record testers
- K. Provide space for analysis of data with recommendations for actions if required

16.5 Example 5: Sample Test Strategy

16.5.1 Introduction

This document provides a record of the test strategy, mapping sensor locations, and the criteria that will be used to select the monitoring sensor locations for the temperature mapping of the cold room. If any changes are found necessary during test execution, a drawing showing the final arrangement with a supporting rationale will be appended to this document.

The mapping will be used to determine the locations for the two permanent independent monitoring sensors. These locations will be based on the product storage locations that typically experience maximum and minimum temperature conditions during simulated normal use. It should be noted that the product stored within the cold room has no humidity requirements; hence humidity will not be monitored or mapped.

⁵ Instrument time response is usually quoted as: "T10–90% or T90–10%." (Referring to rise and fall times respectively) when the instrument is subject to a step change in the input, i.e., the temperature recording system needs to respond faster that conditions are likely to change in order to get an accurate reading – though note that the material stored will have a response time as well – but when testing the goal is to get accurate information to get a complete understanding of how the equipment is performing."

The cold room is equipped with duty standby refrigeration systems – these units changeover automatically on a fixed schedule, and the standby system cuts in in the event of an alarm from the duty system, which shuts down.

Note: It may be possible to include a door position sensor with the mapping system to aid with data analysis, otherwise the more traditional approach of noting times on test record sheets can be used.

16.5.2 Approach – Cold Room

The following points were considered:

16.5.2.1 External Heat Gains and Losses

The major heat gains and losses to the cold room are due to the influences of the external environment, with the most significant being through the walls and ceiling.

These gains will be consistent as the structure is uniform and enclosed within the warehouse – a CRT area, with the following exceptions during operation:

- The influence will vary during loading/unloading as the door will be open, allowing cold air to leave the room, and the infiltration of warm air. This will be controlled using a vinyl strip curtain.

16.5.2.2 Internal Heat Gains and Losses

Lighting – the internal lighting is low energy LED lamps; these are left on during the working day.

Product – the product stored in the cold room is supplied at the storage temperature, so will not put a significant load onto the system. The cold room is dedicated for product storage.

Equipment – the load will be put into the cold room using an electric stacking fork truck - a relatively small and intermittent load.

People – the cold room is only accessed by either two people or the truck driver when it is being loaded or unloaded. The heat gains are therefore small and transient.

Refrigeration – the cold room is cooled by one of two refrigeration systems, using two cooling coils per system placed diametrically opposite at the ends of the access corridors, so the air distribution is the same but in the reverse direction for the two systems.

The systems are designed so that no defrost is required.

16.5.3 Sensor Placement

The only areas considered during mapping are where product will be stored, i.e., in the rack location. The sensor locations proposed have been generated considering the following:

- The heat gains will be through the walls and ceiling. There is less risk of temperature extremes for locations in the central racking.
- The location of the unit coolers and the associated airflow

The sensor locations proposed are shown in Figure 16.7. A drawing giving the actual locations will be included with the mapping results.

16.5.4 Load Testing

The unit will be mapped using two load scenarios:

Empty Chamber

The chamber will be mapped empty.

Note that the refrigeration units have a constant flow rate. With the room empty there are no constraints on the airflow direction, hence the airspeed will be at its lowest. When the room is full of product the airflow will be at a higher speed as the same volume of air will need to travel through a smaller area.

Simulated Full Chamber

There will be little impact to the temperatures in the center of the cold room due to load because of the surrounding cold airflow, but adding the load to the racking located against the outer walls will act as an insulator, reducing the heat transfer, with thermal mass increasing that effect.

The area shown shaded in Figure 16.7 will be filled with empty boxes mounted on a pallet, allowing air to flow in the same way the load will, with the boxes sized to simulate the size of the largest stored volume.

The side nearest the door will also have thermal mass added to the boxes in the form of water bottles; this area was chosen as it will represent the most challenge with the use of the cold room door.

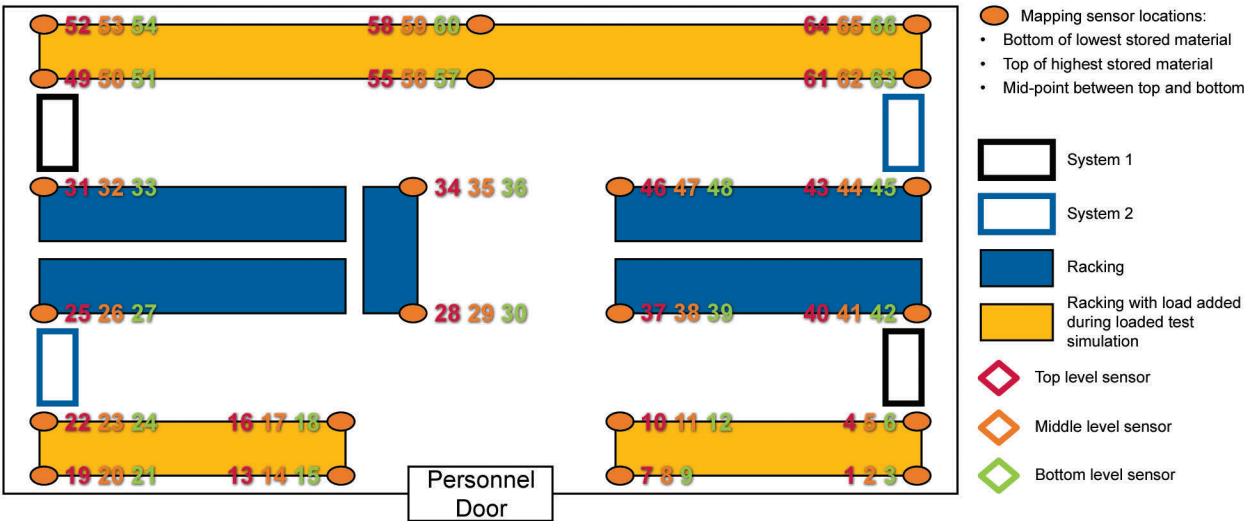
This arrangement will simulate the worst-case thermal mass and physical size, providing a worst-case simulation. See Figure 16.7.

Note that the thermal mass of the material to be stored will be relatively low, so the use of the water provides a reasonable simulation.

Summer and Winter Conditions

Typically for a controlled temperature environment it is necessary to carry out summer and winter temperature mapping. The cold room in this instance is located in a conditioned space with an air gap between the cold room wall and any external wall hence, the effect of the external conditions is limited. The independent temperature monitoring system will give an accurate indication of the internal conditions in the warmest/coldest locations (these will not change as the cold room environment is the same) so any impact of potential refrigeration system capacity changes due to seasonal changes will be seen. The refrigeration systems are sized for the seasonal extremes so the risk of any impact is considered low. It is therefore not considered necessary to map during summer/winter conditions.

Figure 16.7: Plan View Showing Proposed Mapping Sensor Layout and Locations for Simulated Load during Loaded Test



Note: External and control sensor omitted for clarity.

16.5.5 Testing – Operational Verification

The following tests are proposed for operational verification.

16.5.5.1 Install the Mapping Sensors

Identify the boundaries of the load and locate mapping sensors as shown in Figure 16.7, with an additional sensor adjacent to the control sensor, and one measuring external weather conditions. The vinyl curtains should be in place with no gaps at the sides or bottom.

The data from the sensor adjacent to the control sensor is for reference purposes. The data from the external temperature sensor is used to show where the conditions at the time of testing are in relation to the specified limits.

A drawing will be provided showing the sensor locations in enough detail to allow the mapping to be repeated if necessary.

16.5.5.2 System Cooldown and Recovery Test

With the room at ambient temperature, operate the duty refrigeration system and use the mapping sensors to determine the cool down time, that is, the time it takes to go from ambient conditions until all the sensors are within the 2°C–8°C range.

After allowing the room to stabilize, shut down the refrigeration systems (both systems), and record the time it takes for the first and last temperature sensor to exceed 8°C.

Repeat the test with the standby refrigeration system.

Note: It is only necessary to test the recovery time once, as it will not change.

During the initial operation of the cold room, it is useful to share the mapping data with the commissioning company to help them optimize the settings for the cold room.

Report results as follows:

- Confirm that temperatures at all locations are within the specification range.
- Location and readings for the minimum and maximum recorded temperatures
- Location and readings for the minimum average temperature and the maximum average temperature

16.5.5.3 Power Failure Simulation Test

Operate the cold room with System 1: operate the system normally for a minimum of 1 hour. Shut off the utility power source to System 1. Ensure that System 2 automatically starts.

Repeat for refrigeration System 2.

Do not open the door during this test.

16.5.5.4 System Empty Temperature Mapping

This test is conducted with the cold room empty. The test should be at least 48 hours (to cover at least 24 hours with each refrigeration system after the system has begun modulating at the set point.)

Report results as follows:

- Temperatures at all locations are within the specification range
- Location and readings for the minimum and maximum recorded temperatures
- Location and readings for the minimum average temperature and the maximum average temperature

16.5.5.5 Operation Empty – Door Opening

During the empty temperature mapping test and the simulated load temperature mapping test simulate operations: the access door must be opened, and the truck and or personnel enter the room to simulate loading/unloading; the door must be held fully open (held at 90 degrees for a hinged door).

The test is completed if a mapping sensor goes outside the acceptable operating range or when 2 hours is reached, whichever comes first.

Report open-door test results as follows:

- Location and readings for the minimum and maximum recorded temperatures
- If the sensors go outside the acceptance criteria record the time and locations
- Location and readings for the minimum average temperature and the maximum average temperature

16.5.5.6 System with Simulated Load Temperature Mapping

This test is conducted with the cold room loaded using a simulated load comprised of boxes of the maximum load dimensions with a pack of drinking water (simulating the worst-case interruption of airflow with the minimum thermal mass). The test should be at least 48 hours (to cover at least 24 hours with each refrigeration system after the system has begun modulating at the set point).

Report results as follows:

- Temperatures at all locations are within the specification range
- Location and readings for the minimum and maximum recorded temperatures
- Location and readings for the minimum average temperature and the maximum average temperature

16.5.5.7 Operation Loaded – Door Opening

During testing the access door must be opened, and the truck and or personnel enter the room to simulate loading/unloading; the door must be held fully open (held at 90 degrees for a hinged door).

The test is completed if a mapping sensor goes outside the acceptable operating range or when 2 hours is reached, whichever comes first.

Note: Run the duty and standby refrigeration systems for a minimum of 24 hours after the systems have begun modulating at the set point.

Report results as follows:

- Temperatures at all locations are within the specification range
- For this reporting, do not include data generated during open-door testing
- Location and readings for the minimum and maximum recorded temperatures
- Location and readings for the minimum average temperature and the maximum average temperature

16.5.6 Interpreting the Mapping Results

The results should be reviewed to ensure that all storage areas remain within the acceptance criteria. If there are localized areas that do not, the design/operation may be revised to address this, or the areas may be blocked off with mesh so that they are not used.

16.5.7 Independent Temperature Monitoring System Sensor Placement

The mapping will be used to define the locations where the typical minimum and maximum temperatures are found. Include appendices showing two methods for using the mapping results to do this.

16.5.8 Seasonal Mapping

Where the internal temperatures vary due to the external temperatures, or where the air conditioning has heating and cooling modes (impacting the air distribution/conditions), the strategy needs to include considerations for seasonal conditions. Typically, the mapping needs to cover seasonal variation. Mapping can be conducted in the summer and winter seasons, determining the worst-case locations (hot and cold). Once these are established, the data can be used to confirm the actual conditions throughout the year.

17 Appendix 7 – Templates and Attachments

17.1 Example Science and Risk-Based Approach for Periodic Review of a Controlled Temperature Chamber

17.1.1 Procedure – Categorize the Controlled Temperature Chamber: Based on Material Stored and Temperature Range

Systems are categorized for the purposes of periodic review considering the system operating temperature range, and the material stored.

- **Stored Material Type A:** for example, API, drug product, stability sample(s), cool/hot packs used to maintain temperature during transportation (if the cool/hot pack temperature is not verified before use)
- **Stored Material Type B:** for example, Work in Progress (WIP), Reference Standards, critical reagents, raw materials
- **Stored Material Type C:** for example, Environmental Monitoring (EM) supplies and reagents, media, test materials

Table 17.1: Review Category Based on Material Stored and Temperature Range

Stored Material Type	Precision Unit: Temperature Range $\leq \pm 6^{\circ}\text{C}$	Standard Unit: Temperature Range $> \pm 6^{\circ}\text{C}$	Conditioned Room: Controlled Room Temperature*
A	1	2	2
B	2	2	3
C	2	3	3
* as defined in USP			

17.1.2 Periodic Review Process

The periodic review process must be carried out as follows:

- Category 1 Controlled Temperature Chambers: periodic review not to exceed three years from the previous review
- Category 2 Controlled Temperature Chambers: periodic review not required; system performance verification must be conducted through a defined maintenance job plan
- Category 3 Controlled Temperature Chambers: system performance verification and periodic review is not required

There are two methods that may be used to perform the periodic review of the Controlled Temperature Chambers.

17.1.2.1 Method One: Periodic Review Report – Risk-Based Approach

This method considers risks to the system performance. An example for the Periodic Review Report is shown below.

Template Report for Validation Periodic Review for Controlled Temperature Chamber Summary

The Controlled Temperature Chamber is used for the storage of...

The periodic review considers the following aspects:

- Internal layout and loading:
 - Are there any changes that could impact the system performance?
- Insulation
 - Are there any changes, or is there any degradation of the insulation capability, of the Controlled Temperature Chamber shell?
- Cooling System
 - Is the cooling system still operating with performance aligned with the as commissioned/qualified state?

The following is an example of a periodic review for a Controlled Temperature Chamber. The actual review should replace the text given in gray italic as necessary.

Example:

The results of this review are contained in the report below.

There is nothing in the review to indicate that the original mapping results will have changed, the data from the monitoring system still shows that the system is operating within the operating limits, it is therefore considered fit for continued use.

Example Assessment Report with Draft Content⁶

Reference	Aspect	Review	Conclusion
1	Internal Layout		
1.1	Racking/Storage Locations	<i>The racking layout and defined storage locations remain as defined in the commissioning and qualification file per review of the current installation and operations SOP-xxx with the C&Q report RPT-yyyy</i>	<i>The locations used to store material remain unchanged, so the effect on the unit airflow and temperature distribution has not changed</i>
1.2	Stored Material	<i>The size and or mass of the stored material that was defined in the test strategy or as used for the commissioning and qualification has not changed</i>	<i>The size and mass of the material used for the load testing of the Controlled Temperature Chamber remains representative of the material currently stored in the Controlled Temperature Chamber, hence the performance will not have changed</i>
2	Insulation		
2.1	Shell	<i>PM number requires an annual thermal scan of the system verifying the surface temperature remains consistent with the as installed readings, and there are no hot or cold spots, or visual inspection and a review of the cycle time of the refrigeration plant.</i>	<i>The thermal scan is consistent with those taken during initial operation confirming that there is no degradation of performance, or a visual inspection and a review of the cycle times show no significant degradation.</i>
2.2	Utility Penetrations	<i>Visually inspect the Controlled Temperature Chamber penetrations looking for any leakage or condensation.</i>	<i>The Controlled Temperature Chamber was inspected – all utility penetrations are in good repair with no sign of leakage or condensation.</i>
2.3	Access Door	<i>The door, door seals, and vinyl internal curtain were inspected.</i>	<i>The door, door seals, and vinyl curtain are in good condition. The vinyl curtain sits on the side of the unit with no air gap and just touches the floor so there is no friction to prevent it closing and no significant air gap.</i>
3	Cooling System		
3.1	Operating History	<i>Review alarm history for the refrigeration units and determine if the frequency is normal/ acceptable.</i>	<i>The alarm history was reviewed, with a total of 12 engineering alarms, and no quality alarms; all engineering alarms were resolved while keeping the unit inside the operational limits.</i>
3.2	Maintenance	<i>Confirm that there is a PM that checks duty standby function, and reviews system operating settings, and that the refrigeration system major components are as installed.</i>	<i>The automatic duty standby changeover function is routinely checked, the refrigeration system has not been changed and the system operating parameters: suction and delivery pressure for both systems are within the manufacturer's tolerances, indicating that the system is operating normally.</i>
3.3	Monitoring	<i>The data from the independent monitoring system shows that the system is operating within the operating limits with no indication of a degradation in performance.</i>	<i>A review of the system monitoring data shows that the system is still operating within the operating limits with no indication of a degradation in performance.</i>
3.4	Alarm Function	<i>Confirm system and independent alarms tested.</i>	<i>Alarms tested: all function correctly at the defined set points, with the delay periods assigned as noted in the attached sheet.</i>
Note: Example text is shown in <i>gray italic</i> .			

⁶ The draft content should be revised to suit the specific application.

- Revise the template report to reflect the findings for the Controlled Temperature Chamber under review
- Record the results of the review into the Periodic Review Report, recording the location of the information, and attaching copies where appropriate
- Define any actions required after review of the information
- The report summary must state if the system is considered suitable for ongoing use
- If it is concluded that the system is not suitable for ongoing use, a nonconformance shall be utilized to define the actions required, and manage their completion
- The Periodic Review Report be reviewed and approved by the System Owner or delegate and Quality Unit

17.1.2.2 Method Two: Periodic Review Report – Remapping

As an alternative to method one, the Controlled Temperature Chamber may be remapped.

Remapping should consider the following aspects:

- Loading/unloading of the unit
- The impact of the load on the unit temperature distribution
- The impact of unit defrost on the unit temperature distribution

The test plan supporting rationale and mapping acceptance criteria should be approved by the System Owner or delegate and Quality Unit.

The results of the mapping shall be used to confirm the number and location of the monitoring probes, or provide the rationale to support modifications to the monitoring system.

Record the results of the remapping into the Review Report. The report summary shall define if the system is considered suitable for ongoing use, and shall be approved by the System Owner or delegate, and Quality Unit.

If it is concluded that the system is not suitable for ongoing use, a nonconformance shall be utilized to define the actions required, and manage their completion.

17.2 Example Job Hazard Analysis Worksheet

JHA Title:	<i>Formaldehyde Disinfection of Enclosure Type Equipment</i>		
JHA Number:			
Project Name:		Site:	
Project Number:		Bldg/Rm#:	
General Contractor:	<i>Contractor Services</i>		Prepared by: <i><First and Last Name></i>
Contractor(s):	<i>N/A</i>		Date: <i><DD-MMM-YYYY></i>
Craft(s):	Field Service Technician	Scheduled Start Date:	Reviewed by: <i><First and Last Name></i>
			Date: <i><DD-MMM-YYYY></i>
Briefly Describe Task:	<i>Perform formaldehyde vapor disinfection of Enclosure Type Equipment</i>		
Required PPE:	<i>Lab coat, safety glasses</i>		
Supplemental PPE:	<i>Tyvek suit, full face respirator with formaldehyde cartridges, gloves</i>		
Required Permits:	<i>None</i>		
Competent Persons(s):	<i><First and Last Name></i>		
General Comments:	<i>Contractor Services workers will set up and perform formaldehyde vapor disinfection of Equipment. Contractor Services will follow procedures outlined in Contractor Services SOP 82 to complete this work.</i> <i>Only Contractor Services personnel will be allowed in the rooms after 5 pm on evening of generation of formaldehyde vapor in the equipment. Disinfection will continue overnight, with neutralization and venting occurring the following morning. Process will generally be completed by 10 am the morning after formaldehyde generation and disinfection.</i> <i>Personal Exposure Level monitors will be set up to detect any formaldehyde leakage from the equipment during the disinfection process. Any leakage detected will be reported on the Decontamination Report. Contractor Services will use short-term monitoring to assure formaldehyde levels in the enclosures have fallen below 0.5 ppm before releasing units back to the client.</i>		
<i>Workers shall review the Company Review Comment Record and comply.</i> <i>Workers shall sign the JHA verifying they have reviewed and will comply with the contents.</i> <i>A copy of the JHA will be kept at the work area available for review.</i>			
Work Operation	Potential Accidents or Hazards	Safe Job Actions Needed	
1. Preliminary Setup	A. Workers not understanding job or not trained to perform job.	A.1 Scope of work will be communicated to all workers scheduled to perform work on the job. A.2 All Contractor Services workers on site to perform job are trained on appropriate SOPs related to job.	

JHA Title:	<i>Formaldehyde Disinfection of Enclosure Type Equipment</i>		
JHA Number:			
Project Name:		Site:	
Project Number:		Bldg/Rm#:	
Work Operation	Potential Accidents or Hazards	Safe Job Actions Needed	
1. Preliminary Setup (continued)	B. Units not ready for disinfection.	B.1 Contractor Services will verify all units scheduled for disinfection are cleared of all equipment and materials. B.2 Contractor Services will verify all units are disconnected from wall/room outlets.	
	C. HVAC system in room not operating and/or no power to room lights and receptacles.	C.1 Contractor Services will contact Company facilities/JLL upon arrival on site to confirm HVAC systems are operating normally. C.2 Contractor Services will check that lights in room are operating normally and wall receptacles are powered up.	
2. Setup	A. Breach of unit enclosure	A.1 Inspect entire enclosure for any damage that would indicate a breach of containment. A.2 Seal any suspected areas of breach of containment with adhesive tape.	
	B. Disinfection equipment not working correctly	B.1 Perform function tests on all equipment, including evaporator pans, extension cords and adsorber cell filters.	
	C. Disinfection of enclosures not set up correctly	C.1 Confirm proper amounts of paraformaldehyde and ammonium bicarbonate are distributed in the appropriate evaporator pans. C.2 Confirm all openings of enclosure are sealed with plastic sheeting and/or adhesive tape. C.3 Have enclosure under -0.005" WC to -0.01" WC negative pressure by connecting enclosures to exhaust via ducting to auxiliary fan or exhaust system, with adsorber cell filter connected in-line to filter all formaldehyde vapor leaving the enclosure.	
3. Formaldehyde vapor generation	A. Personnel exposed to formaldehyde vapor	A.1 Hang signs on all enclosures undergoing disinfection and on all doors entering the room to prevent non-Contractor Services personnel from entering the lab or modifying disinfection setup.	

JHA Title:	<i>Formaldehyde Disinfection of Enclosure Type Equipment</i>		
JHA Number:			
Project Name:		Site:	
Project Number:		Bldg/Rm#:	
Work Operation	Potential Accidents or Hazards	Safe Job Actions Needed	
3. <i>Formaldehyde vapor generation (continued)</i>	A. <i>Personnel exposed to formaldehyde vapor (continued)</i>	A.2 Contractor Services will continuously monitor room during vapor generation in the sealed enclosures for any formaldehyde leakage into the room. A.3 If concentration exceeds 0.2 ppm at any time, Contractor Services personnel will don all PPE and will disconnect paraformaldehyde pan and turn on the ammonium bicarbonate pan to neutralize the formaldehyde vapor.	
4. <i>Exposure/contact period</i>	A. <i>Lab or personnel exposed to formaldehyde vapor</i>	A.1 Signs will be posted on all doors entering the room overnight cautioning personnel from entering the room or modifying disinfection in any way. A.2 If no formaldehyde leakage is detected during generation (see 3 above) Contractor Services will leave site in steady state with enclosures completely sealed and under negative pressure. Contractor Services will be available for contact via cell phone # posted around the site.	
	B. <i>Loss of power to HVAC system or loss of electrical power</i>	B.1 Enclosures are completely sealed with in-line adsorber cell connected only to opening from enclosures, preventing any formaldehyde vapor from escaping.	
5. <i>Neutralization and ventilation</i>	A. <i>Loss of power to HVAC system or loss of electrical power</i>	A.1 Contractor Services will be on site during neutralization and ventilation process. Contractor Services personnel will take steps to minimize any exposure to lab of any residual formaldehyde by donning all PPE and preventing any non-Contractor Services personnel from entering the lab in the event of power loss. A.2 If ventilation is already in process, Contractor Services will wait on site until restoration of power to evacuate ammonium bicarbonate residue in enclosures.	

JHA Title:	<i>Formaldehyde Disinfection of Enclosure Type Equipment</i>		
JHA Number:			
Project Name:		Site:	
Project Number:		Bldg/Rm#:	
Work Operation	Potential Accidents or Hazards	Safe Job Actions Needed	
6. <i>Re-establishment of preexisting conditions</i>	A. <i>Exposure to lab of formaldehyde vapor or ammonium bicarbonate</i>	A.1 Contractor Services personnel will clean and wipe down all exposed surfaces exposed to formaldehyde vapor during the disinfection process following procedures outlined in Contractor Services SOP 82 to eliminate chemical residue. A.2 Contractor Services will monitor interior of enclosures to assure formaldehyde concentration is below 0.5 ppm, at which time enclosures can be released back to Company.	
Note: Example text is shown in <i>gray italic</i>. <Gray italicized text in brackets> provides sample text that should be modified.			

18 Appendix 8 – References

1. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 5 – Commissioning and Qualification*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, June 2019, www.ispe.org.
2. *ISPE Good Practice Guide: Cold Chain Management*, International Society for Pharmaceutical Engineering (ISPE), First Edition, May 2011, www.ispe.org.
3. USP <659> Packaging and Storage Requirements, US Pharmacopeial Convention, www.usp.org.
4. International Council for Harmonisation (ICH), ICH Harmonised Tripartite Guideline, *Stability Testing of New Drug Substances and Products – Q1A(R2)*, Step 4, 6 February 2003, www.ich.org.
5. USP General Notices and Requirements: Applying to Standards, Test, Assays, and Other Specifications of the United States Pharmacopeia, Section 7.20 Rounding Rules, US Pharmacopeial Convention, www.usp.org.
6. HPRA, IA-G0046-6 Guide to Good Distribution Practice of Medicinal Products for Human Use, Health Products Regulatory Authority (HPRA), 9 March 2021, www.hpra.ie.
7. HPRA, IA-G0011-3 Guide to Control and Monitoring of Storage and Transportation Temperature Conditions for Medicinal Products and Active Substances, Health Products Regulatory Authority (HPRA), 2 October 2020, www.hpra.ie.
8. EMA, Guidelines of 5 November 2013 on Good Distribution Practice of medicinal products for human use (Text with EEA relevance) (2013/C 343/01), European Commission, *Official Journal of the European Union*, <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:C:2013:343:0001:0014:EN:PDF>.
9. Health Canada, GUI-0069, Guidelines for environmental control of drug products during storage and transportation, Health Canada, 24 August 2020, www.canada.ca/en.html.
10. PIC/S Guide: PE 011-1 Good Distribution Practice for Medicinal Products, Pharmaceutical Inspection Convention Pharmaceutical Inspection Co-Operation Scheme (PIC/S), June 2014, www.picscheme.org.
11. WHO Technical Report Series, No. 961 – Forty-fifth Report, WHO Expert Committee on Specifications for Pharmaceutical Preparations, World Health Organization (WHO), 2011, www.who.int.
12. WHO, WHO/IVB/19.09, How to Temperature Map Cold Chain Equipment and Storage Areas, World Health Organization (WHO), December 2019, www.who.int.
13. WHO, WHO/IVB/18.05, Evidence Brief, Cold room temperature mapping studies, World Health Organization (WHO), <https://apps.who.int/iris/bitstream/handle/10665/272857/WHO-IVB-18.05-eng.pdf?ua=1>.
14. USP <1079> Risks and Mitigation Strategies for the Storage and Transportation of Finished Drug Products, US Pharmacopeial Convention, www.usp.org.
15. USP <1118> Monitoring Devices – Time, Temperature and Humidity, US Pharmacopeial Convention, www.usp.org.
16. ISO/IEC 17025:2017 General requirements for the competence of testing and calibration laboratories, International Standards Organization (ISO)/International Electrotechnical Commission (IEC), www.iso.org.

17. Taylor, J.L., Recommendations on the Control and Monitoring of Storage and Transportation Temperatures of Medicinal Products, 2001, <http://temptrust.mesalabs.com/wp-content/uploads/sites/47/2012/05/MHRAcon007569.pdf>.
18. Seevers, R.H., Hofer, J., Harber, P., Ulrich, D.A., and Bishara, R., The use of mean kinetic temperature (MKT) in the handling, storage, and distribution of temperature sensitive pharmaceuticals, *Pharmaceutical Outsourcing*, 2009, Vol. 10, pp. 30-39.
19. JTM K 01 Temperature and Humidity Chambers – Test and Indication Method for Performance, Japan Testing Machinery Association (JTM), <http://www.jtma.jp>.
20. JTM K 03 Environmental Temperature and Humidity Rooms – Test and Indication Method for Performance, Japan Testing Machinery Association (JTM), <http://www.jtma.jp>.
21. JTM K 05 High Temperature Chambers – Test and Indication Method for Performance, Japan Testing Machinery Association (JTM), <http://www.jtma.jp>.
22. FD X15-140 Measurement of Air Moisture – Climatic and Thermostatic Chambers – Characterization and Verification, Association Francaise de Normalisation, 2013, <http://infostore.saiglobal.com/emea/details.aspx?ProductID=1639789>.
23. DIN 12880 Electrical Laboratory Devices – Heating Ovens and Incubators, Deutsches Institut für Normung e.V. (DIN), 2007, <http://www.din.de/en/standards/din-12880/97210882>.
24. IEC 60068 Environmental Testing Series, International Electrotechnical Commission (IEC), www.iec.ch.
25. International Council for Harmonisation (ICH), ICH Harmonised Tripartite Guideline, *Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients – Q7/Q7A*, Step 4, 10 November 2000, www.ich.org.
26. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 15: Qualification and Validation, http://ec.europa.eu/health/documents/eudralex/vol-4/index_en.htm.
27. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 15: Qualification and Validation, 1 May 2021, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.
28. *ISPE GAMP® Good Practice Guide Series*, International Society for Pharmaceutical Engineering (ISPE), www.ispe.org.
29. *ISPE Good Practice Guide: Project Management for the Pharmaceutical Industry*, International Society for Pharmaceutical Engineering (ISPE), First Edition, November 2011, www.ispe.org.
30. *ISPE Concept Paper: Controlled Temperature Chamber Mapping*, International Society for Pharmaceutical Engineering (ISPE), Packaging Community of Practice (COP) April 2012, www.ispe.org.
31. IEC 60584, Thermocouples – Part 1: EMF specifications and tolerances, Third Edition, 2013, International Electrotechnical Commission (IEC), www.iec.ch.
32. IEC 60751:2008: Industrial platinum resistance thermometers and platinum temperature sensors, International Electrotechnical Commission (IEC), www.iec.ch.
33. *ISPE Good Practice Guide: Maintenance*, International Society for Pharmaceutical Engineering (ISPE), Second Edition, January 2021, www.ispe.org.

34. *ISPE Good Practice Guide: Decommissioning of Pharmaceutical Equipment and Facilities*, International Society for Pharmaceutical Engineering (ISPE), First Edition, June 2017, www.ispe.org.
35. International Air Transportation Association (IATA), www.iata.org.
36. Envirotainer® container RAP e2 Technical Specification, aircargopedia.com.
37. ATA's Technology & Maintenance Council (TMC), <http://tmc.trucking.org>.
38. UN, ATP as amended on 6 July 2020, Agreement on the International Carriage of Perishable Foodstuffs and on the Special Equipment to be Used for such Carriage, United Nations Economic Commission for Europe, Inland Transport Committee, United Nations (UN), 2020, <https://unece.org>.
39. ISO 1496-2:2018 Series 1 freight containers — Specification and testing — Part 2: Thermal containers, International Standards Organization (ISO), www.iso.org.
40. National Standards Authority of Ireland (NSAI) – National Metrology Laboratory (NML), <http://www.nsai.ie/nml.aspx>.
41. Haycocks, N.R, Goldschmidt, N.A., Thomsen, U., "Temperature and Humidity Requirements in Pharmaceutical Facilities," *Pharmaceutical Engineering*, September/October 2021, Vol. 41, No. 5, pp. 61-67, www.ispe.org.
42. IEST-RP-CC006.3 Testing Cleanrooms, Institute of Environmental Sciences and Technology (IEST), <http://www.iest.org/Standards-RPs/Recommended-Practices/IEST-RP-CC006>.
43. 21 CFR Part 11 – Electronic Records; Electronic Signatures, Code of Federal Regulations, US Food and Drug Administration (FDA), www.fda.gov.
44. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 11 – Computerized Systems, June 2011, http://ec.europa.eu/health/documents/eudralex/vol-4/index_en.htm.
45. "Definition of Peltier Effect," *Merriam-Webster.com Dictionary*, Accessed 11 Nov 2021, <https://www.merriam-webster.com/dictionary/Peltier%20effect>.

19 Appendix 9 – Glossary

19.1 Acronyms and Abbreviations

AC	Air Conditioner
AHU	Air Handling Unit
API	Active Pharmaceutical Ingredient
ATA	American Trucking Association
ATP	Agreement on the International Carriage of Perishable Foodstuffs
BAS	Building Automation System
BMS	Building Management System
BTU	British Thermal Unit
CA	Critical Aspect
CDE	Critical Design Element
C&Q	Commissioning and Qualification
CEIV Pharma	Center of Excellence for Independent Validators in Pharmaceutical Logistics
CFD	Computational Fluid Dynamics
CMS	Continuous Monitoring System
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CRT	Controlled Room Temperature
CTC	Controlled Temperature Chamber
CTSS	Controlled Temperature Shipping Solution
DQ	Design Qualification
ECM	Engineering Change Management
EM	Environmental Monitoring
EMA	European Medicines Agency (EU)
EMS	Environmental Monitoring System
FAT	Factory Acceptance Testing
FDA	Food and Drug Administration (US)
FMEA	Failure Mode and Effect Analysis
GB	Guard Band
GDP	Good Distribution Practice
GEP	Good Engineering Practice
GMP	Good Manufacturing Practice
GSP	Good Supply Practice

GxP	Good “x” Practice
HPRA	Health Products Regulatory Authority (Ireland)
HVAC	Heating, Ventilation, and Air Conditioning
IATA	International Air Transportation Association
ICH	International Council for Harmonisation
IEC	International Electrotechnical Commission
IFC	Issue for Construction
I/O	Input/Output
IV	Installation Verification
IQ	Installation Qualification
JHA	Job Hazard Analysis
MKT	Mean Kinetic Temperature
MU	Measurement Uncertainty
NTC	Negative Temperature Coefficient
OOS	Out of Specification
OQ	Operational Qualification
OV	Operational Verification
PLC	Programmable Logic Controller
PM	Preventive Maintenance
PPE	Personal Protection Equipment
PQ	Performance Qualification
PTC	Positive Temperature Coefficient
PVC	Polyvinyl Chloride
RH	Relative Humidity
RTD	Resistance Temperature Detector
SAT	Site Acceptance Testing
SCADA	Supervisory Control And Data Acquisition
SME	Subject Matter Expert
SOP	Standard Operating Procedure
SRA	System Risk Assessment
TM	Traceability Matrix
TMC	Technology & Maintenance Council (US)
ULD	Unit Load Device
UPS	Uninterruptible Power Supply
URS	User Requirements Specification
USP	United States Pharmacopeia

WHO	World Health Organization
WIP	Work in Progress

19.2 Definitions

Commissioning (*ISPE Baseline Guide: C&Q (Second Edition)* [1])

A well planned, documented, and managed engineering approach to the start-up and turnover of facilities, systems, and equipment to the end user, that results in a safe and functional environment that meets established design requirements and stakeholder expectations.

Design Authority

Architect, engineering company, specialist supplier

Mapping Probes

Mapping probes are located in the storage locations to confirm that all locations remain in the defined limits during operation.

Monitoring Probes

Monitoring probes are the independent means of verifying that the system remains operating within the qualified range.

Qualification (*ISPE Baseline Guide: C&Q (Second Edition)* [1])

A process to demonstrate and document that the critical manufacturing facilities, systems, utilities, and equipment are suitable for the intended purpose.

Peltier Effect [45]

The production or absorption of heat at the junction of two metals on the passage of a current.

Rounding of Temperature Values (USP [5])

When rounding is required, consider only one digit in the decimal place to the right of the last place in the limit expression. If this digit is smaller than 5, it is eliminated and the preceding digit is unchanged. If this digit is equal to or greater than 5, it is eliminated and the preceding digit is increased by 1.

Temperature Mapping

Temperature mapping is the process of measuring and documenting temperatures at multiple predefined locations within the storage locations over a specified study duration to verify the capability of a controlled temperature chamber to maintain predetermined temperature specifications.



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